

Endpoint crossings for random walks with memory

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Abstract

Kagey asks when the middle bin of a Galton board is as likely as its ends if the ball tends to keep its direction. For a Markov walk that switches with probability q , a companion paper shows that the crossing satisfies $nq_n \approx \log n$. Here we ask the same question for 2 kinds of longer memory. In the elephant random walk each step copies a uniformly chosen earlier step and flips it with probability ε . We prove that the crossing is unique, that $x_n = n\varepsilon_n$ satisfies $x_n + \log x_n = 2 \log n - \log 8 + O((\log n)^2/n)$, and that x_n is twice the Markov value minus $\log(2\pi)$ up to $O(1/\log n)$. For large n the law at the crossing is not unimodal, and bin $n - m$ has probability close to $x_n/(2m^2)$ when $x_n \log^2 x_n \ll m \ll n$. For alternating runs with $\mathbb{P}(L \geq k) = k^{-\alpha}$, $1 < \alpha < 2$, and a stationary start, the ratio of the end to the middle is of order $n^{1-\alpha+1/\alpha}$. So for large n the ends win exactly when α is below the golden ratio. This threshold was known as a heuristic for Lévy walks. A remark and an appendix treat aging walks.

1 Introduction

Problem 131 of Kagey's Open Problem Collection [1] asks about a Galton board where the ball tends to keep going the same way. In particular it asks when the middle bin and the ends are equally likely. We count n steps, so the bins are Kagey's row $n + 1$, and we write A_n for the number of right steps. Unless we say otherwise, the first step is left or right with probability $\frac{1}{2}$ each. So the law of A_n is symmetric about $n/2$. For even n put $h = n/2$, $C_n = \mathbb{P}(A_n = h)$ and $E_n = \mathbb{P}(A_n = n) = \mathbb{P}(A_n = 0)$. Our companion paper [2], which we call Paper A, treats the Markov walk. This walk repeats its last step with probability $p = 1 - q$. Paper A shows that there is exactly one p_n with $C_n = E_n$ and that

$$n(1 - p_n) = \log n - \frac{1}{2} \log \log n + \frac{1}{2} \log \frac{\pi}{8} + o(1).$$

So a Markov walk needs about $\log n$ switches before the middle catches up with the ends. We write $q_n = 1 - p_n$ and $z_n = nq_n$. A second companion paper [3], Paper B, asks the same question for a Markov chain with more than 2 directions.

The Markov walk remembers only its last step. We compare it with 2 walks that have a longer memory. The elephant random walk of Schütz and Trimper [4] remembers its whole past. Each step copies a uniformly chosen earlier step and flips it with probability ε . In the second walk the steps come in runs of alternating direction, and the runs after the first one have i.i.d. lengths L with $\mathbb{P}(L \geq k) = k^{-\alpha}$ for $k \geq 1$. Its memory is the time since the last switch. The first run is either a fresh run or, for $\alpha > 1$, the remaining part of a run in progress. We call these the fresh and the stationary start. A remark and Appendix D treat a third kind of memory, aging, where the walk switches at step k with probability c/k . Table 1 compares the answers.

Walk	Rule	Crossing	Center at the crossing	Shape at the crossing
Markov [2]	switch with probability q	$nq_n = \log n - \frac{1}{2} \log \log n + \frac{1}{2} \log \frac{\pi}{8} + o(1)$	order $(\log n)^{1/2}/n$	the middle and the ends are the largest bins, bins 1 and $n-1$ the smallest
Elephant (Sections 2, 3)	copy a uniform earlier step, flip it with probability ε	$x_n = 2 \log n - \log \log n - \log 16 + o(1)$, and $x_n = 2nq_n - \log(2\pi) + o(1)$	$\sim 4x_n/n^2 \sim 8 \log n/n^2$	each end below its neighbor, modes about $x_n \log x_n$ from the ends (for very large n , using one computed sign), the center below all interior bins outside a window (all of them if Conjecture 3.12 holds)
Heavy-tailed runs (Section 4)	alternating runs with $\mathbb{P}(L \geq k) = k^{-\alpha}$	$R_n = E_n/C_n \sim C(\alpha)n^{1-\alpha+1/\alpha}$ for $1 < \alpha < 2$ and a stationary start, so the ends win for large n exactly when $\alpha < \varphi = (1 + \sqrt{5})/2$. With a fresh start they win for $\alpha < 1$ and lose for $\alpha > 1$.	order $n^{-1/\alpha}$ for $1 < \alpha < 2$	a stable bulk of width $n^{1/\alpha}$ and 2 end atoms [10, 12] (the full profile is not studied here)

Table 1: The 3 walks near the point where the middle bin and the ends are equally likely. For the Markov and elephant walks, n is the number of steps and the crossing is a value of the switch or flip probability, with $x_n = n\varepsilon_n$ for the elephant walk. For the heavy-tailed walk the exponent α is fixed and we compare E_n with C_n as n grows.

Results. For the elephant walk, C_n/E_n is strictly increasing in ε , so the crossing ε_n exists and is unique (Proposition 2.4). The derivative of C_n at $\varepsilon = 0$ is exactly $4/(n+2)$ (Proposition 2.5). Theorem 2.6 gives explicit bounds for C_n . In particular $\log C_n = \log \frac{4\varepsilon}{n+2} + 2\varepsilon \log n + O(\varepsilon)$ when $\varepsilon(1 + \log n) \leq \frac{1}{100}$. Hence $x_n = n\varepsilon_n$ satisfies $x_n + \log x_n = 2 \log n - \log 8 + O((\log n)^2/n)$ (Theorem 2.7). The reason for the factor 2 is that in the elephant walk a single early flip already moves a large family of steps. So the middle is reached with probability of order ε/n , and the ends must fall to about n^{-2} before they lose, instead of about n^{-1} for the Markov walk. Since $E_n \approx \frac{1}{2}e^{-n\varepsilon}$, the crossing is near $2 \log n$ and not $\log n$. Using the third-order expansion of Paper A, Theorem 2.8 shows that $x_n - (2z_n - \log(2\pi)) = c_*/\log n + O((\log \log n)^2/(\log n)^2)$ with $c_* = \frac{1}{2} \log(2\pi) - \frac{1}{4}$. By Theorem 2.9 the second coefficient of C_n in ε divided by the first is $2 \log n + K_2 + o(1)$, where $K_2 = 2\gamma - 2 - \log 2$ and γ is Euler's constant.

At the crossing and for large n , each end is below its neighbor (Theorem 3.5 and Corollary 3.6), so the law is not unimodal. With $x = n\varepsilon$, the number of steps that differ from the first step is close to a compound Poisson law (Theorem 3.7), with a Landau limit near $x \log x$ on the scale x (Theorem 3.8). For very large n , and using one numerically computed sign, the same theorem puts the modes near $x \log x - 0.2228x$. We also show $\mathbb{P}(A_n = n - m) \sim x/(2m^2)$ for $x \log^2 x \ll m \ll n$ (Theorem 3.9). At the crossing and for large n , the center is below every interior bin outside a window of width of order $n^{3/4}(\log n)^{3/4}$ (Theorem 3.11). Conjecture 3.12 would close this window, and we checked it for every $10 \leq n \leq 600$ and for 5 larger n .

For heavy-tailed runs with $1 < \alpha < 2$ and any first run, Theorem 4.4 gives C_n to leading order, and it is of order $n^{-1/\alpha}$. With a stationary start the end is of order $n^{1-\alpha}$, so $R_n =$

$E_n/C_n \sim C(\alpha)n^{1-\alpha+1/\alpha}$ (Corollary 4.5). The reason is that the end needs one first run that covers all n steps, while the center needs the renewal processes of $+$ and $-$ runs to meet, and these fluctuate on the scale $n^{1/\alpha}$. The exponent vanishes at the golden ratio $\varphi = (1 + \sqrt{5})/2$, and there $R_n \rightarrow C(\varphi) = 0.7761317206\dots$, so the center still wins. Theorems 4.6 and 4.7 treat $\alpha \geq 2$ and $0 < \alpha < 1$. Remark 4.9 gives computed crossover sizes for α just below φ , and Remark 4.10 discusses $\alpha = 5/3$, the value used for anomalous heat conduction. Remark 4.11 treats aging. The crossing is exactly $c = 1$ when the first step is fixed, and for a fair first step it is $c_n = 1 - \log 2/(\log n + \gamma + 2 - 2\log 2) + O((\log n)^{-3})$.

Prior work. The steps of the elephant walk split into clusters, since a step either copies an earlier step exactly or starts a new cluster with a fair sign. This is Kürsten's coupling with bond percolation on random recursive trees [5], and Baur and Bertoin [6] relate the walk to Pólya-type urns. Our regime $n\varepsilon \asymp \log n$ lies in the strongly supercritical percolation regime of Baur [7] (see also Bertoin [8]). Those results describe cluster sizes and do not cover the center probability, which is a large deviation of the root cluster. Guérin, Laulin, Raschel and Simon [9, proof of Theorem 1.1] prove that the law with a fixed first step is unimodal for every n and every memory parameter. Their other results fix the memory parameter as n grows. We did not find an asymptotic formula for the center probability in the literature when $\varepsilon \rightarrow 0$ as n grows.

For heavy-tailed runs, Godrèche and Luck [10] give the transform of the occupation time of an alternating renewal process and the scalings of the center and of both edges. They also give Lamperti's law and its critical value α_c , which they credit to Baldassarri, Bouchaud, Dornic and Godrèche. Zaburdaev, Denisov and Klafter [12] and Dhar, Saito and Derrida [13] describe the profile of a Lévy walk. Wang, Li and Hänggi [14] already observe that equal decay of the central peak and the side peaks gives the golden ratio, as a heuristic for the continuum model of heat conduction. Here we prove the lattice statement with a local limit at the center, keep the 2 starts apart, and compute $C(\alpha)$ and the crossover sizes. For aging, Dietz and Sethuraman [23] proved the limit law of A_n/n , Engländer and Volkov [24] gave a second proof, and Engländer, Volkov and Wang [25] found the scaling limit of the walk (see Remark 4.11 and Appendix D).

Organization. Sections 2 and 3 treat the crossing and the shape of the elephant walk. Section 4 treats heavy-tailed runs and ends with the remark on aging. Section 5 describes the computations and lists open questions. Appendices A, B and C hold the longer proofs for Sections 2 to 4, and Appendix D proves the aging crossing.

2 The elephant walk and its crossing

In this section we reduce the elephant walk to a chain with a fixed first step, prove that the crossing is unique, and find where it is. Throughout, n is even whenever C_n appears, and $h = n/2$.

Definition 2.1 (elephant walk). Fix $0 < \varepsilon < 1$. Let X_1 be uniform on $\{\pm 1\}$. For $t \geq 1$ let U_t be uniform on $\{1, \dots, t\}$, and put $X_{t+1} = X_{U_t}$ with probability $1 - \varepsilon$ and $X_{t+1} = -X_{U_t}$ with probability ε . All these choices are independent. We call the event $X_{t+1} = -X_{U_t}$ a flip. As before, A_n is the number of $i \leq n$ with $X_i = +1$, $C_n = \mathbb{P}(A_n = h)$ and $E_n = \mathbb{P}(A_n = n)$. The crossing ε_n is the value of ε with $C_n = E_n$, and $x_n = n\varepsilon_n$.

In the notation of [4] the memory parameter is $p = 1 - \varepsilon$, and [9] write $a = 2p - 1 = 1 - 2\varepsilon$. We write $\mathbb{P}_+$ and $\mathbb{E}_+$ for the law and expectation given $X_1 = +1$. Let M_t be the number of $i \leq t$ with $X_i \neq X_1$, and put $r_t(b) = \varepsilon + (1 - 2\varepsilon)b/t$ and $f_n(m) = \mathbb{P}_+(M_n = m)$.

Lemma 2.2. *The process $(M_t)_{t \geq 1}$ is a Markov chain with $M_1 = 0$ and $\mathbb{P}(M_{t+1} = b+1 \mid M_t = b) = r_t(b) = 1 - \mathbb{P}(M_{t+1} = b \mid M_t = b)$, and it is independent of X_1 . For $0 \leq k \leq n$, $\mathbb{P}(A_n = k) = \frac{1}{2}[f_n(n-k) + f_n(k)]$. In particular $C_n = f_n(h)$, $E_n = \frac{1}{2}(1 - \varepsilon)^{n-1}$, and the law of A_n is symmetric about h .*

Proof. Given $X_1, \dots, X_t$, the copied step X_{U_t} differs from X_1 with probability M_t/t . So $X_{t+1} \neq X_1$ with conditional probability $(1 - \varepsilon)M_t/t + \varepsilon(1 - M_t/t) = r_t(M_t)$. This depends only on M_t and not on X_1 , which proves the first claim. On $\{X_1 = +1\}$ we have $A_n = n - M_n$, and on $\{X_1 = -1\}$ we have $A_n = M_n$. Averaging over X_1 gives the formula for $\mathbb{P}(A_n = k)$, which does not change under $k \mapsto n - k$. Finally $M_n \leq n - 1$, so $f_n(n) = 0$, and $f_n(0) = \prod_{t=1}^{n-1} (1 - r_t(0)) = (1 - \varepsilon)^{n-1}$. $\square$

Remark 2.3. For $\varepsilon \leq \frac{1}{2}$ we can write $r_t(b) = (1 - 2\varepsilon)\frac{b}{t} + 2\varepsilon \cdot \frac{1}{2}$. So each new step copies a uniformly chosen earlier step exactly with probability $1 - 2\varepsilon$, and it is a fresh fair sign with probability 2ε . The steps then split into clusters, each made of a fresh sign and every step copied from it directly or through other copies, and the cluster signs are i.i.d. and fair. These are the clusters of Bernoulli bond percolation with retention parameter $1 - 2\varepsilon$ on a random recursive tree [5]. We do not use this remark in the proofs.

Proposition 2.4. *For every $n \geq 2$ and $1 \leq k \leq n - 1$, the ratio $f_n(k)/f_n(0)$ is strictly increasing in $\varepsilon \in (0, 1)$. Hence $\mathbb{P}(A_n = k)/E_n$ is strictly increasing in ε for every $1 \leq k \leq n - 1$. In particular C_n/E_n is strictly increasing, the crossing ε_n exists and is unique, and $\varepsilon_n < \frac{1}{2}$. In fact $C_n \geq 2E_n$ for $\varepsilon \geq \frac{1}{2}$.*

Divide the probability of every path of M by the probability of the path that never flips. Then each step contributes a factor that only grows with ε .

Proof. The probability of a path of M is a product of $n - 1$ factors, namely $r_s(b_s)$ for an up step at time s and $1 - r_s(b_s)$ otherwise, where b_s is the state at time s . We divide it by $f_n(0) = (1 - \varepsilon)^{n-1}$. With $y = b_s/s \in [0, 1)$ the factors become

$$\frac{r_s(b_s)}{1 - \varepsilon} = \frac{\varepsilon + (1 - 2\varepsilon)y}{1 - \varepsilon} \quad \text{and} \quad \frac{1 - r_s(b_s)}{1 - \varepsilon} = 1 - y \frac{1 - 2\varepsilon}{1 - \varepsilon},$$

with derivatives $(1 - y)/(1 - \varepsilon)^2 > 0$ and $y/(1 - \varepsilon)^2 \geq 0$ in ε . Both factors are positive, since $r_s(b_s) = \varepsilon(1 - y) + (1 - \varepsilon)y$ lies strictly between 0 and 1. So the ratio for each path is nondecreasing in ε , and it is strictly increasing if the path has an up step. Summing over the paths with $k \geq 1$ up steps proves the first claim. The second follows, since $\mathbb{P}(A_n = k)/E_n = [f_n(n - k) + f_n(k)]/f_n(0)$ by Lemma 2.2. As $\varepsilon \rightarrow 0$ we have $C_n \rightarrow 0$ and $E_n \rightarrow \frac{1}{2}$. For $\varepsilon \geq \frac{1}{2}$ every $r_s(b)$ lies in $[1 - \varepsilon, \varepsilon]$, so each path of M that ends at h has probability at least $(1 - \varepsilon)^{n-1} = 2E_n$. Hence $C_n \geq 2E_n$. The ratio C_n/E_n is continuous and strictly increasing, so it equals 1 at exactly one ε , and this ε lies in $(0, \frac{1}{2})$. $\square$

Next we look at the first flip. For $1 \leq s \leq n$ let $V_s(b) = \mathbb{P}_+(M_n = h \mid M_s = b)$. For $1 \leq t \leq h$ let

$$\pi_t = \frac{t \prod_{i=1}^{t-1} (h - i)}{\prod_{i=1}^t (n - i)}, \quad c_1 = \sum_{t=1}^h \pi_t, \quad w_t = \frac{\pi_t}{c_1},$$

so that $(w_t)_{1 \leq t \leq h}$ is a probability vector.

Proposition 2.5. *Let $n \geq 2$ be even.*

$$(a) \ C_n = \sum_{t=1}^h \varepsilon(1-\varepsilon)^{t-1} V_{t+1}(1).$$

(b) C_n is a polynomial in ε with $C_n(0) = 0$. At $\varepsilon = 0$ we have $V_{t+1}(1) = \pi_t$, and $C'_n(0) = c_1 = \frac{4}{n+2}$ exactly.

(c) For each fixed t , $w_t \rightarrow t2^{-t-1}$ as $n \rightarrow \infty$. For every even n , $\sum_t w_t t = 3 - \frac{6}{h+2}$ and $\sum_t w_t(t+1)^3 \leq 124$.

Proof. (a) Under $\mathbb{P}_+$ the first flip happens at step $t+1$ exactly when $M_s = 0$ for $s \leq t$ and $M_{t+1} = 1$. Since $r_s(0) = \varepsilon$, this has probability $(1-\varepsilon)^{t-1}\varepsilon$, and by the Markov property the chain then ends at h with probability $V_{t+1}(1)$. At time $t+1$ there are t steps equal to X_1 , and this number never decreases. So $V_{t+1}(1) = 0$ for $t > h$. Paths with no flip end at $0 \neq h$. Since $C_n = f_n(h)$, this proves (a).

(b) Every transition probability is affine in ε , so C_n and $V_{t+1}(1)$ are polynomials in ε . By (a), $C_n(0) = 0$ and $C'_n(0) = \sum_t V_{t+1}(1)|_{\varepsilon=0}$. At $\varepsilon = 0$ we have $r_s(b) = b/s$, so M is Pólya's urn. At time $t+1$ it holds t balls of one color and 1 of the other, and it makes $n-t-1$ more draws. By the Beta-binomial law of Pólya's urn, $h-1$ of these draws are of the minority color with probability $\binom{n-t-1}{h-1} B(h, h)/B(1, t) = t \binom{n-t-1}{h-1} (h-1)!^2 / (n-1)!$, which equals π_t . The Chu–Vandermonde identity $\sum_t \binom{t}{1} \binom{n-1-t}{h-1} = \binom{n}{h+1}$ then gives $c_1 = \binom{n}{h+1} (h-1)!^2 / (n-1)! = n/(h(h+1)) = 4/(n+2)$.

(c) Write $\pi_t = \frac{t}{n-t} \prod_{i=1}^{t-1} \frac{h-i}{n-i}$. For fixed t each fraction $\frac{h-i}{n-i}$ tends to $\frac{1}{2}$, so $n\pi_t \rightarrow t2^{1-t}$. Since $nc_1 \rightarrow 4$, we get $w_t \rightarrow t2^{-t-1}$. For the moments, w_t is proportional to $\binom{t}{1} \binom{n-1-t}{h-1}$, and Chu–Vandermonde gives $\sum_t \binom{t}{j} \binom{n-1-t}{h-1} = \binom{n}{h+j}$. Since $t^2 = \binom{t}{1} + 2\binom{t}{2}$, we get $\sum_t w_t t = 1 + 2\binom{n}{h+2}/\binom{n}{h+1} = 1 + \frac{2(h-1)}{h+2} = 3 - \frac{6}{h+2}$. Similarly $t(t+1)^3 = 8\binom{t}{1} + 38\binom{t}{2} + 54\binom{t}{3} + 24\binom{t}{4}$, and $\binom{n}{h+j} \leq \binom{n}{h+1}$ for $j \geq 1$. So $\sum_t w_t(t+1)^3 \leq 8 + 38 + 54 + 24 = 124$. $\square$

To first order in ε the walk flips once, and after that it is Pólya's urn. If the flip comes at step $t+1$, the urn starts with t balls of one color and 1 of the other, and π_t is the chance that it ends with h of each. For large n the minority fraction is close to a Beta(1, t) variable, whose density at $\frac{1}{2}$ is $t2^{1-t}$. Summing $t2^{1-t}$ over t gives the 4 in $c_1 \approx 4/n$.

To go beyond first order we follow the path after the first flip. For $1 \leq t \leq h$ we write $\mathbb{E}_t^{\text{br}}$ for the expectation over paths from $M_{t+1} = 1$ to $M_n = h$ whose $h-1$ up steps are placed uniformly at random among the $n-t-1$ remaining steps. This is Pólya's urn from state $(t+1, 1)$ conditioned to end at h (Lemma A.1). For $1 \leq b \leq u-1$ let

$$\theta_u(b) = \frac{n(u-2b)^2}{2(n-u)b(u-b)}, \quad A_t = \sum_{u=t+1}^{n-1} \mathbb{E}_t^{\text{br}} \theta_u(M_u).$$

The second-order bridge sum $B_t \geq 0$ is defined in Proposition A.5, and the lower bound J_n in (3). We also put $\kappa = 2\varepsilon/(1-4\varepsilon)$, $H_k = \sum_{j \leq k} 1/j$ and $\mathcal{G}(t) = (t+11)(\log \frac{(t+1)^2}{4t} + \frac{2}{3}) - H_t + \frac{2}{3t}$.

Theorem 2.6. *Let $n \geq 2$ be even.*

(a) For $0 < \varepsilon < \frac{1}{4}$,

$$C_n \leq U_n := \varepsilon c_1 e^{\kappa H_{n-1}} \sum_{t=1}^h w_t (1-\varepsilon)^{t-1} e^{\kappa \mathcal{G}(t)}.$$

(b) For $0 < \varepsilon \leq \frac{1}{2}$ and $1 \leq t \leq h$, $\log V_{t+1}(1) \geq \log \pi_t + \varepsilon A_t - \varepsilon^2 B_t$. Hence $C_n \geq J_n$.

(c) For even $n \geq 64$ and $0 < \varepsilon \leq \frac{1}{100(1+\log n)}$,

$$-11\varepsilon - \varepsilon^2(84 \log n + 1.05 \cdot 10^5) \leq \log C_n - \log \frac{4\varepsilon}{n+2} - 2\varepsilon \log n \leq 566\varepsilon.$$

In particular $|\log C_n - \log \frac{4\varepsilon}{n+2} - 2\varepsilon \log n| \leq 566\varepsilon$ there.

Proof. Parts (a) and (b) follow from Propositions A.4 and A.5 together with Proposition 2.5(a). Part (c) is proved in Appendix A. $\square$

The constants in (c) are crude. By Theorem 2.9, the coefficient of ε in $\log(C_n/(c_1\varepsilon))$ is $2 \log n + K_2 + o(1)$ with $K_2 \approx -1.54$. At the crossing the bounds (a) and (b) are close to C_n (Table 2). The exception is $U_{50}/C_{50} = 69.5$, since the upper bound needs $\kappa \log n$ to be small.

After the first flip the chain M at time s is Pólya's urn with βs extra balls of each color, where $\beta = \varepsilon/(1-2\varepsilon)$ (Lemma A.1). These extra balls pull the minority fraction toward $\frac{1}{2}$, and along a path that ends at the center the pull adds about $2\varepsilon/s$ to the log-probability at time s . Summing over $s \leq n$ gives the factor $n^{2\varepsilon}$ in (c).

We now locate the crossing. By Theorem 2.6(c), $C_n \approx 4x/n^2$ when $\varepsilon = x/n$ and x is of order $\log n$. Also $E_n = \frac{1}{2}(1-x/n)^{n-1} \approx \frac{1}{2}e^{-x}$. Setting them equal gives $x + \log x \approx 2 \log n - \log 8$. So let $L_n = 2 \log n - \log 8$ and $G(x) = x + \log x$, and let g_n be the root of $G(g_n) = L_n$.

Theorem 2.7. *As $n \rightarrow \infty$ over even n , the following hold.*

- (a) $x_n + \log x_n = L_n - \eta_n$ with $\eta_n = \frac{2x_n \log n + x_n^2/2}{n} + O\left(\frac{\log n}{n}\right)$. So $0 < g_n - x_n = O((\log n)^2/n)$ for large n .
- (b) $x_n = g_n - \frac{2g_n \log n + g_n^2/2}{n(1+1/g_n)} + O\left(\frac{\log n}{n}\right)$.
- (c) $x_n = 2 \log n - \log \log n - \log 16 + O\left(\frac{\log \log n}{\log n}\right)$.

Proof. (a) Put $F(x) = \log C_n(x/n) - \log E_n(x/n)$ for $0 < x < n$. By Proposition 2.4, F is strictly increasing and x_n is its only zero. Let $n \geq 6 \cdot 10^4$ and $1 \leq x \leq 2L_n$. Then $x \leq 4 \log n$ and $400 \log n(1 + \log n) \leq n$, so $\varepsilon = x/n \leq \frac{1}{100(1+\log n)}$ and Theorem 2.6(c) applies. For $0 \leq y \leq \frac{1}{2}$ we have $-y - \frac{y^2}{2} - y^3 \leq \log(1-y) \leq -y - \frac{y^2}{2}$. We use this with $y = x/n$ in $E_n = \frac{1}{2}(1-y)^{n-1}$, together with $0 \leq \log(1 + \frac{2}{n}) \leq \frac{2}{n}$ and $x^2 \leq 2n$. Then

$$F(x) = G(x) - L_n + \frac{2x \log n - x + x^2/2}{n} + R_n(x), \quad |R_n(x)| \leq \frac{570x + 3}{n}.$$

Hence $|F(x) - G(x) + L_n| \leq \rho_n$ on $[1, 2L_n]$, where $\rho_n = (2L_n(2 \log n + L_n + 571) + 3)/n = O((\log n)^2/n)$. Since $G' \geq 1$ and $G(g_n) = L_n$, this gives $F(x) < 0$ for $1 \leq x < g_n - \rho_n$ and $F(x) > 0$ for $g_n + \rho_n < x \leq 2L_n$. As $L_n - \log L_n \leq g_n \leq L_n$, for large n both ranges are nonempty, and so $|x_n - g_n| \leq \rho_n$. In particular $x_n \rightarrow \infty$. Now $F(x_n) = 0$ gives $G(x_n) = L_n - \eta_n$ with $\eta_n = (2x_n \log n + x_n^2/2)/n - x_n/n + R_n(x_n)$, and the last 2 terms are at most $(571x_n + 3)/n = O(\log n/n)$ in absolute value. This is (a), except for the sign. Since $x_n \rightarrow \infty$, the term $2x_n \log n/n$ is larger than $(571x_n + 3)/n$ for large n . Then $\eta_n > 0$, so $G(x_n) < G(g_n)$ and $x_n < g_n$.

(b) By the mean value theorem, $x_n - g_n = -\eta_n/G'(\xi)$ for some ξ between x_n and g_n . Here $G'(\xi) = 1 + 1/\xi = 1 + 1/g_n + O(1/n)$, since $|\xi - g_n| = O((\log n)^2/n)$ and $g_n \geq \log n$ for large n .

n	x_n	g_n	$x_n - x_{2\text{nd}}$	$x_n/\log n$	J_n/C_n	U_n/C_n	$D(n)$
50	3.9404868665	4.288636	+0.3452	1.007	0.915	69.5	-0.02123
400	7.6292168962	7.843768	+0.0621	1.273	0.9905	1.770	-0.03319
1600	10.2405649147	10.340051	+0.0179	1.388	0.99868	1.188	+0.02921
3200	11.5468344808	11.610464	+0.0097	1.431	0.99954	1.0996	+0.04786
12800	14.1588452626	14.182921	+0.0028	1.497	0.99995	1.0289	+0.06416
51200	16.7784677365	16.786946	+0.0008	1.547	n/c	n/c	+0.06522

Table 2: Exact crossings of the elephant walk (to 10 decimal places), the root g_n of $G(g) = L_n$, the error of the second-order formula, the explicit bounds of Theorem 2.6 at the crossing, and the difference $D(n)$ of Theorem 2.8. Here $x_{2\text{nd}}$ is the formula of Theorem 2.7(b) without its error term. The tabulated J_n keeps only the terms $t \leq 60$ of (3), so it is still a lower bound for C_n . The entries marked n/c were not computed.

As $\eta_n = O((\log n)^2/n)$, we get $x_n - g_n = -\eta_n/(1 + 1/g_n) + O((\log n)^2/n^2)$. Replacing x_n by g_n in the main term of η_n changes it by $O((\log n)^3/n^2)$. This gives (b).

(c) Since $g_n = L_n - \log g_n$ and $L_n - \log L_n \leq g_n \leq L_n$, we have $\log g_n = \log L_n + O(\log L_n/L_n)$. Also $\log L_n = \log 2 + \log \log n + O(1/\log n)$. So $g_n = 2 \log n - \log \log n - \log 16 + O(\log \log n/\log n)$, and $x_n - g_n = O((\log n)^2/n)$ by (a). $\square$

The O -constants in (a) and (b) are explicit once $n \geq 6 \cdot 10^4$, but we have not turned them into an explicit n beyond which $g_n > x_n$. Table 2 gives exact crossings on the grid $n = 50 \cdot 2^k$, computed from the recursion for f_n in Lemma 2.2 (Section 5). The second-order formula (b) is already within 0.07 of x_n at $n = 400$. On the other hand $x_n/\log n$ is only 1.55 at $n = 51200$, so the limit 2 in (c) is approached slowly.

Now we compare with the Markov walk. Paper A [2] shows that $z = z_n$ solves $z + \frac{1}{2} \log z = \log n + \frac{1}{2} \log \frac{\pi}{8} + o(1)$. Doubling gives $2z + \log(2z) = 2 \log n + \log \frac{\pi}{4} + o(1)$, which has the same form as the equation for x_n in Theorem 2.7. The right sides differ by $\log(2\pi)$.

Theorem 2.8. *Let p_n be the central crossing of the Markov walk with n steps in [2], and let $z_n = n(1 - p_n)$. As $n \rightarrow \infty$ over even n ,*

$$D(n) := x_n - (2z_n - \log(2\pi)) = \frac{c_*}{\log n} + O\left(\frac{(\log \log n)^2}{(\log n)^2}\right), \quad c_* = \frac{1}{2} \log(2\pi) - \frac{1}{4} = 0.668939 \dots$$

Proof. Write $L = \log n$ and $c_0 = \frac{1}{2} \log \frac{\pi}{8}$. The third-order expansion in the uniform Bessel approximation theorem of [2] says that

$$z_n = L + d + \frac{e}{L} + O\left(\frac{(\log L)^2}{L^2}\right), \quad d = -\frac{1}{2} \log L + c_0, \quad e = \frac{1}{4} \log L - \frac{1}{2} c_0 + \frac{1}{8}.$$

Then $\log z_n = \log L + d/L + O((\log L)^2/L^2)$. Since $e + d/2 = \frac{1}{8}$, we get $z_n + \frac{1}{2} \log z_n = L + c_0 + \frac{1}{8L} + O((\log L)^2/L^2)$. Put $Y = 2z_n$ and $\ell = \log(2\pi)$. Since $2c_0 + \log 2 = \log \frac{\pi}{4}$, this gives $Y + \log Y = 2L + \log \frac{\pi}{4} + \frac{1}{4L} + O((\log L)^2/L^2)$. Also $Y = 2L - \log L + O(1)$, so $\log(1 - \ell/Y) = -\ell/(2L) + O(\log L/L^2)$. Using $\log \frac{\pi}{4} - \ell = -\log 8$, we get

$$G(Y - \ell) = Y + \log Y - \ell + \log\left(1 - \frac{\ell}{Y}\right) = L_n + \frac{1}{4L} - \frac{\ell}{2L} + O\left(\frac{(\log L)^2}{L^2}\right) = L_n - \frac{c_*}{L} + O\left(\frac{(\log L)^2}{L^2}\right).$$

By Theorem 2.7(a), $G(x_n) = L_n - \eta_n$ with $\eta_n = O(L^2/n)$. Both x_n and $Y - \ell$ are $2L + O(\log L)$, so $G' = 1 + O(1/L)$ between them. The mean value theorem then gives $x_n - (Y - \ell) =$

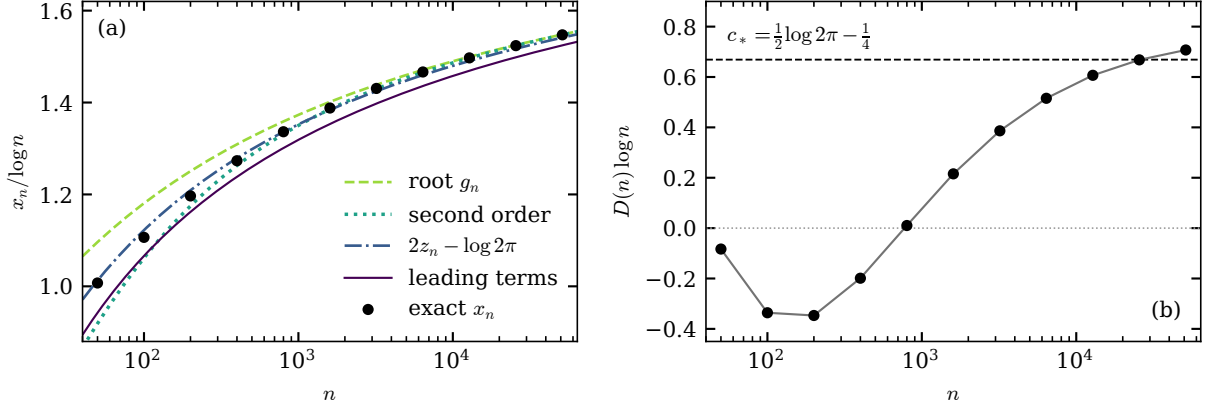


Figure 1: (a) The scaled elephant crossing $x_n / \log n$. The dots are the exact roots for $n = 50 \cdot 2^k$. The curves are the root g_n of $x + \log x = 2 \log n - \log 8$, the second-order formula of Theorem 2.7(b) without its error term, the Markov crossing through $2z_n - \log(2\pi)$, and the leading terms $2 \log n - \log \log n - \log 16$ of Theorem 2.7(c). (b) The difference $D(n) = x_n - (2z_n - \log(2\pi))$ of Theorem 2.8 times $\log n$. By that theorem it tends to c_* (dashed), with an error of order $(\log \log n)^2 / \log n$, so the approach is slow.

$c_*/L + O((\log L)^2/L^2) + O(L^2/n)$, and the last term is absorbed. Since $Y - \ell = 2z_n - \log(2\pi)$, this is the claim. $\square$

We computed z_n from the exact law of the Markov walk [2]. The difference $D(n)$ is negative up to $n = 400$ and changes sign near $n = 800$. At $n = 3200, 12800$ and 51200 it is $0.0479, 0.0642$ and 0.0652 , against $c_*/\log n = 0.083, 0.071$ and 0.062 . So the asymptotic regime is reached slowly (Figure 1).

Theorem 2.6(c) does not identify the $O(\varepsilon)$ term. For small ε we can find it exactly. Write $C_n = c_1\varepsilon + c_2\varepsilon^2 + O(\varepsilon^3)$ as $\varepsilon \rightarrow 0$ (Proposition 2.5(b)) and put $k_2(n) = c_2/c_1$. As in Section 1, γ is Euler's constant.

Theorem 2.9. (a) For every even $n \geq 2$, $k_2(n) = \sum_{t=1}^h w_t [A_t - (t-1)]$.

(b) For each fixed $t \geq 1$, as $n \rightarrow \infty$ over even n , $A_t - 2 \log n \rightarrow A_t^\infty := 2(\gamma - 1 - \log 2) + (t-2)H_{t-1}$.

(c) As $n \rightarrow \infty$ over even n ,

$$k_2(n) - 2 \log n \rightarrow K_2 := \sum_{t \geq 1} t 2^{-t-1} [A_t^\infty - (t-1)] = 2\gamma - 2 - \log 2 = -1.5387158508 \dots$$

Proof. Part (a) follows from Proposition 2.5(a) by differentiating the bridge formula of Proposition A.5 at $\varepsilon = 0$. Parts (b) and (c) are proved in Appendix A. $\square$

From the Taylor coefficients of C_n we get $k_2(n) - 2 \log n = -1.585878, -1.546572$ and -1.541005 at $n = 1600, 12800$ and 51200 . So, informally, $C_n \approx \frac{4\varepsilon}{n+2} n^{2\varepsilon} e^{K_2\varepsilon}$ when $\varepsilon \log n$ is small.

Remark 2.10. Let $\hat{x}_n$ solve

$$\log \frac{4x}{n(n+2)} + \frac{x}{n} k_2(n) = \log \frac{1}{2} + (n-1) \log \left(1 - \frac{x}{n}\right).$$

The lower bound behind Theorem 2.6 gives $x_n \leq \hat{x}_n + O((\log n)^3/n^2)$, and we prove this at the end of Appendix A. A matching upper bound would need $\log C_n \leq \log(\varepsilon c_1) + \varepsilon k_2(n) + O(\varepsilon^2(\log n)^k)$ for some k , which we do not have. The rest of this remark is numerical only. Let $k_3(n)$ be the coefficient of ε^2 in $\log(C_n/(c_1\varepsilon))$, computed from the Taylor coefficients of C_n . It lies between -8.8 and -8.7 for $12800 \leq n \leq 51200$. If we add $\frac{x_n^2}{n^2}k_3(n)$ to the left side of the equation, the root differs from x_n by $5.3 \cdot 10^{-7}$ at $n = 3200$ and by $3.6 \cdot 10^{-10}$ at $n = 51200$. We predicted x_{51200} in this way before computing it.

3 The elephant walk near its crossing

We now look at the whole law of A_n near the crossing. Unless stated otherwise, $n \geq 64$, $0 < \varepsilon \leq \frac{1}{4}$ and $x = n\varepsilon \geq 1$. Recall from Lemma 2.2 that $\mathbb{P}(A_n = n - m) = \frac{1}{2}[f_n(m) + f_n(n - m)]$ for $0 \leq m \leq n$. Some numerical statements below also use odd n , and there x_n is the value of x at which the 2 central bins are as likely as the ends (it is unique by Proposition 2.4). For a mode $a^* \geq n/2$ of the law of A_n we put $m^* = n - a^*$, and m_h is the largest mode of f_n . Figure 2 shows the law at the crossing for $n = 3200$.

3.1 Families and the first flip

For $k_0 \geq 1$ let $(M_k^{[k_0]})_{k \geq k_0}$ be the Markov chain with $M_{k_0}^{[k_0]} = 0$ and the transition probabilities $r_k(b)$ of M . So $M^{[1]}$ has the law of M . We write $Y_d = M_d^{[1]}$, and $Z_t^{(j)} = M_t^{[j-1]}$ for $j \geq 2$ and $t \geq j - 1$. In a random recursive tree on $[n] = \{1, \dots, n\}$, vertex $k + 1$ is joined to a uniform vertex of $[k]$, independently for $k = 1, \dots, n - 1$. We write T_v for the subtree of v , that is, v and its descendants. We also call T_v the family of v . A single flip at v turns every step of its family.

Lemma 3.1 (tree and marks). *Let T be the tree on $[n]$ in which the parent of $k + 1$ is U_k . For $2 \leq v \leq n$ let $\mu_v = 1$ if step v flips the step it copies, and $\mu_v = 0$ otherwise. Then T is a random recursive tree, the marks μ_v are i.i.d. Bernoulli(ε) and independent of T , and $X_v = X_1(-1)^{\sigma(v)}$, where $\sigma(v)$ is the number of marks on the path from v to the root, the root excluded. So $M_n = \#\{v : \sigma(v) \text{ odd}\}$. If the marks on $2, \dots, k_0$ are set to 0, then the number of odd vertices in $[k]$, $k \geq k_0$, is a copy of the chain $M^{[k_0]}$.*

The tree picture is due to Kürsten [5], who uses it with bond percolation on the tree.

Lemma 3.2 (subtree sizes). *Let T be a random recursive tree on $[n]$ and let $2 \leq v \leq n$. Then*

$$\beta_v(d) := \mathbb{P}(|T_v| = d) = \binom{n-d-1}{v-2} / \binom{n-1}{v-1}, \quad 1 \leq d \leq n - v + 1,$$

and $\beta_v(d)$ is nonincreasing in d . For $1 \leq d \leq n - 1$,

$$\sum_{v=2}^n \beta_v(d) = \frac{n}{d(d+1)}, \quad \sum_{v=2}^n (v-2) \beta_v(d) = \frac{2n(n-d-1)}{d(d+1)(d+2)}.$$

So the expected number of subtrees with exactly d vertices is $n/(d(d+1))$, and a single flip at the root of such a subtree turns all d of its steps. Hence the part of $f_n(m)$ of first order in ε is exactly $x/(m(m+1))$, for every $1 \leq m \leq n - 1$. Indeed, 2 or more marks have probability $O(\varepsilon^2)$, and if v is the only marked vertex then the odd vertices are those of T_v . So $f_n(m) = \varepsilon \sum_v \beta_v(m) + O(\varepsilon^2)$ at fixed n . The higher-order terms are not small when m is close to n .

Lemma 3.3 (the first flip). *Let $V = \min\{v \geq 2 : \mu_v = 1\}$, with $V = \infty$ if there is no mark, and let $D = |T_V|$. Then $\mathbb{P}(V = j, D = d) = \varepsilon(1 - \varepsilon)^{j-2}\beta_j(d)$, and for $m \geq 1$*

$$f_n(m) = \sum_{j=2}^n \varepsilon(1 - \varepsilon)^{j-2} \sum_{d=1}^{n-j+1} \beta_j(d) \mathbb{P}(d - Y_d + Z_{n-d}^{(j)} = m),$$

where Y_d and $Z_{n-d}^{(j)}$ are independent.

In words, the first flip turns its whole family except the steps of that family that flip back, and the rest of the tree is a walk whose first $j - 1$ steps agree with X_1 .

Lemma 3.4 (unimodality given the first step). *For every $k_0 \geq 1$ and $k \geq k_0$ the law of $M_k^{[k_0]}$ is unimodal. In particular f_n is unimodal.*

For $k_0 = 1$ this is shown in the proof of Theorem 1.1 of Guérin, Laulin, Raschel and Simon [9], for every n and every memory parameter. Lemmas 3.1 to 3.4 are proved in Appendix B, where we also repeat their short induction in our notation.

3.2 The end and its neighbor

Bin $n - 1$ needs a single flip that is never copied, and a flip at step j is never copied with probability about j/n . Summing $\varepsilon j/n$ over j gives about $x/2$. So the end should fall below its neighbor when x passes 2.

Theorem 3.5 (endpoint dip). *Let $n \geq 3$ and $0 < \varepsilon \leq \frac{1}{2}$, and put $\rho_1 = \mathbb{P}(A_n = n - 1)/E_n$. Then*

$$\frac{x}{2(1 - \varepsilon)} < \rho_1 \leq \frac{x(1 + 1/n)^2}{2 - 3\varepsilon} + \frac{\varepsilon}{1 - \varepsilon}.$$

Hence the end is strictly below its neighbor whenever $x \geq 2(1 - \varepsilon)$, and strictly above it whenever the right side is below 1.

Proof. By Lemma 2.2, $\rho_1 = (f_n(1) + f_n(n - 1))/f_n(0)$ with $f_n(0) = (1 - \varepsilon)^{n-1}$. On the event $M_n = 1$ the chain first moves up from time $j - 1$ to time j , for some $2 \leq j \leq n$, and then stays at 1. Since $r_t(0) = \varepsilon$ and $1 - r_k(1) = (1 - \varepsilon)(1 - c/k)$ with $c = (1 - 2\varepsilon)/(1 - \varepsilon) \in [0, 1]$,

$$f_n(1) = \varepsilon(1 - \varepsilon)^{n-2} \sum_{j=2}^n P_j, \quad P_j = \prod_{k=j}^{n-1} \left(1 - \frac{c}{k}\right).$$

Since $c \leq 1$, $P_j \geq \prod_{k=j}^{n-1} (1 - 1/k) = (j - 1)/(n - 1)$, and these lower bounds sum to $n/2$. So $f_n(1)/f_n(0) \geq x/(2(1 - \varepsilon))$. Next, $1 - c/k \leq e^{-c/k}$ and $\sum_{k=j}^{n-1} 1/k \geq \log(n/j)$ give $P_j \leq (j/n)^c$. As $(t/n)^c$ is nondecreasing in t ,

$$\sum_{j=2}^n P_j \leq \int_2^{n+1} \left(\frac{t}{n}\right)^c dt \leq \frac{n(1 + 1/n)^{c+1}}{c + 1} \leq \frac{n(1 + 1/n)^2(1 - \varepsilon)}{2 - 3\varepsilon},$$

using $c \leq 1$ and $1/(c + 1) = (1 - \varepsilon)/(2 - 3\varepsilon)$. So $f_n(1)/f_n(0) \leq x(1 + 1/n)^2/(2 - 3\varepsilon)$. Finally, $M_n = n - 1$ needs an up move at every time. The first has probability $r_1(0) = \varepsilon$, and the one at time $k \geq 2$ has probability $r_k(k - 1) = 1 - \varepsilon - (1 - 2\varepsilon)/k \in [\frac{1}{2}, 1 - \varepsilon]$. So $0 < f_n(n - 1) \leq \varepsilon(1 - \varepsilon)^{n-2}$. This makes the lower bound strict and gives the term $\varepsilon/(1 - \varepsilon)$. $\square$

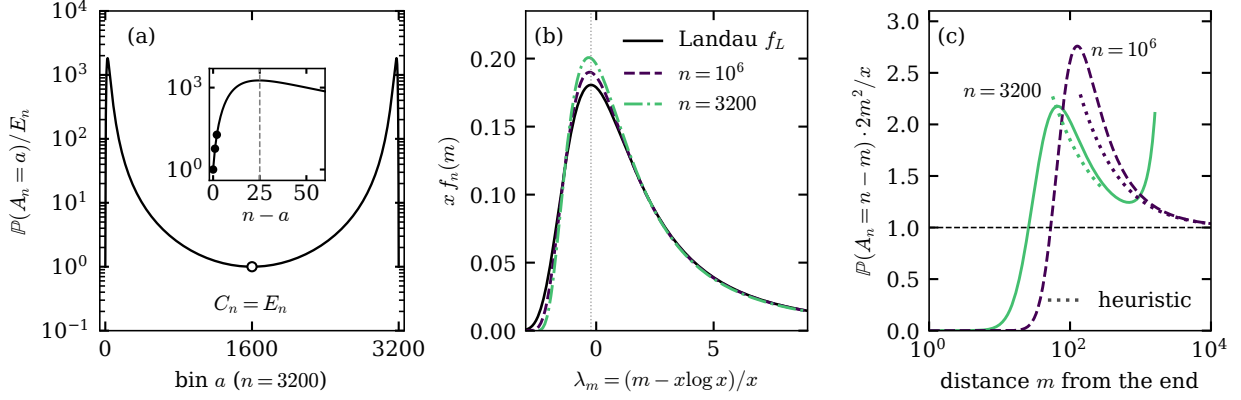


Figure 2: The elephant walk at its crossing. (a) The law of A_n at $n = 3200$ and its crossing $x = 11.5468$, divided by the end E_n . The center equals the end, and the law rises from each end to a mode and falls to the center. The inset shows the last 60 bins, with dots at bins n , $n - 1$ and $n - 2$. The end is below its neighbor, and the modes are 25 steps from each end (dashed). (b) The law of M_n , scaled as $x f_n(m)$, against $\lambda_m = (m - x \log x)/x$, with the Landau density f_L . The dotted line is its mode λ_0 . (c) The tail $\mathbb{P}(A_n = n - m)$ divided by $x/(2m^2)$. The dashed line is the limit 1 of Theorem 3.9. The rise of the $n = 3200$ curve near $m = n/2$ comes from the mirror term $f_n(n - m)$. The dotted curves are the heuristic correction $1 + x(2 \log m + 2\gamma - 3)/m$, drawn for $2x \log x \leq m \leq n/x$. At $n = 10^6$ we use $x = 22.4407$, the root g_n . The second-order formula of Theorem 2.7(b) puts the crossing at 22.4399.

At the crossing for $n = 3200$, where $x = 11.5468$, the bounds give $5.7943 < \rho_1 \leq 5.8121$, and the exact value is 5.8048.

Corollary 3.6 (the dip at the crossing). *Let $n \geq 4$ be even and suppose that $x_n \geq 2(1 - \varepsilon_n)$. Then at the crossing $E_n = C_n < \mathbb{P}(A_n = n - 1)$, and the law of A_n is not unimodal.*

Proof. By Proposition 2.4, $\varepsilon_n < \frac{1}{2}$, so Theorem 3.5 gives $\mathbb{P}(A_n = n - 1) > E_n = C_n$. Suppose the law is nondecreasing on $\{0, \dots, c\}$ and nonincreasing on $\{c, \dots, n\}$. It is symmetric about h , so we may replace c by $n - c$ and assume $c \leq h$. Then $\mathbb{P}(A_n = a) \leq C_n$ for every $a \geq h$, including $a = n - 1$, which is a contradiction. $\square$

The hypothesis $x_n \geq 2(1 - \varepsilon_n)$ holds for every $8 \leq n \leq 600$, at every computed crossing on the grid $n = 50 \cdot 2^k \leq 51200$, and for all large n by Theorem 2.7. It fails for every $n \leq 7$. We also checked the dip directly at the crossing for every $4 \leq n \leq 600$ and for $n = 1600, 3200, \dots, 25600$. So for $4 \leq n \leq 7$ the dip is known only from this direct check. For $n = 2$ and 3 the neighbor is a central bin and ties with the end. Note that the law given the first step is unimodal by Lemma 3.4. So the law of A_n is a mixture of 2 unimodal laws with different modes, as is the limit law in [9, eq. (1)].

3.3 The bulk and the tail

Let CP_n be the law of $\sum_{d=1}^{n-1} dN_d$, where the N_d are independent Poisson variables with means $x/(d(d+1))$. Its generating function is $\exp\{x \sum_{d < n} (s^d - 1)/(d(d+1))\}$, and for $n = \infty$ it is $\exp\{x\chi(s)\}$ with $\chi(s) = (1-s) \log(1-s)/s$. Most minority steps come from flipped families that do not overlap. By Lemma 3.2 the expected number of flipped vertices whose subtree has size d is

exactly $x/(d(d+1))$, so the total size of the flipped families is close to a compound Poisson sum. Two flips in the same family are rare when $x^2 \log n/n$ is small.

Theorem 3.7 (compound Poisson bulk). *For $n \geq 2$ and $0 < \varepsilon \leq \frac{1}{4}$,*

$$\sup_{m \geq 0} |f_n(m) - \text{CP}_n(m)| \leq \delta_B := \frac{x^2 \log n}{n} + \frac{4x^2}{n} e^{4x^2/n} + \frac{4x^2(H_n^2 + H_n)}{n}.$$

The proof is in Appendix B. For $x^2 \leq n$ the bound is $O(x^2 \log^2 n/n)$. It is additive, so at the crossing it says nothing for $n \leq 10^6$ (δ_B is about 0.46 at $n = 10^6$). Numerically CP_n is much closer to f_n than this bound says. At $n = 10^6$ and $x = 22.4407$ the largest value of $|f_n(m)/\text{CP}_n(m) - 1|$ over $1 \leq m \leq 10^4$ is $3.5 \cdot 10^{-4}$. Near $m = n/2$ the relative error is larger. At $n = 1600$ and 3200 the ratio f_n/CP_n falls to 0.931 and 0.956 at $m = n/2$.

Let f_L be the Landau density, defined by $\int_{\mathbb{R}} e^{-u\lambda} f_L(\lambda) d\lambda = u^u$ for $u > 0$. Its mode is $\lambda_0 = -0.2227829813$. We put $\lambda_m = (m - x \log x)/x$. The Landau law appears because families of size d are flipped at rate about x/d^2 . The families of size up to about x add up to nearly their mean $x \log x$, and the few families of size of order x give fluctuations of order x . So the bulk of M_n sits near $x \log x$ and has width of order x .

Theorem 3.8 (Landau limit). *Let $n \geq 2$ and $x \geq 1$.*

(a) *For all integers $m \geq 0$,*

$$|x \text{CP}_n(m) - f_L(\lambda_m)| \leq E_L := e^{2x/n} \left[\frac{99.7 \log x + 517}{x} + \frac{4x}{n} \right] + \frac{2}{\pi^2} e^{-\pi^2 x/2}.$$

(b) *If also $\varepsilon \leq \frac{1}{4}$, then $\sup_m |x f_n(m) - f_L(\lambda_m)| \leq E_L + x \delta_B$. So $(M_n - x \log x)/x$ converges in law to the Landau law when $x \rightarrow \infty$ and $x^3 \log^2 n/n \rightarrow 0$, in particular at the crossing.*

(c) *Let $\varepsilon \leq \frac{1}{4}$, and let $\kappa_L > 0$ satisfy $f_L(\lambda_0) - f_L(\lambda) \geq \kappa_L \min\{1, (\lambda - \lambda_0)^2\}$ for all real λ . Put $E' = E_L + x \delta_B + 16x^2 H_n/n^2 + \|f_L''\|_{\infty}/(8x^2)$. If $E' < \kappa_L/2$ and $m_h \leq n/4$, then every mode m of f_n satisfies $|m - x \log x - \lambda_0 x| \leq x \sqrt{2E'/\kappa_L}$, and so does every m^* .*

The proof is in Appendix B. A constant κ_L as in (c) exists if $f_L''(\lambda_0) < 0$. Indeed, Taylor's theorem then gives $c, \delta > 0$ with $f_L(\lambda_0) - f_L(\lambda) \geq c(\lambda - \lambda_0)^2$ for $|\lambda - \lambda_0| \leq \delta$. The Landau law is stable, so f_L is unimodal [18], and the difference is at least $c\delta^2$ for $|\lambda - \lambda_0| \geq \delta$. So $\kappa_L = c \min\{1, \delta^2\}$ works. We computed $f_L''(\lambda_0) = -0.0789$ by quadrature with 25 digits, but we did not certify its sign, so the existence of κ_L rests on this computation. Given κ_L , part (c) gives $m^* = x \log x - 0.2228x + O(\sqrt{x \log x})$ at the crossing for all large n , but not effectively.

The best constant is numerically $\kappa_L \approx 0.0262$, and then $E' < \kappa_L/2$ needs $x \gtrsim 1.3 \cdot 10^5$, that is, $\log n \gtrsim 6.5 \cdot 10^4$ at the crossing. So at the sizes we can compute the evidence for the mode is numerical. At every tested crossing m^* equals the mode of CP_n . It is $m^* = 21, 25, 29, 33, 38$ at $n = 1600, 3200, 6400, 12800, 25600$, and $m^* = 47$ and 64 at $n = 10^5$ and 10^6 with $x = g_n$ in place of x_n . A heuristic expansion one order further, which we have not proved, gives $x \log x - 0.222783x - 0.535021 \log x + 0.549151 + o(1)$ for the mode of CP_{∞} . At $x = 10 \cdot 2^k$ for $0 \leq k \leq 10$ it differs from the computed mode, interpolated between integers, by at most 0.28. The difference is largest at $x = 10$ and shrinks as x grows.

Theorem 3.9 (the tail). *Let $x = n\varepsilon \geq 1$ depend on n with $x \log^2 n/n \rightarrow 0$, and let $\omega_n \rightarrow \infty$ with $\omega_n x \log^2(x+1) = o(n)$. There is $\eta_n \rightarrow 0$, depending only on n , x and ω_n , such that for all large n and all m with $\omega_n x \log^2(x+1) \leq m \leq n/2$,*

$$\frac{x}{2}(1 - \eta_n) \left[\frac{1}{m(m+1)} + \frac{1}{(n-m)(n-m+1)} \right] \leq \mathbb{P}(A_n = n-m) \leq \frac{x(1 + \eta_n)}{2m(m+1)} + \frac{1}{2}f_n(\lceil n/2 \rceil),$$

and $f_n(\lceil n/2 \rceil) \leq \frac{4x}{n^2}(1 + \eta_n)$. In particular $\mathbb{P}(A_n = n-m) = \frac{x}{2m^2}(1 + o(1))$ uniformly for $\omega_n x \log^2(x+1) \leq m \leq \delta'_n n$, for any $\delta'_n \rightarrow 0$. For $m = un$ with $u \in [u_0, \frac{1}{2}]$ and $u_0 > 0$ fixed,

$$\frac{x}{2n^2}(u^{-2} + (1-u)^{-2})(1 - o(1)) \leq \mathbb{P}(A_n = n-m) \leq \frac{x}{2n^2}(u^{-2} + 4)(1 + o(1)).$$

The proof is in Appendix B, from Theorems B.5, B.6 and B.8. The constant is $\frac{1}{2}$ because, for $m \ll n$, bin $n-m$ is reached through a family of size m only when $X_1 = +1$. For example, at $n = 10^6$, $x = 22.4407$ and $m = 10^4$ the exact ratio $\mathbb{P}(A_n = n-m)/(x/m^2)$ is 0.51889. The explicit bounds behind the theorem are weak at the sizes we can compute. At the crossing, the lower bound of Theorem B.5 says something only for $164 \leq m \leq 400$ at $n = 1600$ and for $118 \leq m \leq 2215$ at $n = 3200$. The upper bound of Theorem B.8 is useful at the center and the crossing only from $n \approx 1.1 \cdot 10^8$, where its factor is about 1.75. The tail is not proved for m between $x \log x$ and $x \log^2 x$, and for m of order n the upper bound on the mirror term $f_n(n-m)$ holds only up to a constant. Heuristically, CP_n suggests $f_n(m)m(m+1)/x \approx 1 + x(2 \log m + 2\gamma - 3)/m$ at moderate m . This factor is still about 2 near $m = 70$ when $n = 1600$, so there the tail looks like x/m^2 and not $x/(2m^2)$.

Remark 3.10 (the center value). For the center alone, Theorem 2.6(c) is sharper. It gives $C_n = \frac{4\varepsilon}{n+2} n^{2\varepsilon} e^{O(\varepsilon)}$, with explicit constants, whenever $\varepsilon(1 + \log n) \leq \frac{1}{100}$. Corollary B.9 gives only $C_n = \frac{4x}{n^2}(1 + O(\sqrt{x \log^2 n/n}))$, and this is the form that the proof of Theorem 3.11 uses.

3.4 The shape at the crossing

Theorem 3.11 (shape at the crossing). *Let n be even and let $x \in [2, n^{1/4}]$ satisfy $C_n = E_n$. There are absolute constants C_w and n_0 such that the following hold for $n \geq n_0$.*

- (a) $\mathbb{P}(A_n = n-1) > E_n$.
- (b) $\mathbb{P}(A_n = a) > C_n$ for every $1 \leq a \leq n-1$ with $|a-h| \geq C_w n^{3/4}(x \log^2 n)^{1/4}$.
- (c) If $E' < \kappa_L/2$ for a constant κ_L as in Theorem 3.8(c), then m^* satisfies the bound of Theorem 3.8(c).

Part (a) is Theorem 3.5, since $x \geq 2 > 2(1-\varepsilon)$. Part (c) is Theorem 3.8(c), since $m_h = O(x \log^2 x) \leq n/4$ for large n by Lemma B.7. Part (b) is proved in Appendix B. By Theorem 2.7 the crossing x_n lies in $[2, n^{1/4}]$ for all large n . We did not compute C_w or n_0 , and n_0 is at least of order 10^8 because the proof uses Corollary B.9. The window has this size because, at distance t from h , the main term of the law exceeds its value at h by a factor of about $1 + 12t^2/n^2$, while Corollary B.9 gives C_n only up to a relative error of order $(x \log^2 n/n)^{1/2}$. Inside the window the proof gives nothing, and closing it needs second differences of the law.

Conjecture 3.12 (convexity between the humps). *For every $n \geq 10$, at $x = x_n$, the second difference $\mathbb{P}(A_n = a-1) - 2\mathbb{P}(A_n = a) + \mathbb{P}(A_n = a+1)$ is strictly positive for $2m^* + 1 \leq a \leq n - 2m^* - 1$.*

With the symmetry of the law, the conjecture makes the law strictly convex on $[2m^*, n - 2m^*]$, so h is its unique minimizer there (the 2 central bins for odd n). For large even n the other bins are covered by Theorem 3.11(b), since $m^* = O(x \log^2 x)$. Indeed, at the distance ℓ_0 of Lemma B.7, which is $O(x \log x)$, the law is at least $x/(4\ell_0(\ell_0 + 1))$ by Theorem B.5 and Lemma B.7, and by Theorem 3.9 it is smaller than this at every distance $m \geq \omega_n x \log^2(x + 1)$, for any ω_n that tends to infinity slowly enough. So the conjecture and Theorem 3.11 together would show that the center is the unique interior minimum for all large even n .

The conjecture holds for every $10 \leq n \leq 600$ and for $n = 1600, 3200, 6400, 12800, 25600$. We predicted the cases $100 \leq n \leq 600$ and $n = 6400, 12800, 25600$ before computing them, and we checked $10 \leq n \leq 99$ and $n = 1600, 3200$ afterwards. The version on the closed range $[2m^*, n - 2m^*]$ is false for $n = 10, 11, 17$ to $22, 30$ to 35 and 48 to 51 , and in each of these cases only the 2 endpoints fail. It holds for every checked $n \geq 52$. Along $n = 1600, 3200, 6400, 12800, 25600$ the second difference is positive on $[a_1, n - a_1]$ with $a_1 = 33, 39, 45, 51, 57$, while $2m^* = 42, 50, 58, 66, 76$. So the margin $2m^* - a_1 = 9, 11, 13, 15, 19$ grows with n .

We can only explain the convexity heuristically. In our computations f_n itself is not convex on $[2m^*, n - 2m^*]$. It has a concave stretch where $n - m$ lies between $m_h + 1$ and about $0.68(xn^2)^{1/3}$. In the full law this stretch is added to the tail $x/(2r^2)$ of the other branch, with $r = n - a$, whose second difference $3x/r^4$ beats the concave curvature of order $x^2/(n^2r^3)$. So the convexity is a cancellation between the 2 branches.

At every crossing where we computed the whole law, the law of A_n has 2 humps. It rises from each end to a mode at distance m^* and falls from there to the center (to the 2 equal central bins for odd n). So for even n its first difference changes sign exactly 3 times. We observed this for every $10 \leq n \leq 600$ and on the grid $n = 50 \cdot 2^k \leq 25600$. Figure 2 shows it at $n = 3200$.

4 Heavy-tailed runs

In this section the walk remembers the time since its last switch. The steps form maximal runs whose directions alternate. The first run has length D , and the later runs have i.i.d. lengths $L_1, L_2, \dots$ distributed as L , where $\bar{F}(k) = \mathbb{P}(L \geq k) = k^{-\alpha}$ for $k \geq 1$ and some $\alpha > 0$. So $p_k = \mathbb{P}(L = k) = k^{-\alpha} - (k + 1)^{-\alpha} > 0$ for every $k \geq 1$. The first direction is fair, and $D, (L_i)$ and the first direction are independent. With a fresh start D has the law of L . For $\alpha > 1$ the stationary start is $\mathbb{P}(D = d) = \bar{F}(d)/\mu$, where $\mu = \mathbb{E}L = \zeta(\alpha)$. This law is stationary. The reason is that the number of steps left in the current run, counting the current one, is a Markov chain. It moves from k to $k - 1$ if $k \geq 2$ and from 1 to a fresh copy of L . Since $\bar{F}(k) = \bar{F}(k + 1) + p_k \bar{F}(1)$, the law $\bar{F}(k)/\mu$ is invariant for this chain. As in Section 1, A_n is the number of $+1$ steps, $C_n = \mathbb{P}(A_n = h)$ for even n , $E_n = \mathbb{P}(A_n = n)$, and we put $R_n = E_n/C_n$. The law of A_n is symmetric about $n/2$. We write $\mathbb{P}_+$ for the law given that the first run is $+$.

Proposition 4.1 (the edge). *For every law of D , $E_n = \frac{1}{2}\mathbb{P}(D \geq n)$. With a fresh start $E_n = \frac{1}{2}n^{-\alpha}$. With the stationary start,*

$$E_n = \frac{\zeta(\alpha, n)}{2\zeta(\alpha)} = \frac{n^{1-\alpha}}{2(\alpha - 1)\zeta(\alpha)} \left(1 + \theta_n \frac{\alpha - 1}{n}\right) \quad \text{for some } \theta_n \in [0, 1],$$

where $\zeta(\alpha, n) = \sum_{k \geq n} k^{-\alpha}$ is the Hurwitz zeta function.

Proof. All n steps are $+1$ exactly when the first direction is $+$ and $D \geq n$. Hence $E_n = \frac{1}{2}\mathbb{P}(D \geq n)$, and $\mathbb{P}(L \geq n) = n^{-\alpha}$. For the stationary start $\mathbb{P}(D \geq n) = \zeta(\alpha, n)/\zeta(\alpha)$. Since $x^{-\alpha}$ is decreasing, $\int_n^\infty x^{-\alpha} dx \leq \zeta(\alpha, n) \leq n^{-\alpha} + \int_n^\infty x^{-\alpha} dx$, and the integral equals $n^{1-\alpha}/(\alpha - 1)$. $\square$

So the edge only sees the first run, while the center depends on all the runs. Let $\tau_m = L_1 + \dots + L_m$ with $\tau_0 = 0$, $u_m(k) = \mathbb{P}(\tau_m = k)$ and $q_m(k) = \mathbb{P}(\tau_{m-1} < k \leq \tau_m) = \sum_{j \geq 1} \bar{F}(j) u_{m-1}(k-j)$ for $m, k \geq 1$. For the delayed versions let D be independent of (L_i) , and put $u_m^D(k) = \mathbb{P}(D + \tau_{m-1} = k)$, $q_1^D(k) = \mathbb{P}(D \geq k)$ and $q_m^D(k) = \mathbb{P}(D + \tau_{m-2} < k \leq D + \tau_{m-1})$ for $m \geq 2$. For the stationary start $u_m^D(k) = \sum_d \frac{\bar{F}(d)}{\mu} u_{m-1}(k-d) = q_m(k)/\mu$.

The $+$ runs and the $-$ runs form 2 independent renewal processes. The walk is in bin a at time n exactly when one of them sits at its level and the other is inside a run that covers its level.

Proposition 4.2 (two renewal processes). *Let D have any law on $\{1, 2, \dots\}$. For $a + b = n$ with $a, b \geq 1$,*

$$\mathbb{P}_+(A_n = a) = \sum_{m \geq 1} \left[u_m(b) q_{m+1}^D(a) + u_m^D(a) q_m(b) \right],$$

and $\mathbb{P}_+(A_n = n) = \mathbb{P}(D \geq n)$. In particular, for even n , $C_n = \sum_{m \geq 1} [u_m(h) q_{m+1}^D(h) + u_m^D(h) q_m(h)]$. For the fresh start this is $C_n = \sum_m u_m(h) [q_m(h) + q_{m+1}(h)]$, and for the stationary start $u_m^D = q_m/\mu$.

Proof. Let the $+$ runs have lengths $D, L_2^+, L_3^+, \dots$ and the $-$ runs $L_1^-, L_2^-, \dots$, all independent with $L_i^\pm$ distributed as L . Let $s_j^\pm$ be the sum of the first j lengths of each sign, with $s_0^\pm = 0$. So s_j^+ has law u_j^D and s_j^- has law u_j . The j -th $+$ run is the set of steps in $(s_{j-1}^+ + s_{j-1}^-, s_j^+ + s_{j-1}^-]$, and the j -th $-$ run is $(s_j^+ + s_{j-1}^-, s_j^+ + s_j^-]$. Step $n = a + b$ lies in exactly one run. If it lies in the first run then $A_n = n$, which happens exactly when $D \geq n$. If it lies in the j -th $+$ run with $j \geq 2$, then $A_n = n - s_{j-1}^-$, so $A_n = a$ means $s_{j-1}^- = b$, and then step n is in this run exactly when $s_{j-1}^+ < a \leq s_j^+$. By independence this case has probability $u_{j-1}(b) q_j^D(a)$. If step n lies in the j -th $-$ run, then $A_n = s_j^+$, so we need $s_j^+ = a$ and $s_{j-1}^- < b \leq s_j^-$, which has probability $u_j^D(a) q_j(b)$. Summing over the disjoint cases, with $m = j - 1$ in the first and $m = j$ in the second, gives the formula. For even n the symmetry gives $C_n = \frac{1}{2} [\mathbb{P}_+(A_n = h) + \mathbb{P}_+(A_n = n - h)] = \mathbb{P}_+(A_n = h)$. $\square$

This identity is not new. It is the coefficient form of the double transform of the occupation time of an alternating renewal process in Godrèche and Luck [10, Section 7]. Computing the u_m with the fast Fourier transform gives C_n in order $n^2 \log n$ operations, and we used this up to $n = 10^5$.

Lemma 4.3 (stable and local limits). *Let $0 < \alpha < 2$ with $\alpha \neq 1$, and put $a_m = m^{1/\alpha}$. For $1 < \alpha < 2$ put $K_\alpha = \Gamma(2 - \alpha)/(\alpha - 1) = -\Gamma(1 - \alpha) > 0$ and $\sigma_\alpha = 2K_\alpha |\cos \frac{\pi\alpha}{2}|$.*

- (i) *If $1 < \alpha < 2$, then $\mathbb{E}e^{i\theta L} = 1 + i\mu\theta + K_\alpha(-i\theta)^\alpha + O(\theta^2)$ as $\theta \rightarrow 0$, with the principal branch.*
- (ii) *If $1 < \alpha < 2$, then $(\tau_m - m\mu)/a_m$ converges in law to a variable G with $\mathbb{E}e^{-sG} = e^{K_\alpha s^\alpha}$ for $s \geq 0$ and $\mathbb{E}e^{i\theta G} = \exp(K_\alpha(-i\theta)^\alpha)$. The law of G is totally skewed to the right and has a bounded, uniformly continuous density g with $g(z) \rightarrow 0$ as $|z| \rightarrow \infty$.*
- (iii) (Gnedenko) *Let g be the density in (ii) or in (v). Then $\sup_k |a_m u_m(k) - g((k - m\mu)/a_m)| \rightarrow 0$ if $1 < \alpha < 2$, and $\sup_k |a_m u_m(k) - g(k/a_m)| \rightarrow 0$ if $0 < \alpha < 1$. In both ranges $c_0 := \sup_{m \geq 1} \sup_k a_m u_m(k) < \infty$.*
- (iv) *If $1 < \alpha < 2$ and G' is an independent copy of G , then $G - G'$ is symmetric α -stable with $\mathbb{E}e^{i\theta(G - G')} = e^{-\sigma_\alpha |\theta|^\alpha}$, and $I_\alpha := \int_{\mathbb{R}} g(z)^2 dz = \Gamma(1 + 1/\alpha)/(\pi \sigma_\alpha^{1/\alpha})$.*

(v) If $0 < \alpha < 1$, then τ_m/a_m converges in law to a variable $G \geq 0$ with $\mathbb{E}e^{-sG} = e^{-\Gamma(1-\alpha)s^\alpha}$ for $s \geq 0$. It has a bounded, uniformly continuous density g , with $g = 0$ on $(-\infty, 0]$ and $g(y) \sim \alpha y^{-1-\alpha}$ as $y \rightarrow \infty$.

Item (iii) is Gnedenko's local limit theorem [15, Theorem 4.2.1], which applies because every $p_k > 0$. The tail in (v) is the first term of the series for the positive stable density [16]. We check the rest for our law of L in Appendix C.

Theorem 4.4 (the center for $1 < \alpha < 2$). *Let $1 < \alpha < 2$ and let D be any a.s. finite first run, in particular either start. As $n \rightarrow \infty$ over even n ,*

$$C_n \sim 2I_\alpha \left(\frac{\mu}{h} \right)^{1/\alpha} = \frac{2\Gamma(1+1/\alpha)}{\pi} \left(\frac{\mu}{K_\alpha |\cos(\pi\alpha/2)| n} \right)^{1/\alpha}.$$

The reason is that both kinds of runs must fill about h steps, which takes about h/μ runs of each kind. Each renewal process is then within about $n^{1/\alpha}$ of its mean, so the 2 meet at a given level with probability of order $n^{-1/\alpha}$. The first run only matters for the edge. In Appendix C we split the sum in Proposition 4.2 by the number of runs and use Lemmas C.1 and C.2.

With a stationary start the first run is the remaining part of a run in progress. It covers all n steps with probability of order $n^{1-\alpha}$, while the center has probability of order $n^{-1/\alpha}$. The 2 are comparable when $1 - \alpha = -1/\alpha$, that is $\alpha^2 - \alpha - 1 = 0$.

Corollary 4.5 (the golden-ratio exponent). *Let $1 < \alpha < 2$ and $e(\alpha) = 1 - \alpha + 1/\alpha$. Then $R_n \sim C(\alpha)n^{e(\alpha)}$ for the stationary start and $R_n \sim C_{\text{fr}}(\alpha)n^{e(\alpha)-1}$ for the fresh start, where*

$$C(\alpha) = \frac{\pi(K_\alpha |\cos \frac{\pi\alpha}{2}|)^{1/\alpha}}{4(\alpha-1)\Gamma(1+\frac{1}{\alpha})\zeta(\alpha)^{1+1/\alpha}}, \quad C_{\text{fr}}(\alpha) = (\alpha-1)\zeta(\alpha)C(\alpha).$$

Since $e(\alpha) > 0$ exactly when $\alpha < \varphi$, with a stationary start the ends win for all large n if $\alpha < \varphi$, and the center wins if $\alpha > \varphi$. At $\alpha = \varphi$,

$$R_n \rightarrow C(\varphi) = \frac{\pi\varphi(\varphi\Gamma(\varphi^{-2})|\cos \frac{\pi\varphi}{2}|)^{1/\varphi}}{4\Gamma(\varphi)\zeta(\varphi)^\varphi} = 0.7761317206\dots,$$

so the center still wins by the factor $1/C(\varphi) = 1.2884$. With a fresh start $e(\alpha) - 1 < 0$, and the center wins for every $1 < \alpha < 2$. More generally, if $\mathbb{P}(D \geq n) \sim c_D n^{-\beta_D}$ with $c_D, \beta_D > 0$, then $R_n \sim c_D n^{1/\alpha - \beta_D} / (4I_\alpha(2\mu)^{1/\alpha})$. In particular the length-biased first run $\mathbb{P}(D = k) = kp_k/\mu$, which is not stationary, multiplies $C(\alpha)$ by α .

Proof. By Proposition 4.1 and Theorem 4.4, with $h = n/2$,

$$R_n = \frac{\mathbb{P}(D \geq n)}{2C_n} \sim \frac{c_D n^{-\beta_D}}{4I_\alpha(2\mu/n)^{1/\alpha}},$$

which is the general formula. By Lemma 4.3(iv), $4I_\alpha(2\mu)^{1/\alpha} = 4\Gamma(1+\frac{1}{\alpha})\mu^{1/\alpha}/(\pi(K_\alpha |\cos \frac{\pi\alpha}{2}|)^{1/\alpha})$. The fresh start has $c_D = 1$ and $\beta_D = \alpha$, which gives $C_{\text{fr}}(\alpha)$. By Proposition 4.1 the stationary start has $c_D = 1/((\alpha-1)\mu)$ and $\beta_D = \alpha-1$, which gives $C(\alpha)$. For the length-biased law, summation by parts gives $\sum_{k \geq n} kp_k = n\bar{F}(n) + \zeta(\alpha, n+1) \sim \alpha n^{1-\alpha}/(\alpha-1)$. So $\mathbb{P}(D \geq n)$ is asymptotically α times its value for the stationary start. Hence c_D is multiplied by α and β_D does not change. Next, $e(\alpha) = -(\alpha^2 - \alpha - 1)/\alpha$, and φ is the positive root of $\alpha^2 - \alpha - 1$. At $\alpha = \varphi$ we have $\alpha - 1 = 1/\varphi$, $1 + 1/\alpha = \varphi$ and $2 - \alpha = \varphi^{-2}$. So $K_\varphi = \varphi\Gamma(\varphi^{-2})$, and substituting these into $C(\alpha)$ gives the closed form. $\square$

α	$e(\alpha)$	$C(\alpha)$	$C_{\text{fr}}(\alpha)$	R_{1600}	R_{10^4}	R_{10^5}
1.3	+0.469231	0.422717	0.498631	17.7593 [1.3180]	38.7537 [1.2171]	106.149 [1.1317]
1.5	+0.166667	0.647971	0.846371	2.14409 [0.9675]	2.9528 [0.9818]	4.37642 [0.9914]
φ	0	0.776132	1.073675	0.661763 [0.8526]	0.697331 [0.8985]	0.727689 [0.9376]
5/3	-1/15	0.835101	1.182237	0.411815 [0.8064]	0.389307 [0.8614]	0.352822 [0.9102]
1.7	-0.111765	0.880046	1.265508	0.298369 [0.7733]	0.262005 [0.8334]	0.215839 [0.8881]
1.9	-0.373684	1.438986	2.266075	0.0449706 [0.4923]	0.0256505 [0.5569]	0.0121612 [0.6242]

Table 3: Constants of Corollary 4.5 and exact values of $R_n = E_n/C_n$ for the stationary start, with $R_n/(C(\alpha)n^{e(\alpha)})$ in brackets.

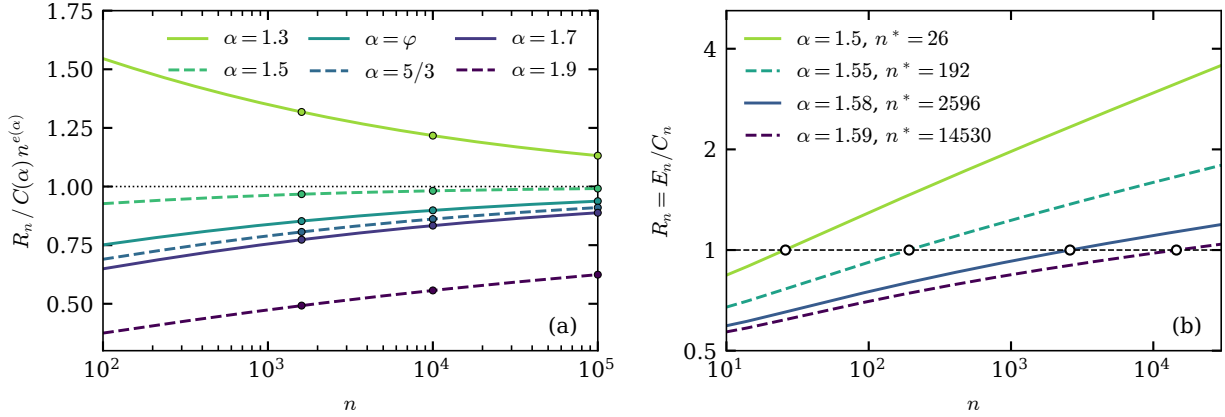


Figure 3: Alternating runs with $\mathbb{P}(L \geq k) = k^{-\alpha}$ and a stationary start. (a) The ratio $R_n = E_n/C_n$ divided by its asymptotic form $C(\alpha)n^{e(\alpha)}$ from Corollary 4.5, for even n from 100 to 10^5 . The dots are the entries of Table 3. (b) R_n for 4 values of α just below the golden ratio. On the computed range $10 \leq n \leq 3 \cdot 10^4$ each curve crosses 1 once, at $n^*(\alpha)$ (circles).

Both scalings behind Corollary 4.5 are in Godrèche and Luck [10]. Wang, Li and Hänggi [14, Section 2.1] get the golden ratio by equating the decay $t^{-1/\mu}$ of the central peak with the decay $t^{1-\mu}$ of the side peaks (their μ is our α), and Corollary 4.5 is the lattice statement with its constant. The golden ratio also appears for the elephant walk, as the log-concavity threshold $(\sqrt{5} - 1)/2 = 1/\varphi$ of [9, Proposition 2.2], but we know of no connection between the 2. The approach to $C(\varphi)$ is slow. At $\alpha = \varphi$ we have $R_{10^5} = 0.72769$, which is $0.938 C(\varphi)$, and R_n increases on the computed grid (Table 3 and Figure 3).

For $\alpha \geq 2$ the sums τ_m have a Gaussian limit, and the center wins with either start.

Theorem 4.6 ($\alpha \geq 2$). *Let D be any a.s. finite first run.*

- (a) *If $\alpha > 2$, let $v = \text{Var } L = 2\zeta(\alpha - 1) - \zeta(\alpha) - \zeta(\alpha)^2$. Then $C_n \sim \sqrt{2\mu/(\pi v n)}$, and with a stationary start $R_n \sim C_2(\alpha)n^{3/2-\alpha}$, where $C_2(\alpha) = \sqrt{\pi v/(2\mu)}/(2(\alpha - 1)\mu)$.*
- (b) *If $\alpha = 2$, then $C_n \sim \sqrt{2\mu/(\pi n \log n)}$, and with a stationary start $R_n \sim \sqrt{\pi \log n/(8\zeta(2)^3 n)}$.*

The proof is in Appendix C. The exponent of R_n is continuous at $\alpha = 2$, since $e(2) = -\frac{1}{2} = \frac{3}{2} - 2$. At $\alpha = 2$, R_n divided by the leading term of (b) is 0.885 at $n = 1600$ and 0.943 at $n = 10^5$, which is consistent with a correction of order $1/\log n$.

For $0 < \alpha < 1$ the runs have infinite mean and only the fresh start makes sense. Then the center is of order $1/n$.

Theorem 4.7 ($0 < \alpha < 1$). *Let $0 < \alpha < 1$, with a fresh start. Then $nC_n \rightarrow \ell_\alpha(\frac{1}{2}) = \frac{2}{\pi} \tan \frac{\pi\alpha}{2}$, where*

$$\ell_\alpha(y) = \frac{\sin \pi\alpha}{\pi} \frac{y^{\alpha-1}(1-y)^{\alpha-1}}{y^{2\alpha} + 2y^\alpha(1-y)^\alpha \cos \pi\alpha + (1-y)^{2\alpha}} \quad (0 < y < 1)$$

is the density of Lamperti's occupation-time law [11, 10]. Hence $R_n \sim \frac{\pi}{4 \tan(\pi\alpha/2)} n^{1-\alpha} \rightarrow \infty$, and the ends win for every $\alpha < 1$. The point $y = \frac{1}{2}$ is a strict local minimum of ℓ_α for $\alpha < \alpha_c$ and a strict local maximum for $\alpha > \alpha_c$, where $\alpha_c = 0.5946116441$ is the root of $\alpha = \cos(\pi\alpha/2)$.

The proof is in Appendix C. The critical value α_c is in [10, Section 7], below eq. (7.6). We do not prove the global shape of ℓ_α (the U and W shapes in [10, 12]), and we do not prove that the center bin at finite n is a local minimum or maximum of the law of A_n . At $n = 16000$ it is a local minimum for $\alpha = 0.58$ and a local maximum for $\alpha = 0.61$, but the relative second differences are only about $4 \cdot 10^{-9}$ and $2 \cdot 10^{-9}$.

Remark 4.8 (the fresh start). With a fresh start the threshold is $\alpha = 1$. The ends win for $\alpha < 1$ by Theorem 4.7, and the center wins for $\alpha > 1$ by Corollary 4.5 and Theorem 4.6. At $\alpha = 1$ we only have a heuristic. Suppose we repeat the proof of Theorem 4.4 with a skewed Cauchy limit, the centering $m(H_{m+1} - 1)$ and the density $1/\pi^2$ at 0 of the difference of 2 copies of the limit. This suggests $R_n \approx \pi^2 \tilde{m}_h / (8h) \sim \pi^2 / (8 \log n)$, where $\tilde{m}_h$ solves $m(H_{m+1} - 1) = h$. We have not controlled the error terms, so the claim that the center wins by a logarithm is not proved. The exact R_n divided by $\pi^2 \tilde{m}_h / (8h)$ is 0.796, 0.850 and 0.886 at $n = 1600$, 10^4 and 10^5 . Below $\alpha = 1$ the crossover can be late. Setting the limit form in Theorem 4.7 equal to 1 gives $n \approx (4 \tan(\pi\alpha/2)/\pi)^{1/(1-\alpha)}$, which is 21 at $\alpha = 0.7$ and $1.1 \cdot 10^9$ at $\alpha = 0.9$. Indeed $R_{16000} = 0.59$ at $\alpha = 0.9$.

Remark 4.9 (crossover sizes). For $\alpha < \varphi$ and a stationary start, $R_n \rightarrow \infty$ by Corollary 4.5. But when α is close to φ , $R_n < 1$ for small n and the crossover is late. For $\alpha = 1.5, 1.55, 1.58, 1.59$ we computed R_n for every even n with $10 \leq n \leq 3 \cdot 10^4$. In each case $R_n - 1$ changes sign once, at $n^*(\alpha) = 26, 192, 2596, 14530$, where $n^*(\alpha)$ is the even n with $R_{n-2} < 1 < R_n$. At $\alpha = 1.60$ a 30-point grid on $[3 \cdot 10^4, 4.2 \cdot 10^5]$ and then bisection over even n give $n^*(1.60) = 372660$. These are the last sign changes in the computed range, and we have no bound that rules out a later one. The leading order $C(\alpha)^{-1/e(\alpha)}$ underestimates them by factors 1.9 to 7.4. Heuristically, setting $C(\alpha)n^{e(\alpha)} = 1$ and using $e(\alpha) \approx (\sqrt{5}/\varphi)(\varphi - \alpha)$ gives $\log n^*(\alpha) = 0.1834/(\varphi - \alpha) + O(1)$ as $\alpha \uparrow \varphi$, where $0.1834 = \varphi |\log C(\varphi)| / \sqrt{5}$. At the 5 values of α above, the $O(1)$ term is 1.7 to 3.0. A proof would need Theorem 4.4, with its correction terms, to hold uniformly in α near φ .

Remark 4.10 (heat conduction). Lévy-walk models of anomalous heat conduction use $\alpha = 5/3$ [13, 12]. Then $e(5/3) = -\frac{1}{15}$, and with a stationary start $R_n \sim 0.8351 n^{-1/15}$. The approach is slow and not monotone. On even n in $[20, 400]$, R_n increases up to $n = 150$, where $R_{150} = 0.42419$, and after that it decreases on the computed grid, with $R_n / (C(5/3)n^{-1/15}) = 0.910$ at $n = 10^5$. In the Lévy-walk model the flights choose their directions independently. So a maximal run is a sum of a geometric number of flights and has the same tail exponent, and the exponent $-\frac{1}{15}$ should carry over with a different constant. We have not proved this. In that model the 2 starts also have to be kept apart. Dhar, Saito and Derrida [13] write the equation for a fresh start, where the end atom is of order $t^{-\alpha}$, but quote the peak size $t^{1-\alpha}$ of the equilibrium start (their β is our α).

The last walk has aging memory. As for $\alpha < 1$, its fraction of $+$ steps has a limit law, so its center is of order $1/n$.

Remark 4.11 (aging). Let the walk take a fair first step and then switch direction at step $k \geq 2$ with probability c/k , where $0 < c < 2$. Dietz and Sethuraman proved that A_n/n converges in law to

Beta(c, c) [23, Example 1.1 and Theorem 1.4]. For $c \leq 1$ their chain is exactly this walk. Engländer and Volkov gave a different proof [24, Theorem 1]. The limit is uniform exactly when $c = 1$ [24, Remark 2]. Fix the first step at $+1$. At $c = 1$, A_n is then exactly uniform on $\{1, \dots, n\}$ for every n (Appendix D). A path with $A_n = s < n$ switches at least once, and a switch at time t multiplies the path probability by the odds $\omega_t = c/(t - c)$, which increase in c . So $\mathbb{P}_+(A_n = s)/\mathbb{P}_+(A_n = n)$ increases strictly in c for $1 \leq s \leq n - 1$ and equals 1 at $c = 1$, and the fixed-start crossing is exactly $c = 1$ for every n and every bin. With a fair first step each end is $\frac{1}{2n}$ at $c = 1$ while every interior bin is $\frac{1}{n}$. By the same argument C_n/E_n increases strictly in c and tends to 0 as $c \rightarrow 0$. So the crossing c_n is unique and below 1. For even n ,

$$c_n = 1 - \frac{\log 2}{\log n + \gamma + 2 - 2 \log 2} + O((\log n)^{-3}).$$

For example, computed to high precision, $c_{200} = 0.8925$ and $c_{800} = 0.9116$. The $\log 2$ comes from the fair start under switching probability c/k , and not from aging. With switching probability $c/(k + 1)$ and a fair start, A_n is exactly uniform on $\{0, \dots, n\}$ at $c = 1$ (Appendix D). A switch at time t then has the odds $c/(t + 1 - c)$, which also increase in c , so this crossing is exactly $c = 1$ for every n . The proof, in Appendix D, compares the end, which is exactly $\frac{1}{2} \prod_{k=2}^n (1 - c/k)$ and of order n^{-c} , with a local limit theorem at the center.

5 Verification and open questions

We computed in Python, in double precision unless stated otherwise. For the elephant walk, a forward dynamic program for the law of M_n and a backward recursion for V_s , folded by the symmetry $V_s(b) = V_s(s - b)$, agree to $1.6 \cdot 10^{-14}$ at $n = 12800$. They also agree with exact rational arithmetic for $n \leq 60$ and with 40-digit arithmetic at $n = 400$. A recursion for Taylor coefficients in ε gives c_1 and $k_2(n)$. For the Markov crossing we count words by their number of switches, and this reproduces the values of p_n in [2]. Section 3 uses exact recursions for f_n and CP_n . Truncating the chain at a level m_0 does not change $f_n(m)$ for $m \leq m_0$, since M_t never decreases. For heavy tails we compared 3 methods. These are the two-renewal formula of Proposition 4.2, a recursion on the first run with exact Hurwitz zeta tails, and a sum over all 2^{n-1} run compositions in 30-digit arithmetic. All 3 agree to $7 \cdot 10^{-16}$ for $n \leq 14$, and the first 2 agree to 10^{-14} up to $n = 3200$. For aging, a forward recursion with nonnegative terms and a transfer matrix evaluated at roots of unity agree to at least 12 digits on the crossings, and small cases were done in exact rationals. The constants of the Landau density come from quadrature with 25 digits. Programs written separately reproduced the main numbers, including values of x_n , of $D(n)$, of R_n and of c_n .

There is one check script for each part of the paper. `verify_elephant_crossing.py` covers Section 2, Table 2 and Appendix A. `verify_elephant_shape.py` covers Section 3 and Appendix B. `verify_heavy_runs.py` covers Section 4, Table 3 and Appendix C, and `verify_aging.py` covers Remark 4.11 and Appendix D. Each script recomputes the numbers printed in its part and compares them with the printed values. Where 2 independent methods were used when a number was first found, the script runs both. `verify_all.py` runs the 4 scripts. The quick run takes a few minutes, and the full run adds the largest cells and takes less than an hour. All 297 checks of the full run pass. The scripts need numpy, scipy and mpmath, and their output is in `paperD/data`.

Many of the exact and finite statements are also proved in Lean 4, in `lean/PaperD` in the Problem 131 folder. For the elephant walk these are Lemma 2.2, Proposition 2.4, the value $C'_n(0) = 4/(n + 2)$ of Proposition 2.5, the 2 sums of Lemma 3.2 (for the urn that gives the subtree sizes), Lemma 3.4 for $k_0 = 1$, Theorem 3.5, Corollary 3.6 and Lemma B.4. For heavy tails they

are Proposition 4.1, the identities for $e(\alpha)$ and K_α behind Corollary 4.5, and the value $\ell_\alpha(\frac{1}{2})$ in Theorem 4.7. For aging they are Propositions D.1 and D.2, Corollary D.3, Lemma D.4 and the finite part of Theorem D.5. The file `MAP.md` in that folder lists the Lean theorems. The asymptotic statements are not in Lean.

The proofs do not use these computations, with 2 exceptions. The existence of κ_L in Theorem 3.8(c) uses the sign of $f_L''(\lambda_0)$, which we computed but did not certify. Also some constants in the proofs are values of explicit series and integrals, which we evaluated numerically. Conjecture 3.12, the checked ranges after Corollary 3.6, and the numerical parts of Remarks 2.10, 4.8, 4.9 and 4.10 rest on computation alone.

Before computing a new case we wrote down, in the ledgers `ledger.D1.md` to `ledger.D4.md` in `paperD/data`, a prediction and a condition that would refute it. Outcomes are recorded below each prediction, refuted predictions were kept, and observations made after a computation are marked as such. Most predictions survived, and 8 failed. The law f_n given the first step is not convex between $2m^*$ and $n - 2m^*$ (item D2-P9), which led us to Conjecture 3.12 for the full law. At $n = 10^6$ and $x = 320$ the mode of f_n is 3 below the mode of CP_n , where we allowed 2 (D2-P11). The endpoint dip already holds at $n = 4$ and 5, where we predicted that it fails (D2-P12). Two scaling laws for the finite- n shift between these 2 modes were both wrong (D2-P15). The form of Conjecture 3.12 on the closed range $[2m^*, n - 2m^*]$ fails for 18 values of $n \leq 51$, which a separate check of $10 \leq n \leq 99$, also written down in advance, found. For heavy tails, one tolerance was too small because of an arithmetic slip in the prediction (D3-P2), and one predicted value of $n^*(1.60)$ was off by 2 because we misread a rounded number (D3-V1). For aging, we predicted $nC_n = d_c(1 + (c - 1)/n + O(n^{-2}))$, with d_c the density of $\text{Beta}(c, c)$ at $\frac{1}{2}$, and at $c = 0.3$ the remainder is larger (D4-P7). None of the failures concerns a statement that we call a theorem.

Several questions remain open. For the elephant walk, the refined crossing of Remark 2.10 needs an upper bound $\log C_n \leq \log(\varepsilon c_1) + \varepsilon k_2(n) + O(\varepsilon^2(\log n)^k)$, which we do not have. We have no explicit n beyond which $g_n > x_n$ (Theorem 2.7). Evaluating the lower bound of Theorem 2.6(c) at g_n suggests that $n \geq 2^{17}$ is enough, and $x_n < g_n$ at every n in Table 2, but we have not proved it. Theorem 3.11 says nothing in a window around the center, and closing it needs second differences of the law, as in Conjecture 3.12. Also open are the tail of Theorem 3.9 for m between $x \log x$ and $x \log^2 x$, a sharp upper bound on the mirror term for m of order n , and effective constants in Theorems 3.8(c) and 3.11, starting with a certified value of κ_L . For heavy tails we have no rate of convergence in Theorem 4.4. The monotonicity of R_n , and the fact that $R_n - 1$ changes sign only once, at $n^*(\alpha)$, are only observed on the computed range (Remark 4.9). The case $\alpha = 1$ is only heuristic (Remark 4.8). We have not studied the whole law for $1 < \alpha < 2$, for example whether the end atoms are isolated maxima. For $\alpha < 1$ we know the shape of ℓ_α only near $y = \frac{1}{2}$, and we do not know whether the center bin at finite n is a local minimum for $\alpha < \alpha_c$ and a local maximum for $\alpha > \alpha_c$. For aging, we checked that at $c = c_n$ the center is the unique interior minimum and bins 1 and $n - 1$ are the largest, for every even n from 4 to 1000 and for $n = 1600, 3200, 6400$, but we have not proved this.

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Data and code availability. The manuscript source, verification scripts, recorded results and the prediction ledgers are in the Problem 131 folder of <https://github.com/apemm/Kagey-Problems>, under `paperD`. The Lean files are in `lean/PaperD` in the same folder.

A Proofs for Section 2

Throughout this appendix $n \geq 2$ is even, $h = n/2$, and we condition on $X_1 = +1$. So $M_1 = 0$, the chain M moves from b to $b+1$ at time t with probability $r_t(b)$, and $V_s(b) = \mathbb{P}_+(M_n = h \mid M_s = b)$. We write $\mathcal{L}_s F(b) = (1 - r_s(b))F(b) + r_s(b)F(b+1)$, so that $V_s = \mathcal{L}_s V_{s+1}$. Both M_t and $t - M_t \geq 1$ are nondecreasing. So from a state $b \geq 1$ the chain stays at states ≥ 1 , and for $b \geq 1$ we have $V_s(b) = 0$ unless b is feasible, that is, $b \in \mathcal{S}_s = \{b \in \mathbb{Z} : 1 \leq b \leq h, 1 \leq s - b \leq h\}$. Note that $\mathcal{S}_n = \{h\}$. Theorem 2.6(a) is Proposition A.4(d), and Theorem 2.6(b) is Proposition A.5.

A.1 Pseudo-count urns

For $c \geq 0$ the Pólya(c) chain moves at time s from b to $b+1$ with probability $p_s^c(b) = \frac{b+c}{s+2c}$ and stays at b otherwise. It is Pólya's urn with weights $b+c$ and $s-b+c$ for the 2 colors, where each draw adds a ball of the color drawn. For $b+c > 0$ and $s-b+c > 0$ let

$$\Phi_s^c(b) = \binom{n-s}{h-b} \frac{B(h+c, h+c)}{B(b+c, s-b+c)},$$

where B is the beta function and the binomial coefficient is 0 unless $0 \leq h-b \leq n-s$. For $1 \leq b \leq s-1$ this is defined for all $c \geq 0$, and it vanishes at infeasible b .

Lemma A.1. *Let $c \geq 0$, $1 \leq s \leq n$, $b+c > 0$ and $s-b+c > 0$.*

- (a) $\Phi_s^c(b)$ is the probability that the Pólya(c) chain at b at time s is at h at time n . So $\Phi_n^c(b) = \mathbf{1}\{b = h\}$, and $\Phi_s^c(b) = p_s^c(b)\Phi_{s+1}^c(b+1) + (1 - p_s^c(b))\Phi_{s+1}^c(b)$ for $s < n$.
- (b) If $s < n$ and $\Phi_s^c(b) > 0$, then $p_s^c(b)\Phi_{s+1}^c(b+1)/\Phi_s^c(b) = r_s^{\text{br}}(b) := \frac{h-b}{n-s}$. So, given that it ends at h , the chain moves up with probability $r_s^{\text{br}}(b)$ for every c , and its remaining $h-b$ up steps occupy a uniformly random set of $h-b$ of the remaining $n-s$ steps.
- (c) If $0 < \varepsilon < \frac{1}{2}$ and $\beta = \frac{\varepsilon}{1-2\varepsilon}$, then $r_s(b) = p_s^{\beta s}(b)$ for $0 \leq b \leq s$. So at time s the chain M moves like the Pólya(βs) chain.
- (d) If $0 < \varepsilon < \frac{1}{2}$ and $0 \leq c \leq \beta s$, then $V_s(b) \geq \Phi_s^c(b)$ for $1 \leq b \leq s-1$. In particular $V_{t+1}(1) \geq \Phi_{t+1}^0(1) = \pi_t$ for $1 \leq t \leq h$.

Proof. (a) Let $x = b+c$, $y = s-b+c$ and $(x)_k = x(x+1)\cdots(x+k-1)$. Each sequence of $n-s$ draws with $h-b$ draws of the first color has probability $(x)_{h-b}(y)_{n-s-h+b}/(x+y)_{n-s} = B(h+c, h+c)/B(x, y)$, and there are $\binom{n-s}{h-b}$ such sequences. The recursion is the Markov property.

(b) Since $\binom{n-s-1}{h-b-1}/\binom{n-s}{h-b} = \frac{h-b}{n-s}$ and $B(x, y)/B(x+1, y) = \frac{x+y}{x}$, we get $\Phi_{s+1}^c(b+1)/\Phi_s^c(b) = \frac{h-b}{n-s} \cdot \frac{s+2c}{b+c}$. Choosing a uniform set of $h-b$ of the $n-s$ steps one step at a time gives the same up-probabilities.

(c) Since $1 + 2\beta = 1/(1-2\varepsilon)$, we have $\frac{b+\beta s}{s+2\beta s} = (1-2\varepsilon)(\frac{b}{s} + \beta) = r_s(b)$.

(d) Let $F_u = \Phi_u^c$ on $\{1, \dots, u-1\}$ for $s \leq u \leq n$. By (a),

$$\mathcal{L}_u F_{u+1}(b) - F_u(b) = (r_u(b) - p_u^c(b))(F_{u+1}(b+1) - F_{u+1}(b)).$$

By (c), $r_u(b) - p_u^c(b) = (\frac{u}{2} - b)(\frac{1}{u+2c} - \frac{1}{u(1+2\beta)})$, and the last factor is ≥ 0 since $c \leq \beta u$. Let $b \in \mathcal{S}_u$. If $b = h$ then $b \geq u/2$ and $F_{u+1}(b+1) = 0$. If $u-b = h$ then $b \leq u/2$ and $F_{u+1}(b) = 0$. Otherwise $F_{u+1}(b+1)/F_{u+1}(b) = \frac{(h-b)(u-b+c)}{(h-u+b)(b+c)}$, which is ≥ 1 for $b \leq u/2$ and ≤ 1 for $b \geq u/2$. In each case the 2 factors above have the same sign or one of them is 0, so $\mathcal{L}_u F_{u+1} \geq F_u$ on $\mathcal{S}_u$. Since $F_n(b) = \mathbf{1}\{b = h\}$ and F_u vanishes off $\mathcal{S}_u$, Lemma A.2(b) below gives $V_u \geq F_u$. Finally $\Phi_{t+1}^0(1) = t \binom{n-t-1}{h-1} (h-1)!^2 / (n-1)! = \pi_t$. $\square$

Lemma A.2. Let $1 \leq s_0 \leq n$, and for $s_0 \leq s \leq n$ let $F_s \geq 0$ be a function on $\{1, \dots, s-1\}$.

- (a) If $F_n(h) \geq 1$ and $\mathcal{L}_s F_{s+1} \leq K_s F_s$ on $\mathcal{S}_s$ for $s_0 \leq s < n$, with constants $K_s \geq 0$, then $V_s \leq (\prod_{u=s}^{n-1} K_u) F_s$ on $\mathcal{S}_s$ for $s_0 \leq s \leq n$.
- (b) If $F_n(b) \leq \mathbf{1}\{b=h\}$, each F_s vanishes off $\mathcal{S}_s$, and $\mathcal{L}_s F_{s+1} \geq F_s$ on $\mathcal{S}_s$ for $s_0 \leq s < n$, then $V_s \geq F_s$ for $s_0 \leq s \leq n$.

Proof. We use backward induction on s , starting from $V_n(b) = \mathbf{1}\{b=h\}$ and using $V_s = \mathcal{L}_s V_{s+1}$ on $\mathcal{S}_s$. In (a), V_{s+1} vanishes at infeasible states and $F_{s+1} \geq 0$, so the induction hypothesis gives $V_s \leq \prod_{u>s} K_u \mathcal{L}_s F_{s+1} \leq \prod_{u\geq s} K_u F_s$ on $\mathcal{S}_s$. In (b) it gives $V_s \geq \mathcal{L}_s F_{s+1} \geq F_s$ on $\mathcal{S}_s$, and $F_s = 0 \leq V_s$ off $\mathcal{S}_s$. $\square$

A.2 The supersolution

Let ψ be the digamma function.

Lemma A.3. Let $2 \leq s \leq n$, $b \in \mathcal{S}_s$ and $\tau \geq 0$. Put $X = b + \tau$, $Y = s - b + \tau$, $\Sigma = X + Y$ and $z = (s - 2b)^2 / \Sigma^2$. Then $\partial_\tau \log \Phi_s^\tau(b) = 2\psi(\Sigma) - \psi(X) - \psi(Y) - 2\psi(n + 2\tau) + 2\psi(h + \tau)$, and

$$\partial_\tau \log \Phi_s^\tau(b) < -\log(1 - z) + \frac{1 + z}{(1 - z)\Sigma} + \frac{2(1 + z)}{3\Sigma^2(1 - z)^2}.$$

Proof. The formula follows from $\log B(x, y) = \log \Gamma(x) + \log \Gamma(y) - \log \Gamma(x + y)$. For $x > 0$, [22, 5.9.15] gives $\psi(x) = \log x - \frac{1}{2x} - 2 \int_0^\infty \frac{t dt}{(t^2 + x^2)(e^{2\pi t} - 1)}$, and the subtracted term is positive and less than $\frac{2}{x^2} \int_0^\infty \frac{t dt}{e^{2\pi t} - 1} = \frac{1}{12x^2}$. So $\log x - \frac{1}{2x} - \frac{1}{12x^2} < \psi(x) < \log x - \frac{1}{2x}$. We use the upper bound for $\psi(\Sigma)$ and the lower bound for $\psi(X)$ and $\psi(Y)$. The logarithmic derivative of the duplication formula [22, 5.5.5] is $2\psi(2y) = \psi(y) + \psi(y + \frac{1}{2}) + 2\log 2$, and ψ is increasing, so $2\psi(n + 2\tau) - 2\psi(h + \tau) > 2\log 2$. Hence the derivative is less than $\log \frac{\Sigma^2}{4XY} + (\frac{1}{2X} + \frac{1}{2Y} - \frac{1}{\Sigma}) + \frac{1}{12}(\frac{1}{X^2} + \frac{1}{Y^2})$. Now $4XY = \Sigma^2(1 - z)$ and $X^2 + Y^2 = \Sigma^2(1 + z)/2$, so the 3 terms are $-\log(1 - z)$, $\frac{X^2 + Y^2}{2XY\Sigma} = \frac{1 + z}{(1 - z)\Sigma}$ and $\frac{X^2 + Y^2}{12X^2Y^2} = \frac{2(1 + z)}{3\Sigma^2(1 - z)^2}$. $\square$

Proposition A.4. Let $0 < \varepsilon < \frac{1}{4}$, $\kappa = \frac{2\varepsilon}{1 - 4\varepsilon}$, $c_s = \kappa(s + 10)$ and $\Sigma_s = s + 2c_s$. For $1 \leq s \leq n - 1$ and $b \in \mathcal{S}_s$ let $D_s(b) = \mathcal{L}_s \Phi_{s+1}^{c_{s+1}}(b) / \Phi_s^{c_s}(b)$.

- (a) Let $c = c_s$, $c' = c_{s+1}$, $d = \frac{s}{2} - b$, $u = \frac{d}{s + 2c'}$, $v = \frac{d}{n - s}$ and $\rho = \frac{s + 2c'}{s(1 + 2\beta)}$. Then $|u| < \frac{1}{2}$ and

$$D_s(b) = \mathcal{J} \frac{\Phi_s^{c'}(b)}{\Phi_s^c(b)}, \quad \mathcal{J} = 1 - (\rho - 1) \frac{u(u + v)}{\frac{1}{4} - u^2}.$$

- (b) $\log D_s(b) \leq \frac{\kappa}{\Sigma_s} + \frac{2\kappa}{3\Sigma_s^2}$.

- (c) For $1 \leq t \leq h$, $V_{t+1}(1) \leq \pi_t \exp(\kappa[H_{n-1} - H_t + \frac{2}{3t}] + c_{t+1}[\log \frac{(t+1)^2}{4t} + \frac{2}{3}])$.

- (d) $C_n \leq U_n$, with U_n as in Theorem 2.6(a).

The slope κ is chosen so that $2(\kappa - \beta)/(1 + 2\beta) = \kappa$, which follows from the formulas for κ and β . In the proof of (b) this makes the gain in $\mathcal{J}$ cancel the main loss from raising the pseudo-count from c to c' , and the offset 10 in c_s controls what is left.

Proof. (a) Write $p' = p_s^{c'}(b)$, $\bar{r} = r_s^{\text{br}}(b)$ and $r = r_s(b)$. By Lemma A.1(a) and (b) with c' , the numerator of $D_s(b)$ divided by $\Phi_s^{c'}(b)$ is $\frac{(1-r)(1-\bar{r})}{1-p'} + \frac{r\bar{r}}{p'}$. Here $p' = \frac{1}{2} - u$, $\bar{r} = \frac{1}{2} + v$ and $r = \frac{1}{2} - \rho u$, since $\frac{1}{2} - r = (1 - 2\varepsilon)\frac{d}{s} = \rho u$. Over the denominator $p'(1 - p') = \frac{1}{4} - u^2$ the numerator is $f(u, v) + f(-u, -v)$ with $f(u, v) = (\frac{1}{2} + \rho u)(\frac{1}{2} - v)(\frac{1}{2} - u)$, and this equals $\frac{1}{4} - \rho u^2 - (\rho - 1)uv$. This gives $\mathcal{J}$. Also $|d| \leq \frac{s}{2} - 1$, so $|u| < \frac{1}{2}$.

(b) Write $c = c_s$, $c' = c + \kappa$ and $\Sigma = \Sigma_s$. Since $c' - \beta s = (\kappa - \beta)s + 11\kappa$, the identity for κ gives $\rho - 1 = \kappa + \nu_s$ with $\nu_s = \frac{22\kappa}{s(1+2\beta)}$. As u and v have the sign of d , $u(u + v) \geq u^2$, so $\mathcal{J} \leq 1 - (\kappa + \nu_s)W'$ with $W' = \frac{z'}{1-z'}$ and $z' = 4u^2 = \frac{(s-2b)^2}{(s+2c')^2}$. Also $\mathcal{J} > 0$ by the expression in (a), since $0 < r < 1$ and $\bar{r}$, $1 - \bar{r}$ are not both 0. So $\log \mathcal{J} \leq -(\kappa + \nu_s)W'$. Next, $\log(\Phi_s^{c'}(b)/\Phi_s^c(b))$ is the integral over $\tau \in [c, c']$ of the derivative in Lemma A.3. There z decreases and Σ increases in τ , and the bound of the lemma increases in z and decreases in Σ , so we may take it at $\tau = c$. Let X , Y and z be as in the lemma at $\tau = c$, so that $X + Y = \Sigma$, and put $W = \frac{z}{1-z}$. Then $\frac{1}{1-z} = 1 + W$, $\frac{1+z}{1-z} = 1 + 2W$ and $-\log(1 - z) \leq W$, so

$$\log D_s(b) \leq \kappa \left[W + \frac{1 + 2W}{\Sigma} + \frac{2(1 + 2W)(1 + W)}{3\Sigma^2} \right] - (\kappa + \nu_s)W'. \quad (1)$$

We use 3 facts. (i) $X, Y \geq 1 + c$, since $b \geq 1$ and $s - b \geq 1$. So $XY \geq (1 + c)\Sigma/2$ and $1 + W = \frac{\Sigma^2}{4XY} \leq \frac{\Sigma}{2(1+c)}$. (ii) $z' = \lambda z$ with $\lambda = (\frac{\Sigma}{\Sigma+2\kappa})^2 \geq 1 - \frac{4\kappa}{\Sigma}$. So $W - W' = \frac{(1-\lambda)z}{(1-z)(1-\lambda z)} \leq (1 - \lambda)W(1 + W) \leq \frac{2\kappa}{1+c}W$ by (i). As $c \geq 11\kappa$, this gives $W' \geq \frac{9}{11}W$. (iii) $\Sigma \geq s$ and $1 + c > \kappa s$. If $W > 0$ then $s \neq 2b$, which together with $b \geq 1$ and $s - b \geq 1$ forces $s \geq 3$, so $\Sigma^2 \geq 3s$.

By (ii) and (iii), $\kappa(W - W') \leq \frac{2\kappa^2}{1+c}W \leq \frac{2\kappa}{s}W$ and $\frac{2\kappa W}{\Sigma} \leq \frac{2\kappa}{s}W$. Since $(1 + 2W)(1 + W) = 1 + 3W + 2W^2$, κ times the last term in the bracket of (1) equals $\frac{2\kappa}{3\Sigma^2} + \frac{2\kappa W}{\Sigma^2} + \frac{4\kappa W^2}{3\Sigma^2}$. Here $\frac{2\kappa W}{\Sigma^2} \leq \frac{2\kappa}{3s}W$ by (iii), and $\frac{4\kappa W^2}{3\Sigma^2} \leq \frac{4\kappa W(1+W)}{3\Sigma^2} \leq \frac{2\kappa W}{3\Sigma(1+c)} \leq \frac{2\kappa}{3s}W$ by (i). With $-\nu_s W' \leq -\frac{9}{11}\nu_s W$ this gives

$$\log D_s(b) \leq \frac{\kappa}{\Sigma} + \frac{2\kappa}{3\Sigma^2} + W \left[\frac{16\kappa}{3s} - \frac{18\kappa}{s(1+2\beta)} \right],$$

and the bracket is negative because $1 + 2\beta < 2$ for $\varepsilon < \frac{1}{4}$.

(c) Apply Lemma A.2(a) with $s_0 = t + 1$, $F_s = \Phi_s^{c_s}$ and $K_s = \exp(\frac{\kappa}{s} + \frac{2\kappa}{3s^2})$, which is allowed by (b) and $\Sigma_s \geq s$. Since $1 \in \mathcal{S}_{t+1}$ and $\sum_{s>t} s^{-2} \leq \frac{1}{t}$, we get $V_{t+1}(1) \leq e^{\kappa[H_{n-1} - H_t + 2/(3t)]} \Phi_{t+1}^{c_{t+1}}(1)$. As $\pi_t = \Phi_{t+1}^0(1)$, the logarithm of $\Phi_{t+1}^{c_{t+1}}(1)/\pi_t$ is the integral over $\tau \in [0, c_{t+1}]$ of the derivative in Lemma A.3, whose bound is largest at $\tau = 0$. There $X = 1$, $Y = t$ and $\frac{1}{1-z} = \frac{(t+1)^2}{4t}$, and the bound equals $\log \frac{(t+1)^2}{4t} + \frac{t^2+1}{2t(t+1)} + \frac{t^2+1}{12t^2} \leq \log \frac{(t+1)^2}{4t} + \frac{2}{3}$.

(d) Insert (c) into Proposition 2.5(a). Since $c_{t+1} = \kappa(t+11)$, the exponent in (c) is $\kappa H_{n-1} + \kappa \mathcal{G}(t)$, and $\pi_t = c_1 w_t$. $\square$

A.3 The bridge formula

By Lemma A.1(b), the law of the path under $\mathbb{E}_t^{\text{br}}$ is the Pólya(c) chain from 1 at time $t + 1$ conditioned on $M_n = h$, for every c . At time u it moves up with probability $r_u^{\text{br}}(M_u)$. We write $\mathbb{P}_t^{\text{br}}$ for the matching probability. For $1 \leq b \leq u - 1$ let

$$\varpi_u^+(b) = 1 + \varepsilon \frac{u - 2b}{b}, \quad \varpi_u^-(b) = 1 - \varepsilon \frac{u - 2b}{u - b}, \quad Q_u(b) = (u - 2b)^2 \left[\frac{r_u^{\text{br}}(b)}{b^2} + \frac{1 - r_u^{\text{br}}(b)}{(u - b)^2} \right],$$

and let $\varpi_u = \varpi_u^+(M_u)$ if $M_{u+1} = M_u + 1$ and $\varpi_u = \varpi_u^-(M_u)$ otherwise.

Proposition A.5. *Let $0 \leq \varepsilon \leq 1$ and $1 \leq t \leq h$. Then $V_{t+1}(1) = \pi_t \mathbb{E}_t^{\text{br}} [\prod_{u=t+1}^{n-1} \varpi_u]$. If $0 < \varepsilon \leq \frac{1}{2}$, then $\log V_{t+1}(1) \geq \log \pi_t + \varepsilon A_t - \varepsilon^2 B_t$, where*

$$B_t = \sum_{u=t+1}^{n-1} \mathbb{E}_t^{\text{br}} Q_u(M_u) \geq 0, \quad (2)$$

and hence $C_n \geq J_n$, where

$$J_n = \varepsilon \sum_{t=1}^h (1 - \varepsilon)^{t-1} \pi_t \exp(\varepsilon A_t - \varepsilon^2 B_t). \quad (3)$$

Proof. For $u \geq t+1$ and $b \in \mathcal{S}_u$ put $R_u(b) = V_u(b)/\Phi_u^0(b)$. By Lemma A.1(a) and (b) with $c = 0$, $\frac{b}{u}\Phi_{u+1}^0(b+1) = r_u^{\text{br}}(b)\Phi_u^0(b)$ and $(1 - \frac{b}{u})\Phi_{u+1}^0(b) = (1 - r_u^{\text{br}}(b))\Phi_u^0(b)$. Dividing $V_u = \mathcal{L}_u V_{u+1}$ by $\Phi_u^0(b)$ gives

$$R_u(b) = (1 - r_u^{\text{br}}(b)) \frac{1 - r_u(b)}{1 - b/u} R_{u+1}(b) + r_u^{\text{br}}(b) \frac{r_u(b)}{b/u} R_{u+1}(b+1),$$

where a term with an infeasible state has bridge probability 0. Also $\frac{r_u(b)}{b/u} = \varpi_u^+(b)$ and $\frac{1-r_u(b)}{1-b/u} = \varpi_u^-(b)$. So $R_u(b) = \mathbb{E}_t^{\text{br}}[\varpi_u R_{u+1}(M_{u+1}) \mid M_u = b]$, and $R_n(h) = 1$. Iterating from $u = t+1$ and using $\Phi_{t+1}^0(1) = \pi_t$ gives the formula.

Now let $0 < \varepsilon \leq \frac{1}{2}$. If $b \leq u/2$ then $\varpi_u^+(b) \geq 1$ and $\varpi_u^-(b) - 1 \in [-\varepsilon, 0]$, and if $b \geq u/2$ the same holds with $+$ and $-$ exchanged. So $\varpi_u - 1 \geq -\frac{1}{2}$. By Jensen's inequality, $\mathbb{E} \prod_u \varpi_u \geq \exp \sum_u \mathbb{E} \log \varpi_u$. Also $\log(1+y) \geq y - y^2$ for $y \geq -\frac{1}{2}$, since the difference vanishes at 0 and its derivative $y(1+2y)/(1+y)$ has the sign of y . With $\bar{r} = r_u^{\text{br}}(b)$ and $u\bar{r} - b = \frac{n(u-2b)}{2(n-u)}$,

$$\mathbb{E}_t^{\text{br}}[\varpi_u - 1 \mid M_u = b] = \varepsilon(u-2b) \frac{u\bar{r} - b}{b(u-b)} = \varepsilon \theta_u(b), \quad \mathbb{E}_t^{\text{br}}[(\varpi_u - 1)^2 \mid M_u = b] = \varepsilon^2 Q_u(b).$$

Summing over u gives the bound, and Proposition 2.5(a) gives $C_n \geq J_n$. $\square$

Under $\mathbb{P}_t^{\text{br}}$ let $Z = M_u - 1$ and $Z' = m - Z$ be the numbers of up steps and of other steps among the $m = u - t - 1$ steps between times $t+1$ and u . So Z is hypergeometric, with m draws from $n - t - 1$ steps of which $h - 1$ are up steps. Let $\Omega_u = \{u/8 \leq M_u \leq 7u/8\}$.

Lemma A.6. *Let $n \geq 16$, $1 \leq t \leq h$ and $t+1 \leq u \leq n-1$.*

$$(a) \quad \mathbb{E}_t^{\text{br}}(u - 2M_u)^2 = \frac{(t-1)^2(n-u)^2}{(n-t-1)^2} + \frac{4(u-t-1)(h-1)(h-t)(n-u)}{(n-t-1)^2(n-t-2)} \leq (t-1)^2 + u.$$

$$(b) \quad \text{If } t \leq \frac{n+2}{4}, \text{ then } A_t \geq 2(H_{n-1} - H_t - 1 - \frac{1}{t}) \geq 2(\log n - \log t - 3).$$

$$(c) \quad \text{If } t \leq \frac{n}{8} \text{ and } u \geq 8(t+1), \text{ then } \mathbb{P}_t^{\text{br}}(\Omega_u^c) \leq 2e^{-0.0943u}.$$

$$(d) \quad \text{If } t \leq \frac{n}{8}, \text{ then } B_t \leq 84(t+1 + \log n) + 683(t+1)^3 + 19100.$$

Proof. (a) Since $n-2 = 2(h-1)$, the mean $\mathbb{E}Z = \frac{m(h-1)}{n-t-1}$ gives $\mathbb{E}(u - 2M_u) = \frac{(t-1)(n-u)}{n-t-1}$, and the second term is $4 \text{Var } Z$. For the bound use $n-u \leq n-t-1$, $4(h-1)(h-t) = (n-t-1)^2 - (t-1)^2$, and $m(n-u) \leq u(n-t-2)$ for $u \geq t+2$.

(b) Since $b(u-b) \leq u^2/4$, $\theta_u(b) \geq \frac{2n(u-2b)^2}{(n-u)u^2}$. Keeping only $4 \text{Var } Z$ in (a) gives $A_t \geq P_t \sum_{u=t+1}^{n-1} \frac{u-t-1}{u^2}$ with $P_t = \frac{2n(n-2)(n-2t)}{(n-t-1)^2(n-t-2)}$. Here $P_1 = \frac{2n}{n-3} > 2$, and for real $t \in [1, \frac{n+2}{4}]$,

$\frac{d}{dt} \log P_t = -\frac{2}{n-2t} + \frac{2}{n-t-1} + \frac{1}{n-t-2} \geq \frac{3}{n-t-1} - \frac{2}{n-2t} \geq 0$, since the last step is equivalent to $4t \leq n+2$. So $P_t \geq 2$. The sum is nonnegative and at least $H_{n-1} - H_t - (t+1) \sum_{u>t} u^{-2} \geq H_{n-1} - H_t - \frac{t+1}{t}$. For the second inequality use $H_{n-1} - H_t \geq \log \frac{n}{t+1} \geq \log n - \log t - \frac{1}{t}$.

(c) Here $m \geq \frac{7}{8}u$, so $\frac{u}{8} \leq \frac{m}{7}$. Since $t \geq 1$, $\mathbb{E}Z \geq \frac{m}{2}$, and since $t \leq \frac{n}{8}$, $\mathbb{E}Z' = \frac{m(h-t)}{n-t-1} \geq \frac{3m}{8}$. Hoeffding's inequality also holds for sampling without replacement [19, Section 6], so $\mathbb{P}(Z - \mathbb{E}Z \leq -\sigma m) \leq e^{-2\sigma^2 m}$, and the same holds for Z' . Now $\{M_u < \frac{u}{8}\} \subset \{Z - \mathbb{E}Z < -\frac{5m}{14}\}$, and since $u - M_u = t + Z'$, also $\{u - M_u < \frac{u}{8}\} \subset \{Z' - \mathbb{E}Z' < -\frac{13m}{56}\}$. So $\mathbb{P}(\Omega_u^c) \leq e^{-2(5/14)^2 m} + e^{-2(13/56)^2 m} \leq 2e^{-0.10778m} \leq 2e^{-0.0943u}$.

(d) Since $\bar{r}(u-b)^2 + (1-\bar{r})b^2 \leq u^2$, we have $Q_u(b) \leq \frac{(u-2b)^2 u^2}{b^2(u-b)^2}$. On Ω_u , $b(u-b) \geq \frac{7u^2}{64}$, so $Q_u \leq \frac{4096}{49} \frac{(u-2b)^2}{u^2} \leq 83.6 \frac{(u-2b)^2}{u^2}$, and by (a) these parts of B_t add up to at most $83.6 \sum_{u>t} (\frac{(t-1)^2}{u^2} + \frac{1}{u}) \leq 83.6(t + H_{n-1}) \leq 84(t+1 + \log n)$. Off Ω_u , $b(u-b) \geq u-1 \geq \frac{u}{2}$ gives $Q_u \leq 4u^2$. For $u < 8(t+1)$ these parts add up to at most $\frac{4}{3}(8(t+1))^3 \leq 683(t+1)^3$. For $u \geq 8(t+1)$, (c) bounds them by $8u^2 e^{-0.0943u}$, and $8 \sum_{u \geq 1} u^2 e^{-0.0943u} < 19100$. $\square$

A.4 Proof of Theorem 2.6(c)

Let $n \geq 64$ be even and $0 < \varepsilon \leq \frac{1}{100(1+\log n)}$. Then $\varepsilon < 0.002$, so $\kappa \leq 2.084\varepsilon$ and $\kappa \log(h+1) \leq 2.084\varepsilon \log n < 0.021 < \frac{\log 2}{24}$. Since $\pi_t = \frac{t}{n-t} \prod_{i=1}^{t-1} \frac{h-i}{n-i}$ and $\frac{h-i}{n-i} \leq \frac{1}{2}$,

$$w_t = \frac{n+2}{4} \pi_t \leq \frac{n+2}{2(n-t)} t 2^{-t} \leq 1.032 t 2^{-t} \quad (1 \leq t \leq h). \quad (4)$$

Upper bound. By Theorem 2.6(a), $(1-\varepsilon)^{t-1} \leq 1$ and $\log(1+y) \leq y$, $\log C_n \leq \log(\varepsilon c_1) + \kappa H_{n-1} + \sum_t w_t (e^{\kappa \mathcal{G}(t)} - 1)$. Since $\log \frac{(t+1)^2}{4t} \geq 0$ and $\frac{2}{3}(t+11) > t \geq H_t$, we have $\mathcal{G}(t) > 0$. Since $\frac{(t+1)^2}{4t} \leq \frac{t+1}{2}$, $\frac{2}{3} < \log 2$ and $H_t \geq \frac{2}{3t}$, also $\mathcal{G}(t) \leq (t+11) \log(t+1) \leq 12t \log(t+1)$. So $0 < \kappa \mathcal{G}(t) \leq \frac{t}{2} \log 2$ for $t \leq h$, and $e^{\kappa \mathcal{G}(t)} - 1 \leq \kappa \mathcal{G}(t) 2^{t/2}$. By (4),

$$\sum_t w_t (e^{\kappa \mathcal{G}(t)} - 1) \leq 1.032 \cdot 2.084\varepsilon \sum_{t \geq 1} t(t+11) \log(t+1) 2^{-t/2} \leq 563.3\varepsilon,$$

since the last sum is $261.9013 \dots \leq 261.91$. Also $\kappa \leq 2\varepsilon(1+8\varepsilon)$ and $H_{n-1} \leq 1 + \log n$, so $\kappa H_{n-1} \leq 2\varepsilon \log n + 2\varepsilon + 16\varepsilon^2(1 + \log n) \leq 2\varepsilon \log n + 2.2\varepsilon$. As $c_1 = \frac{4}{n+2}$, the upper bound holds with $565.5\varepsilon \leq 566\varepsilon$.

Lower bound. Let $T = \lfloor n/8 \rfloor \geq 8$. By Proposition 2.5(a), Proposition A.5 for $t \leq T$, Lemma A.1(d) for $t > T$ and Jensen's inequality for the probability vector (w_t) ,

$$\log C_n \geq \log(\varepsilon c_1) + \sum_t w_t (t-1) \log(1-\varepsilon) + \sum_{t \leq T} w_t (\varepsilon A_t - \varepsilon^2 B_t).$$

By the inequality $\log(1+y) \geq y - y^2$ used above and Proposition 2.5(c), the first sum is at least $-(\varepsilon + \varepsilon^2) \sum_t w_t (t-1) \geq -2\varepsilon - 2\varepsilon^2$. Next $T \leq \frac{n+2}{4}$, so $A_t \geq 2(\log n - \log t - 3)$ for $t \leq T$ by Lemma A.6(b). Let $w_{\leq T} = \sum_{t \leq T} w_t$. By Jensen's inequality $\sum_t w_t \log t \leq \log \sum_t w_t t \leq \log 3$, so

$$\sum_{t \leq T} w_t \varepsilon A_t \geq 2\varepsilon(\log n - 3)w_{\leq T} - 2\varepsilon \log 3 \geq 2\varepsilon \log n - 2\varepsilon(1 - w_{\leq T}) \log n - 6\varepsilon - 2\varepsilon \log 3.$$

By (4), $1 - w_{\leq T} \leq 1.032 \sum_{t>T} t 2^{-t} = 1.032(T+2)2^{-T}$, and $n < 8(T+1)$. As $(T+2)2^{-T} \log(8T+8)$ decreases for $T \geq 8$, $2\varepsilon(1 - w_{\leq T}) \log n \leq 2.064 \cdot 10 \cdot 2^{-8} \log 72 \varepsilon < 0.35\varepsilon$. Finally Lemma A.6(d)

and Proposition 2.5(c) give $\sum_{t \leq T} w_t B_t \leq 84(3 + 1 + \log n) + 683 \cdot 124 + 19100 = 84 \log n + 104128$. Collecting terms,

$$\log C_n - \log \frac{4\varepsilon}{n+2} - 2\varepsilon \log n \geq -(2 + 0.35 + 6 + 2 \log 3) \varepsilon - \varepsilon^2(84 \log n + 104130),$$

which gives the lower bound since $2 + 0.35 + 6 + 2 \log 3 < 11$. For the last claim, the hypothesis on ε gives $\varepsilon^2(84 \log n + 1.05 \cdot 10^5) \leq \varepsilon \frac{84 \log n + 1.05 \cdot 10^5}{100(1 + \log n)}$. The fraction decreases in n , and its value at $n = 64$ is $204.2 \dots$. Since $11 + 205 < 566$, the two-sided bound follows. $\square$

A.5 Proof of Theorem 2.9

Part (a). By Proposition 2.5(a) and (b), $c_2 = \sum_t [-(t-1)\pi_t + \partial_\varepsilon V_{t+1}(1)|_{\varepsilon=0}]$. In Proposition A.5 the bridge law does not depend on ε , and each ϖ_u is affine in ε and equals 1 at $\varepsilon = 0$. So by the conditional mean computed in its proof, $\partial_\varepsilon V_{t+1}(1)|_{\varepsilon=0} = \pi_t \sum_u \mathbb{E}_t^{\text{br}}[(\varpi_u - 1)/\varepsilon] = \pi_t A_t$. Divide by c_1 .

Part (b), the limit. Fix t and let $n \geq 16(t+1)$. For $1 \leq b \leq u-1$,

$$\theta_u(b) = \frac{n}{n-u} \left[\frac{2(u-2b)^2}{u^2} + \vartheta_u(b) \right], \quad \vartheta_u(b) = \frac{(u-2b)^4}{2b(u-b)u^2} \geq 0,$$

since $\frac{1}{2b(u-b)} = \frac{2}{u^2} + \frac{(u-2b)^2}{2b(u-b)u^2}$. By Lemma A.6(a), the first part of A_t is

$$\sum_{u=t+1}^{n-1} \frac{2n \mathbb{E}_t^{\text{br}}(u-2M_u)^2}{(n-u)u^2} = \sum_{u=t+1}^{n-1} \frac{2n(t-1)^2(n-u)}{(n-t-1)^2u^2} + P_t \sum_{u=t+1}^{n-1} \frac{u-t-1}{u^2},$$

with P_t as in the proof of Lemma A.6(b). The terms of the first sum are at most $8(t-1)^2/u^2$ and tend to $2(t-1)^2/u^2$, so the first sum tends to $2(t-1)^2\psi'(t+1)$, where $\psi'(t+1) = \sum_{u>t} u^{-2}$. The second sum is $H_{n-1} - H_t - (t+1)(\psi'(t+1) - \psi'(n)) = \log n + \gamma - H_t - (t+1)\psi'(t+1) + o(1)$, and $P_t = 2 + O(1/n)$. So the first part of A_t is $2 \log n + 2\gamma - 2H_t + 2((t-1)^2 - t-1)\psi'(t+1) + o(1)$.

The second part is $\sum_{u=t+1}^{n-1} \frac{n}{n-u} \mathbb{E}_t^{\text{br}} \vartheta_u(M_u)$. For fixed u the law of $M_u - 1$ tends to $\text{Bin}(u-t-1, \frac{1}{2})$ and $\frac{n}{n-u} \rightarrow 1$, so each term tends to $\mathbb{E} \vartheta_u(\bar{M}_u)$, where $\bar{M}_u = 1 + N_u$ with $N_u \sim \text{Bin}(u-t-1, \frac{1}{2})$. We dominate the terms uniformly in n . We have $u - 2M_u - \mathbb{E}(u - 2M_u) = -2(Z - \mathbb{E}Z)$. The last $n-u$ steps also form a uniform sample, so $h - M_u$ is hypergeometric with $n-u$ draws and $u - 2M_u - \mathbb{E}(u - 2M_u) = 2(h - M_u - \mathbb{E}(h - M_u))$. By Hoeffding's comparison with sampling with replacement [19, Theorem 4], applied to a convex fourth power, the fourth central moment of each is at most that of the binomial law with the same mean, which is $kpq(1+3(k-2)pq) \leq k^2/2$ for k draws. With $|\mathbb{E}(u - 2M_u)| \leq t$ and $(x+y)^4 \leq 8(x^4+y^4)$, this gives $\mathbb{E}_t^{\text{br}}(u - 2M_u)^4 \leq 64 \min(u, n-u)^2 + 8t^4$. Also $\frac{n}{n-u} \leq 2u$, because $u(n-u) \geq n-1 \geq \frac{n}{2}$, and $\frac{n}{n-u} \min(u, n-u)^2 \leq 2u^2$, which is clear for $u \leq \frac{n}{2}$ and follows from $n(n-u) \leq 2u^2$ for $u > \frac{n}{2}$. On Ω_u we have $2M_u(u - M_u)u^2 \geq \frac{7}{32}u^4$, so the part of the u -th term on Ω_u is at most $\frac{32}{7u^4}(128u^2 + 16t^4u)$. Off Ω_u , $M_u(u - M_u) \geq \frac{u}{2}$ gives $\vartheta_u \leq u$, so that part is at most $2u^2 \mathbb{P}_t^{\text{br}}(\Omega_u^c)$. This is at most $4u^2 e^{-0.0943u}$ for $u \geq 8(t+1)$ by Lemma A.6(c), and at most $2u^2$ for the finitely many smaller u . The bound is summable and does not depend on n . By dominated convergence,

$$A_t - 2 \log n \rightarrow \tilde{A}_t := 2\gamma - 2H_t + 2((t-1)^2 - t-1)\psi'(t+1) + \sum_{u>t} \mathbb{E} \vartheta_u(\bar{M}_u).$$

Since $\mathbb{E}(u - 2\bar{M}_u)^2 = (t-1)^2 + (u-t-1)$, the same computation with $\frac{n}{n-u}$ replaced by 1 and n by a cutoff U shows that $\tilde{A}_t = \lim_{U \rightarrow \infty} (\sum_{u=t+1}^U \bar{\theta}_u - 2 \log U)$, where $\bar{\theta}_u = \mathbb{E} \frac{(u-2\bar{M}_u)^2}{2\bar{M}_u(u-\bar{M}_u)}$.

Part (b), the closed form. Let $m = u - t - 1$, $N = N_u$, $y = \frac{1+x}{2}$ and $\sigma(x) = \sum_{j=0}^{t-2} x^j$, with $\sigma = 0$ for $t = 1$. Since $(u - 2b)^2 = u^2 - 4b(u - b)$ and $\frac{1}{b(u-b)} = \frac{1}{u}(\frac{1}{b} + \frac{1}{u-b})$, we have $\bar{\theta}_u = \frac{u}{2}\mathbb{E}[\frac{1}{1+N} + \frac{1}{u-1-N}] - 2$, and $u - 1 - N$ has the law of $t + N$ because $N \stackrel{d}{=} m - N$. For $b \geq 1$, $\mathbb{E}\frac{1}{b+N} = \int_0^1 x^{b-1} \mathbb{E}x^N dx = \int_0^1 x^{b-1} y^m dx$. Hence

$$\bar{\theta}_u = u \int_0^1 \frac{1+x^{t-1}}{2} y^m dx - 2 = \left[\frac{2t}{m+1} - \left(1 + \frac{t}{m+1}\right) 2^{-m} \right] - \mathcal{T}_m, \quad \mathcal{T}_m = u \int_0^1 (1-y)\sigma(x)y^m dx,$$

using $\frac{1+x^{t-1}}{2} = 1 - (1-y)\sigma(x)$, $\int_0^1 y^m dx = \frac{2(1-2^{-m-1})}{m+1}$ and $u = m+1+t$. Split $u = (m+1) + t$ in $\mathcal{T}_m$. Since $\frac{d}{dx}y^{m+1} = \frac{m+1}{2}y^m$, integration by parts gives $(m+1) \int_0^1 x^j \frac{1-x}{2} y^m dx = -\delta_{j0}2^{-m-1} + \int_0^1 y^{m+1}((j+1)x^j - jx^{j-1}) dx$ for $j \geq 0$, and the integrands add up to $(t-1)x^{t-2}$ over $0 \leq j \leq t-2$. With $I(j, k) = \int_0^1 x^j y^k dx$ and $(1-y)\sigma(x) = \frac{1-x^{t-1}}{2}$,

$$\mathcal{T}_m = -[t \geq 2]2^{-m-1} + (t-1)I(t-2, m+1) + \frac{t}{2} \int_0^1 (1-x^{t-1})y^m dx.$$

Sum over $0 \leq m \leq m_0$, so that $U = m_0 + t + 1$. Since $\sum_{m \leq m_0} y^m = \frac{2(1-y^{m_0+1})}{1-x}$,

$$\begin{aligned} \sum_{u=t+1}^U \bar{\theta}_u &= 2tH_{m_0+1} - \sum_{m=0}^{m_0} \left(1 + \frac{t}{m+1}\right) 2^{-m} + [t \geq 2](1 - 2^{-m_0-1}) \\ &\quad - (t-1) \sum_{k=0}^{m_0} I(t-2, k+1) - t \int_0^1 \sigma(x)(1-y^{m_0+1}) dx. \end{aligned}$$

Let $m_0 \rightarrow \infty$. First $\sum_{m \geq 0} (1 + \frac{t}{m+1}) 2^{-m} = 2 + 2t \log 2$, and $\int_0^1 \sigma(1-y^{m_0+1}) dx \rightarrow \int_0^1 \sigma = H_{t-1}$. For $t \geq 2$ write $\sum_{k \leq m_0} I(t-2, k+1) = 2 \int_0^1 x^{t-2} y \frac{1-y^{m_0+1}}{1-x} dx$ and $x^{t-2} = 1 - (1-x^{t-2})$. The substitution $x = 2y - 1$ gives $\int_0^1 y \frac{1-y^{m_0+1}}{1-x} dx = \sum_{j=2}^{m_0+2} \frac{1-2^{-j}}{j} = H_{m_0+2} - \frac{1}{2} - \log 2 + o(1)$, and by dominated convergence $\int_0^1 (1-x^{t-2}) y \frac{1-y^{m_0+1}}{1-x} dx \rightarrow \int_0^1 \sum_{j=0}^{t-3} x^j \frac{1+x}{2} dx = \frac{1}{2}(H_{t-2} + H_{t-1} - 1)$. So $\sum_{k \leq m_0} I(t-2, k+1) = 2H_{m_0+2} - 2 \log 2 - H_{t-2} - H_{t-1} + o(1)$. Collecting terms, with $2tH_{m_0+1} - 2(t-1)H_{m_0+2} = 2H_{m_0+1} + o(1)$,

$$\sum_{u=t+1}^U \bar{\theta}_u = 2H_{m_0+1} - 2 - 2 \log 2 + [t \geq 2] + (t-1)H_{t-2} - H_{t-1} + o(1).$$

Here $[t \geq 2] + (t-1)H_{t-2} - H_{t-1} = (t-2)H_{t-1}$ for all $t \geq 1$ (for $t = 1$ the term $(t-1)H_{t-2}$ is absent), and $H_{m_0+1} = \log U + \gamma + o(1)$. So $\tilde{A}_t = 2(\gamma - 1 - \log 2) + (t-2)H_{t-1} = A_t^\infty$.

Part (c). Since $\sum_t w_t = 1$, $k_2(n) - 2 \log n = \sum_{t \leq h} w_t [A_t - 2 \log n - (t-1)]$. Both sides of Proposition A.4(c) equal π_t at $\varepsilon = 0$, and $\kappa'(0) = 2$. Comparing right derivatives at 0 and using part (a) gives $A_t \leq 2(H_{n-1} - H_t) + \frac{4}{3t} + 2(t+11)(\log \frac{(t+1)^2}{4t} + \frac{2}{3})$, so $A_t - 2 \log n \leq 2\gamma + \frac{4}{3} + 2(t+11) \log(t+1)$ as in Section A.4. Also $A_t - 2 \log n \geq -2 \log t - 6$ for $t \leq \frac{n+2}{4}$ by Lemma A.6(b), and $A_t \geq 0$ gives $A_t - 2 \log n \geq -2 \log(4t)$ for $t > \frac{n+2}{4}$. With (4), for $n \geq 64$ the t -th term is at most $C(1+t \log(t+1))t2^{-t}$ in absolute value, with C independent of n . By part (b), Proposition 2.5(c) and dominated convergence, $k_2(n) - 2 \log n \rightarrow \sum_{t \geq 1} t2^{-t-1} [A_t^\infty - (t-1)] = K_2$.

To sum the series, note $\sum_t t2^{-t-1} = 1$ and $\sum_t t(t-1)2^{-t-1} = 2$. Let $\mathcal{H}(x) = \sum_{k \geq 0} H_k x^k = \frac{\ell}{1-x}$ with $\ell = -\log(1-x)$, and $D = \frac{d}{dx}$. With $k = t-1$,

$$\sum_{t \geq 1} t(t-2)2^{-t-1} H_{t-1} = \frac{1}{4} \sum_{k \geq 0} (k^2 - 1) H_k 2^{-k} = \frac{1}{4} [(xD)^2 \mathcal{H} - \mathcal{H}]_{x=1/2}.$$

Here $(xD)^2\mathcal{H} = x\left[\frac{1+\ell}{(1-x)^2} + \frac{x}{(1-x)^3} + \frac{2x(1+\ell)}{(1-x)^3}\right]$ equals $8 + 6\log 2$ at $x = \frac{1}{2}$, and $\mathcal{H}(\frac{1}{2}) = 2\log 2$. So the sum is $2 + \log 2$, and $K_2 = 2(\gamma - 1 - \log 2) + (2 + \log 2) - 2 = 2\gamma - 2 - \log 2$. $\square$

A.6 Proof of the bound in Remark 2.10

Let $n \geq 64$, $0 < \varepsilon \leq \frac{1}{2}$ and $T = \lfloor n/8 \rfloor$. We repeat the lower bound of Section A.4 but keep A_t . With $\log(1 - \varepsilon) \geq -\varepsilon - \varepsilon^2$, $\sum_t w_t(t - 1) \leq 2$ and $\sum_t w_t[A_t - (t - 1)] = k_2(n)$, this gives

$$\log C_n \geq \log(\varepsilon c_1) + \varepsilon k_2(n) - \varepsilon \delta_n - \varepsilon^2(84 \log n + 1.05 \cdot 10^5), \quad \delta_n = \sum_{t>T} w_t A_t.$$

By the upper bound for A_t in the proof of Theorem 2.9(c) and (4), $\delta_n = O(n^2 \log n 2^{-n/8})$. Let $F(x) = \log C_n(\frac{x}{n}) - \log E_n(\frac{x}{n})$, and let $\hat{F}(x)$ be the left side minus the right side of the equation for $\hat{x}_n$. Since $\log(\varepsilon c_1) = \log \frac{4x}{n(n+2)}$ and $\log E_n = \log \frac{1}{2} + (n-1) \log(1 - \frac{x}{n})$, we get $F(x) \geq \hat{F}(x) - \rho_n(x)$ for $0 < x \leq \frac{n}{2}$, with $\rho_n(x) = \frac{x}{n} \delta_n + \frac{x^2}{n^2}(84 \log n + 1.05 \cdot 10^5)$. For large n , $k_2(n) > 0$ by Theorem 2.9(c). Then $\hat{F}$ is strictly increasing on $(0, n)$ with $\hat{F}'(x) = \frac{1}{x} + \frac{k_2(n)}{n} + \frac{n-1}{n-x} \geq 1$ on $[1, n)$, it tends to $+\infty$ as $x \rightarrow n$, and $\hat{F}(1) = -2\log n + O(1) < 0$. So $\hat{x}_n$ is its only zero, and $\hat{x}_n > 1$. By Proposition 2.4, $x_n < \frac{n}{2}$ and $F(x_n) = 0$. If $x_n > \hat{x}_n$, then $x_n - \hat{x}_n \leq \hat{F}(x_n) - \hat{F}(\hat{x}_n) \leq \rho_n(x_n)$. Since $x_n = O(\log n)$ by Theorem 2.7, we get $x_n \leq \hat{x}_n + O((\log n)^3/n^2)$. $\square$

B Proofs for Section 3

We keep the conventions of Section 3. Unless stated otherwise, $n \geq 64$, $0 < \varepsilon \leq \frac{1}{4}$ and $x = n\varepsilon \geq 1$. Since M is independent of X_1 , $f_n(m) = \mathbb{P}(M_n = m)$. We write $\beta_v(s) = \mathbb{P}(|T_v| \geq s)$, with $\beta_v(s) = 1$ for $s \leq 1$. Every path of a chain $M^{[k_0]}$ is nondecreasing in time, since its steps are 0 or 1. Recall that in a random recursive tree on $[n]$ vertex $k+1$ is joined to a uniform vertex of $[k]$. So the tree is uniform over the $(n-1)!$ increasing trees on $[n]$, that is, the trees rooted at 1 in which labels increase away from the root.

B.1 Trees, subtrees and the first flip

Lemma B.1 (splitting). *Let T be a random recursive tree on $[n]$, let $2 \leq v \leq n$ and let $S \subseteq [n]$ with $\min S = v$. Given that T_v has vertex set S , the restrictions of T to S and to $[n] \setminus S$, relabeled in increasing order, are independent random recursive trees. They are independent of the parent of v , which is uniform on $[v-1]$.*

Proof. The subtree T_v has vertex set S exactly when every $u \in S \setminus \{v\}$ has its parent in S and every $u \notin S$ with $u \geq 2$ has its parent outside S . Indeed, then the ancestors of a vertex of S stay in S until they reach v , whose parent is below $\min S$, and the ancestors of a vertex outside S never enter S . The converse is clear. So these trees are given by free and independent choices. The i -th smallest element of S , for $i \geq 2$, picks one of the $i-1$ smaller elements of S . The vertex v picks any element of $[v-1]$. The i -th smallest element of $[n] \setminus S$, for $i \geq 2$, picks one of the $i-1$ smaller elements of $[n] \setminus S$. All increasing trees on $[n]$ are equally likely, so the conditional law is uniform on this product of choices, which is the claim. $\square$

Proof of Lemma 3.1. Vertex $k+1$ is joined to U_k , which is uniform on $[k]$ and independent of the earlier choices, so T is a random recursive tree. The marks are the flip indicators of steps $2, \dots, n$, so they are i.i.d. Bernoulli(ε) and independent of T . We use induction on k . The path

from $k + 1$ to the root is $k + 1$ followed by the path from U_k . So $\sigma(k + 1) = \mu_{k+1} + \sigma(U_k)$, and $X_{k+1} = X_{U_k}(-1)^{\mu_{k+1}} = X_1(-1)^{\sigma(k+1)}$. Now set $\mu_2 = \dots = \mu_{k_0} = 0$. Then the vertices $1, \dots, k_0$ are even. If b of the vertices in $[k]$ are odd, $k \geq k_0$, then vertex $k + 1$ is odd with conditional probability $(1 - \varepsilon)\frac{b}{k} + \varepsilon(1 - \frac{b}{k}) = r_k(b)$, whatever else is known about $[k]$. So the odd count is a copy of $M^{[k_0]}$. $\square$

Proof of Lemma 3.2. Fix v . For $k \geq v$, if T_v contains s of the vertices in $[k]$, then $k + 1$ joins T_v with probability s/k . This is Pólya's urn started at time v with 1 ball inside and $v - 1$ outside. A given order of i joins among the $n - v$ later vertices has probability

$$\frac{i!(v-1)v \cdots (n-i-2)}{v(v+1) \cdots (n-1)} = \frac{(v-1)i!(n-i-2)!}{(n-1)!}.$$

We multiply by $\binom{n-v}{i}$ and put $d = i + 1$. This gives $\beta_v(d) = \frac{(v-1)(n-v)!(n-d-1)!}{(n-v-d+1)!(n-1)!}$, which equals $\binom{n-d-1}{v-2} / \binom{n-1}{v-1}$. By Pascal's rule $\binom{n-s}{v-1} - \binom{n-s-1}{v-1} = \binom{n-s-1}{v-2}$ the tail telescopes. So, for $1 \leq s \leq n-1$,

$$\bar{\beta}_v(s) = \binom{n-s}{v-1} / \binom{n-1}{v-1} = \prod_{i=1}^{v-1} \left(1 - \frac{s-1}{n-i}\right) \leq e^{-(v-1)(s-1)/(n-1)}, \quad \beta_v(s) = \frac{v-1}{n-s} \bar{\beta}_v(s), \quad (5)$$

where the product vanishes as soon as one factor does. Where they are defined, the ratios are

$$\frac{\beta_v(d)}{\beta_v(d-1)} = 1 - \frac{v-2}{n-d}, \quad \frac{\bar{\beta}_v(s-1)}{\bar{\beta}_v(s)} = 1 + \frac{v-1}{n-s-v+2}. \quad (6)$$

So β_v is nonincreasing. Also $\mathbb{E}|T_v| = n/v$, since the fraction of $[k]$ that lies in T_v is a martingale in $k \geq v$ that starts at $1/v$. For the 2 sums put $i = n - v$. Then

$$(v-1) \frac{(n-v)!}{(n-v-d+1)!} = (d-1)! \binom{i}{d-1} \binom{n-1-i}{1},$$

$$(v-1)(v-2) \frac{(n-v)!}{(n-v-d+1)!} = 2(d-1)! \binom{i}{d-1} \binom{n-1-i}{2}.$$

By the Chu–Vandermonde identity, $\sum_i \binom{i}{d-1} \binom{n-1-i}{1} = \binom{n}{d+1}$ and $\sum_i \binom{i}{d-1} \binom{n-1-i}{2} = \binom{n}{d+2}$. Multiplying by $(n-d-1)!/(n-1)!$ gives $\frac{n}{d(d+1)}$ and $\frac{2n(n-d-1)}{d(d+1)(d+2)}$. $\square$

Proof of Lemma 3.3. Marks and tree are independent, so $\mathbb{P}(V = j, |T_j| = d) = \varepsilon(1 - \varepsilon)^{j-2} \beta_j(d)$. Condition on $V = j$ and on the vertex set S of T_j , with $|S| = d$. Then the marks on $j + 1, \dots, n$ are still i.i.d. Bernoulli(ε), and by Lemma B.1 the trees on S and on $[n] \setminus S$ are independent random recursive trees. The vertices $1, \dots, j - 1$ carry no mark, so the parent of j is even and j is odd. A vertex $u \in S$ is therefore odd exactly when the path from u up to j , with j excluded, has an even number of marks. By Lemma 3.1, applied to the tree on S rooted at j , the number of $u \in S$ for which this number is odd is distributed as Y_d . So S contains $d - Y_d$ odd vertices. The path from a vertex outside S to the root stays outside S . The tree on $[n] \setminus S$ has the unmarked vertices $1, \dots, j - 1$ first and i.i.d. marks after them, so by Lemma 3.1 its odd count is distributed as $M_{n-d}^{[j-1]} = Z_{n-d}^{(j)}$. The 2 counts are independent, and their law depends on S only through d . Summing over j and d gives the formula, since $M_n = 0$ when there is no mark. $\square$

Only the law of each pair $(Y_d, Z_{n-d}^{(j)})$ enters Lemma 3.3. So from now on we take $(Y_d)_{d \geq 1}$ to be one path of $M^{[1]}$ and $(Z_t^{(j)})_{t \geq j-1}$ one path of $M^{[j-1]}$, independent of each other and of (V, D) . Then $D - Y_D + Z_{n-D}^{(V)}$ on $\{V < \infty\}$, and 0 otherwise, has the law of M_n . Given $V = j$, D has law β_j .

Proof of Lemma 3.4. We follow the induction in the proof of Theorem 1.1 of [9]. Let $\phi_k(c) = \mathbb{P}(M_k^{[k_0]} = c)$, with $\phi_k(c) = 0$ outside $\{0, \dots, k - k_0\}$, and put $d_k(c) = \phi_k(c+1) - \phi_k(c)$ for $c \geq 0$. From $\phi_{k+1}(c) = (1 - r_k(c))\phi_k(c) + r_k(c-1)\phi_k(c-1)$ and $r_k(c+1) + r_k(c-1) = 2r_k(c)$ we get

$$\begin{aligned} d_{k+1}(c) &= (1 - r_k(c+1))d_k(c) + r_k(c-1)d_k(c-1) \quad (c \geq 1), \\ d_{k+1}(0) &= (1 - r_k(1))d_k(0) + \left(\varepsilon - \frac{1-2\varepsilon}{k}\right)\phi_k(0). \end{aligned}$$

Note that $0 \leq r_k(b) \leq 1$ for $0 \leq b \leq k$, and $d_k(c) = 0$ for $c > k - k_0$. So in the first identity each coefficient is nonnegative whenever it multiplies a nonzero term. We show by induction on k that there is c_k with $d_k(c) \geq 0$ for $c < c_k$ and $d_k(c) \leq 0$ for $c \geq c_k$. For $k = k_0$ take $c_{k_0} = 0$. Suppose it holds for k . The first identity keeps the sign of $d_{k+1}(c)$ for $1 \leq c \leq c_k - 1$ and for $c \geq c_k + 1$. If $c_k = 0$ we may take $c_{k+1} \in \{0, 1\}$. Let $c_k \geq 1$, so $d_k(0) \geq 0$. If $\varepsilon(k+2) < 1$, then $\varepsilon - \frac{1-2\varepsilon}{i} < 0$ for all $i \leq k$, and the second identity with $d_{k_0}(0) = -1$ and $\phi_i(0) > 0$ gives $d_i(0) < 0$ for $k_0 \leq i \leq k$. This contradicts $d_k(0) \geq 0$. So $\varepsilon(k+2) \geq 1$, the last term is nonnegative, and $d_{k+1}(0) \geq 0$. Then $c_{k+1} = c_k$ or $c_k + 1$ works, according to the sign of $d_{k+1}(c_k)$. $\square$

B.2 A priori bounds

Lemma B.2 (tail). *Let $0 < \varepsilon < 1$, $k_0 \geq 1$ and $N \geq k_0$. For every integer $z \geq 1$,*

$$\mathbb{P}(M_N^{[k_0]} \geq z) \leq \frac{\varepsilon N H_z}{z}, \quad \mathbb{E}M_N^{[k_0]} \leq \varepsilon N \log \frac{N}{k_0}.$$

In particular $\mathbb{P}(M_n \geq z) \leq x H_z / z$, and $\mathbb{P}(Z_t^{(j)} \geq z) \leq x H_z / z$ for $t \leq n$.

Proof. We use the tree of Lemma 3.1 on $[N]$, with marks only on $v > k_0$. Every odd vertex lies in the subtree of a marked vertex, so $M_N^{[k_0]} \leq \sum_{v \text{ marked}} |T_v|$. If no marked subtree has z or more vertices, the sum equals the sum of $|T_v| \mathbf{1}\{|T_v| < z\}$. By the union bound, Markov's inequality and the independence of marks and tree,

$$\mathbb{P}(M_N^{[k_0]} \geq z) \leq \varepsilon \sum_{v=2}^N \mathbb{P}(|T_v| \geq z) + \frac{\varepsilon}{z} \sum_{v=2}^N \mathbb{E}[|T_v|; |T_v| < z].$$

By Lemma 3.2 the first sum is $\sum_{d=z}^{N-1} \frac{N}{d(d+1)} \leq \frac{N}{z}$ and the second is $\sum_{d=1}^{z-1} \frac{N}{d+1} = N(H_z - 1)$. For the mean, $\mathbb{E}M_N^{[k_0]} \leq \varepsilon \sum_{v=k_0+1}^N \mathbb{E}|T_v| = \varepsilon \sum_{v=k_0+1}^N \frac{N}{v} \leq \varepsilon N \log \frac{N}{k_0}$. $\square$

Lemma B.3 (late start). *Let $j \geq 2$, $N \geq j - 1$ and $z \geq 0$. Then*

$$\mathbb{P}(M_N^{[j-1]} \geq z) \leq \exp\left\{-\frac{j-1}{2} \left(\frac{z}{N} - \varepsilon \log \frac{2N}{j-1}\right)\right\}.$$

Proof. Write $C_k = M_k^{[j-1]}$ and $\nu = (j-1)/2$. For $k > \nu$ let $\theta_k = \log \frac{k}{k-\nu}$, so that $e^{\theta_k} - 1 = \frac{\nu}{k-\nu}$. Since $r_k(c) \leq \varepsilon + c/k$ and $1 + v \leq e^v$,

$$\mathbb{E}[e^{\theta_{k+1}C_{k+1}} \mid C_k = c] \leq \exp\left\{c\left(\theta_{k+1} + \frac{\nu}{k(k+1-\nu)}\right) + \frac{\varepsilon\nu}{k+1-\nu}\right\}.$$

Now $\theta_k - \theta_{k+1} = \log(1+w)$ with $w = \frac{\nu}{(k-\nu)(k+1)}$, and $\log(1+w) \geq \frac{w}{1+w} = \frac{\nu}{k(k+1-\nu)}$. So $\mathbb{E}e^{\theta_{k+1}C_{k+1}} \leq e^{\varepsilon\nu/(k+1-\nu)} \mathbb{E}e^{\theta_k C_k}$ for $k \geq j-1$. Starting from $C_{j-1} = 0$,

$$\mathbb{E}e^{\theta_N C_N} \leq \exp\left\{\varepsilon\nu \sum_{k=j}^N \frac{1}{k-\nu}\right\} \leq \exp\left\{\varepsilon\nu \log \frac{N-\nu}{\nu}\right\} \leq \exp\left\{\varepsilon\nu \log \frac{2N}{j-1}\right\}.$$

Markov's inequality with $\theta_N \geq \nu/N$ gives the claim. $\square$

Lemma B.4 (the first flipped family). *For $1 \leq d \leq n-1$,*

$$\frac{x}{d(d+1)} \left(1 - \frac{2x(n-d-1)}{n(d+2)}\right) \leq \mathbb{P}(D=d) \leq \frac{x}{d(d+1)}, \quad \mathbb{P}(D \geq d) \leq x \left(\frac{1}{d} - \frac{1}{n}\right).$$

Proof. By Lemma 3.3, $\mathbb{P}(D=d) = \sum_j \varepsilon(1-\varepsilon)^{j-2} \beta_j(d)$, and $1 - (j-2)\varepsilon \leq (1-\varepsilon)^{j-2} \leq 1$. Apply the 2 sums of Lemma 3.2, and sum the upper bound over $d' \geq d$. $\square$

B.3 Proof of Theorem 3.7

Let $\Psi_n(\theta) = \sum_{d=1}^{n-1} \frac{e^{i\theta d} - 1}{d(d+1)}$, so that CP_n has characteristic function $\widehat{\text{CP}}_n(\theta) = e^{x\Psi_n(\theta)}$. Here only $n \geq 2$ and $\varepsilon \leq \frac{1}{4}$ are assumed.

Step 1. Let $Z_+ = \sum_{v \geq 2} \mu_v |T_v|$. Subtrees are nested or disjoint. If no marked vertex has a marked strict descendant, the marked subtrees are disjoint, each of their vertices has exactly one mark on its path, and the other vertices have none. Then $M_n = Z_+$. Hence

$$\mathbb{P}(M_n \neq Z_+) \leq \varepsilon^2 \mathbb{E} \sum_{v=2}^n (|T_v| - 1) = \varepsilon^2 \sum_{v=2}^n \left(\frac{n}{v} - 1\right) \leq \frac{x^2 \log n}{n}.$$

Step 2. Fix $\theta \in [-\pi, \pi]$. Given T , $\mathbb{E}[e^{i\theta Z_+} \mid T] = \prod_{v \geq 2} (1 + a_v)$ with $a_v = \varepsilon(e^{i\theta|T_v|} - 1)$. Here $|a_v| \leq 2\varepsilon \leq \frac{1}{2}$ and $\text{Re } a_v \leq 0$. Put $W = \sum_v a_v$. Since $|\log(1+a) - a| \leq |a|^2$ for $|a| \leq \frac{1}{2}$ and $|e^w - 1| \leq |w|e^{|w|}$,

$$\left| \prod_v (1 + a_v) - e^W \right| \leq e^{\text{Re } W} \cdot 4\varepsilon^2(n-1) e^{4\varepsilon^2(n-1)} \leq \frac{4x^2}{n} e^{4x^2/n}.$$

Step 3. Write $W = \varepsilon F$ with $F = \sum_{v \geq 2} (e^{i\theta|T_v|} - 1)$. By Lemma 3.2, $\mathbb{E}W = x\Psi_n(\theta) = \log \widehat{\text{CP}}_n(\theta)$. Put $\delta = W - \mathbb{E}W$. Since $\text{Re } W \leq 0$ and $\text{Re } \mathbb{E}W \leq 0$, $e^W - e^{\mathbb{E}W} - e^{\mathbb{E}W} \delta = \delta^2 \int_0^1 (1-s) e^{\mathbb{E}W + s\delta} ds$ has modulus at most $\frac{1}{2}|\delta|^2$. As $\mathbb{E}\delta = 0$, $|\mathbb{E}e^W - e^{\mathbb{E}W}| \leq \frac{1}{2}\varepsilon^2 \text{Var } F$, where $\text{Var } F = \mathbb{E}|F - \mathbb{E}F|^2$.

Step 4. F is a function of the independent parents $U_1, \dots, U_{n-1}$. Let $F^{(k)}$ be F with U_k replaced by an independent copy U'_k . The Efron–Stein inequality [20] gives $\text{Var } F \leq \frac{1}{2} \sum_k \mathbb{E}|F - F^{(k)}|^2$. (We use it for a complex function of independent variables that are not identically distributed. It follows by writing $F - \mathbb{E}F$ as the sum of the martingale differences $\zeta_k = \mathbb{E}[F - \mathbb{E}_k F \mid U_1, \dots, U_k]$, where $\mathbb{E}_k$ averages over U_k alone, and by $\mathbb{E}|\zeta_k|^2 \leq \mathbb{E}|F - \mathbb{E}_k F|^2 = \frac{1}{2}\mathbb{E}|F - F^{(k)}|^2$.) Replacing

U_k moves the subtree of $k+1$ from U_k to U'_k . So $|T_v|$ changes only for vertices $v \geq 2$ on the path from U_k or from U'_k to the root, and these paths do not depend on U_k , since $U_k, U'_k \leq k$. There are at most $\text{dp}(U_k) + \text{dp}(U'_k)$ such vertices, where $\text{dp}(u)$ is the depth of u , and each term of F moves by at most 2. Hence $\mathbb{E}|F - F^{(k)}|^2 \leq 8\mathbb{E}[\text{dp}(U_k)^2 + \text{dp}(U'_k)^2] = 16\mathbb{E}\text{dp}(U_k)^2$. Since $\text{dp}(u+1) = 1 + \text{dp}(U_u)$, the generating function $g_u(s) = \mathbb{E}s^{\text{dp}(u)}$ satisfies $g_2(s) = s$ and $u g_{u+1}(s) = (u-1+s)g_u(s)$. So $g_u(s) = \prod_{i=1}^{u-1} \frac{i-1+s}{i}$, $\text{dp}(u)$ is a sum of independent Bernoulli($1/i$) variables with $i < u$, and $\mathbb{E}\text{dp}(u)^2 \leq H_{u-1}^2 + H_{u-1}$. As U_k is independent of the depths of $1, \dots, k$, we get $\text{Var } F \leq \frac{1}{2}(n-1) \cdot 16(H_n^2 + H_n) \leq 8n(H_n^2 + H_n)$. So the bound of Step 3 is at most $4x^2(H_n^2 + H_n)/n$.

Step 5. By Fourier inversion on $\mathbb{Z}$, $|\mathbb{P}(Z_+ = m) - \text{CP}_n(m)| \leq \sup_\theta |\mathbb{E}e^{i\theta Z_+} - \widehat{\text{CP}}_n(\theta)|$, which is at most the sum of the bounds of Steps 2 to 4. With Step 1 this proves Theorem 3.7. $\square$

B.4 Local and cumulative bounds

Theorem B.5 (local lower bound). (a) For $2 \leq m \leq n/4$,

$$f_n(m) \geq \frac{x}{m(m+1)} [1 - e_1(m)],$$

$$e_1(m) = \frac{2x}{m+2} + \frac{xH_m}{m} + 2\varepsilon H_{m+1} + 11\varepsilon H_{2m} + 4(m+2)e^{-(m-1)/9}.$$

(b) For $n/4 < m \leq n-16$, with $r = n-m$,

$$f_n(m) \geq \frac{x}{m(m+1)} [1 - e_2(m)],$$

$$e_2(m) = \frac{2x}{m+2} + \frac{xH_m}{m} + \frac{8xH_n}{r} + \frac{3xH_n}{m+2} + \tau(r),$$

where $\tau(r) = \frac{n}{r}(0.7r + 31)e^{-(r-8)/40}$.

We write $e(m)$ for $e_1(m)$ if $m \leq n/4$ and for $e_2(m)$ if $m > n/4$.

Proof. (a) Fix $2 \leq j \leq J := \lfloor n/8 \rfloor$ and put $\Gamma_d = d - Y_d + Z_{n-d}^{(j)}$ for $1 \leq d \leq n-j+1$. With the paths chosen after Lemma 3.3, the inner sum in Lemma 3.3 is $Q_j(m) := \mathbb{E} \sum_d \beta_j(d) \mathbf{1}\{\Gamma_d = m\}$, and every step of Γ lies in $\{-1, 0, 1\}$. On the event $\{Z_{n-1}^{(j)} \leq m-1, Y_{2m} \leq m\}$ we have $\Gamma_1 = 1 + Z_{n-1}^{(j)} \leq m$, and at $d^\circ = m + Y_{2m} \leq 2m$ we have $\Gamma_{d^\circ} \geq d^\circ - Y_{d^\circ} \geq d^\circ - Y_{2m} = m$, since Y is nondecreasing. Note that $2m+j \leq n$, so these indices are in range. So $\Gamma_{d_1} = m$ for some $d_1 \leq d^\circ$, and as β_j is nonincreasing the sum is at least $\beta_j(m + Y_{2m})$. By independence,

$$Q_j(m) \geq \beta_j(m) F_j G, \quad F_j = \mathbb{E} \left[\frac{\beta_j(m + Y_{2m})}{\beta_j(m)}; Y_{2m} \leq m \right], \quad G = \mathbb{P}(Z_{n-1}^{(j)} \leq m-1),$$

with $F_j, G \in [0, 1]$, so $Q_j(m) \geq \beta_j(m)(F_j + G - 1)$. By (6), $\beta_j(m+y)/\beta_j(m) = \prod_{i=1}^y (1 - \frac{j-2}{n-m-i})$. For $y \leq m$ each fraction is at most $\frac{2(j-2)}{n} \leq \frac{1}{4}$, and $1-u \geq e^{-4u/3}$ for $0 \leq u \leq \frac{1}{4}$. So $\beta_j(m+y) \geq \beta_j(m)(1 - \frac{8y(j-2)}{3n})$. Lemma B.2 gives $\mathbb{P}(Y_{2m} > m) \leq 2\varepsilon H_{m+1}$, $\mathbb{E}Y_{2m} \leq 2\varepsilon m H_{2m}$ and $\mathbb{P}(Z_{n-1}^{(j)} \geq m) \leq xH_m/m$. Hence

$$Q_j(m) \geq \beta_j(m) \left[1 - 2\varepsilon H_{m+1} - \frac{xH_m}{m} \right] - \frac{16\varepsilon m H_{2m}}{3n} (j-2) \beta_j(m).$$

We multiply by $\mathbb{P}(V = j) \leq \varepsilon$ and sum over $2 \leq j \leq J$. By Lemma 3.2 the last terms add up to at most $\frac{16\varepsilon m H_{2m}}{3n} \cdot \frac{2x(n-m-1)}{m(m+1)(m+2)} \leq \frac{x}{m(m+1)} 11\varepsilon H_{2m}$. For the first terms we may assume that the bracket is nonnegative, since otherwise $e_1(m) > 1$ and there is nothing to prove. By Lemma B.4,

$$\sum_{j=2}^J \mathbb{P}(V = j) \beta_j(m) \geq \frac{x}{m(m+1)} \left(1 - \frac{2x}{m+2}\right) - \varepsilon \sum_{j>J} \beta_j(m).$$

By (5), $\beta_j(m) \leq \frac{j-1}{n-m} e^{-(j-1)a}$ with $a = \frac{m-1}{n-1} \leq \frac{1}{4}$. Since $1 - e^{-a} \geq \frac{7}{8}a$,

$$\sum_{i \geq J} i e^{-ai} = e^{-aJ} \left[\frac{J}{1 - e^{-a}} + \frac{e^{-a}}{(1 - e^{-a})^2} \right] \leq e^{-aJ} \left[\frac{8J}{7a} + \frac{64}{49a^2} \right],$$

and $aJ \geq \frac{m-1}{9}$ because $\lfloor n/8 \rfloor \geq \frac{n-1}{9}$ for $n \geq 64$. With $n - m \geq \frac{3}{4}n$ and $8J \leq n$ this gives

$$\varepsilon \sum_{j>J} \beta_j(m) \leq \frac{x}{m(m+1)} \left[\frac{4m(m+1)}{21(m-1)} + \frac{256m(m+1)}{147(m-1)^2} \right] e^{-(m-1)/9} \leq \frac{x}{m(m+1)} 4(m+2) e^{-(m-1)/9},$$

where we used $\frac{m(m+1)}{m-1} \leq m+4$ and $\frac{m(m+1)}{(m-1)^2} \leq 6$ for $m \geq 2$. Finally $ab \geq a + b - 1$ for $a, b \leq 1$ combines the factors into $1 - e_1(m)$.

(b) The same argument works with $J = w = \lfloor r/8 \rfloor \geq 2$, the event $\{Z_{n-1}^{(j)} \leq m-1, Y_{m+w} \leq w\}$ and $d^\circ = m + Y_{m+w} \leq m + w$. Note that $m + w + j \leq n$. For $y \leq w$ and $j \leq J$ we have $n - m - i \geq \frac{7}{8}r$, so each fraction is at most $\frac{8(j-2)}{7r} \leq \frac{1}{7}$. As $1 - u \geq e^{-7u/6}$ for $0 \leq u \leq \frac{1}{7}$, $\beta_j(m+y) \geq \beta_j(m)(1 - \frac{4y(j-2)}{3r})$. Since $w+1 \geq r/8$, Lemma B.2 gives $\mathbb{P}(Y_{m+w} > w) \leq \frac{8xH_n}{r}$ and $\mathbb{E}Y_{m+w} \leq xH_n$. The correction term is at most

$$\frac{4xH_n}{3r} \varepsilon \sum_j (j-2) \beta_j(m) \leq \frac{x}{m(m+1)} \cdot \frac{8xH_n(r-1)}{3r(m+2)} \leq \frac{x}{m(m+1)} \cdot \frac{3xH_n}{m+2}.$$

For the dropped terms, $a = \frac{m-1}{n-1} \geq \frac{1}{5}$ since $n \geq 16$. Then $\frac{1}{1-e^{-a}} \leq 5.6$ and $\frac{e^{-a}}{(1-e^{-a})^2} \leq 31$, so $\sum_{i \geq J} i e^{-ai} \leq e^{-J/5} (5.6J + 31)$. With $m(m+1) \leq n^2$, $5.6J \leq 0.7r$ and $J \geq \frac{r-7}{8}$ the dropped part is at most $\frac{x}{m(m+1)} \tau(r)$. $\square$

A lower bound with $e(m) \geq 1$ is true but empty. The term $4(m+2)e^{-(m-1)/9}$ is at least 1 exactly for $m \leq 48$, and $\tau(r) \geq 1$ for r up to about $8 + 40 \log(0.7n)$, for example $r \leq 294$ at $n = 1600$. This is why the bound is empty at small sizes (see the discussion after Theorem 3.9).

Theorem B.6 (cumulative upper bound). *Let $6 \leq m \leq n/2$ with $m \geq 16x \log m$. Then*

$$\begin{aligned} \mathbb{P}(M_n \geq m) &\leq x \left(\frac{1}{m} - \frac{1}{n} \right) + \frac{x}{m} \delta_c(m), \\ \delta_c(m) &= \frac{x}{m} \left[36 \log m + 256 H_m \log \frac{m}{x} + 33 \right] + \frac{m}{x} e^{\varepsilon + 0.11x - 0.093m}. \end{aligned}$$

Proof. Let $J = \lfloor n/4 \rfloor$ and $Z_j = Z_{n-1}^{(j)}$. On $\{V = j\}$ we have $M_n = D - Y_D + Z_{n-D}^{(j)} \leq D + Z_j$, since the path of $Z^{(j)}$ is nondecreasing, and D is independent of Z_j with law β_j . On $\{V > J\}$ there is no mark on $2, \dots, J$, so M_n is distributed as $M_n^{[J]}$ by Lemma 3.1. Hence

$$\mathbb{P}(M_n \geq m) \leq \sum_{j=2}^J \mathbb{P}(V = j) \mathbb{E} \bar{\beta}_j(m - Z_j) + \mathbb{P}(V > J) \mathbb{P}(M_n^{[J]} \geq m).$$

(i) For $0 \leq z \leq m/8$, (6) gives $\bar{\beta}_j(m-z) \leq \bar{\beta}_j(m)\varrho_j^z$ with $\varrho_j = 1 + \frac{j-1}{n-m-j+2} \leq 1 + \frac{4(j-1)}{n}$. Since $\varrho^z - 1 \leq z(\varrho - 1)\varrho^z$,

$$\mathbb{E} \bar{\beta}_j(m - Z_j) \leq \bar{\beta}_j(m)(1 + (\varrho_j - 1)\varrho_j^{m/8} \mathbb{E} Z_j) + \mathbb{P}(Z_j > m/8).$$

(ii) The main term is $\sum_j \mathbb{P}(V = j) \bar{\beta}_j(m) \leq \mathbb{P}(D \geq m) \leq x(\frac{1}{m} - \frac{1}{n})$ by Lemma B.4.

(iii) For the shift term we use $\mathbb{P}(V = j) \leq \varepsilon$, $\bar{\beta}_j(m) \leq e^{-(j-1)(m-1)/(n-1)}$ from (5), $\varrho_j^{m/8} \leq e^{(j-1)m/(2n)}$, and $\mathbb{E} Z_j \leq x \log \frac{n}{j-1}$ from Lemma B.2. As $m \geq 6$, $\frac{m-1}{n-1} - \frac{m}{2n} \geq \frac{m-2}{2n} \geq \frac{m}{3n}$. With $i = j - 1$ and $a = \frac{m}{3n} \leq \frac{1}{6}$ the shift term is at most

$$\frac{4\varepsilon x}{n} \sum_{i=1}^n i \log \frac{n}{i} e^{-ai} \leq \frac{4\varepsilon x}{n} \left[\log \frac{m}{3} \sum_{i \geq 1} i e^{-ai} + \sum_{i \leq 1/a} i \log \frac{1}{ai} \right].$$

Here $\sum_i i e^{-ai} \leq a^{-2}$. The function $t \mapsto t \log \frac{1}{at}$ rises and then falls on $(0, 1/a]$, with maximum $\frac{1}{ea}$, so the last sum is at most $\int_0^{1/a} t \log \frac{1}{at} dt + \frac{1}{ea} = \frac{1}{4a^2} + \frac{1}{ea} \leq \frac{0.32}{a^2}$. Since $\log \frac{m}{3} + 0.32 \leq \log m$, the shift term is at most $\frac{36x^2 \log m}{m^2}$.

(iv) For the big- Z term let $K = 32 \log(m/x)$ and $j_K = \lfloor Kn/m \rfloor + 1$. Note that $m \geq 16x \log 6 > 28x$, so $K \geq 2$. For $j \leq j_K$, Lemma B.2 with $z = \lfloor m/8 \rfloor + 1$ gives $\mathbb{P}(Z_j > m/8) \leq 8xH_m/m$, and these j contribute at most $\varepsilon \frac{Kn}{m} \cdot \frac{8xH_m}{m} = \frac{x}{m} \cdot \frac{256xH_m \log(m/x)}{m}$. For $j > j_K$ we have $\frac{2(n-1)}{j-1} \leq \frac{2m}{K} \leq m$, so $\varepsilon \log \frac{2(n-1)}{j-1} \leq \frac{x \log m}{n} \leq \frac{m}{16n}$. Then Lemma B.3 with $N = n - 1$ gives $\mathbb{P}(Z_j \geq m/8) \leq e^{-(j-1)m/(32n)}$. Summing over $j - 1 \geq j_K \geq Kn/m$ gives at most $\varepsilon \frac{e^{-K/32}}{1 - e^{-m/(32n)}} \leq \varepsilon \left(\frac{32n}{m} + 1 \right) \frac{x}{m} \leq \frac{x}{m} \cdot \frac{33x}{m}$.

(v) Finally $\mathbb{P}(V > J) = (1 - \varepsilon)^{J-1} \leq e^{-\varepsilon(3n/16-1)}$, since $J \geq 3n/16$. Lemma B.3 with $j - 1 = J \in [\frac{3n}{16}, \frac{n}{4}]$ and $N = n$ gives $\mathbb{P}(M_n^{[J]} \geq m) \leq \exp\{-\frac{3m}{32} + \frac{x}{8} \log \frac{32}{3}\}$. So the last term is at most $e^{\varepsilon+0.109x-0.09375m} \leq \frac{x}{m} \cdot \frac{m}{x} e^{\varepsilon+0.11x-0.093m}$. Adding (ii) to (v) proves the theorem. $\square$

The constants are crude. When $x \approx 20$, $\delta_c(m)$ is small only for m of order 10^6 and beyond.

Lemma B.7 (mode bound). (a) If $\ell \leq n/4$ and $f_n(\ell) \geq \frac{x}{2\ell(\ell+1)}$, then $m_h \leq (\ell + 1)(1 + 2H_\ell)$.

(b) Let $\ell_0 = \lceil 8x(3 + \log x) \rceil + 100$, and suppose that $\ell_0 \leq n/4$ and $13\varepsilon(1 + H_n) \leq \frac{1}{8}$. Then $e_1(\ell) \leq \frac{1}{2}$ for $\ell_0 \leq \ell \leq n/4$, and $m_h \leq m_1(x) := (\ell_0 + 1)(1 + 2H_{\ell_0})$.

Proof. (a) Suppose $m_h \geq \ell$. By Lemma 3.4, f_n is nondecreasing on $[0, m_h]$, so $\mathbb{P}(M_n \geq \ell) \geq (m_h - \ell + 1)f_n(\ell)$. By Lemma B.2, $\mathbb{P}(M_n \geq \ell) \leq xH_\ell/\ell$. So $m_h - \ell + 1 \leq 2(\ell + 1)H_\ell$.

(b) Let $\ell_0 \leq \ell \leq n/4$. First $\frac{2x}{\ell+2} \leq \frac{2x}{\ell_0} \leq \frac{2x}{24x} = \frac{1}{12}$. Next, $\ell_0 \leq 8x(3 + \log x) + 101 \leq e^5 x^2$ for $x \geq 1$, so $H_{\ell_0} \leq 1 + \log \ell_0 \leq 2(3 + \log x)$. As H_ℓ/ℓ decreases, this gives $\frac{xH_\ell}{\ell} \leq \frac{xH_{\ell_0}}{\ell_0} \leq \frac{1}{4}$. The ε terms are at most $13\varepsilon H_n \leq \frac{1}{8}$. The last term decreases for $\ell \geq 7$, and $\ell_0 \geq 124$, so it is at most $4 \cdot 126 e^{-123/9} < 0.01$. So $e_1(\ell) \leq \frac{1}{2}$, and (a) with $\ell = \ell_0$ gives the bound on m_h . $\square$

This bound is crude. For example $m_1(10.24) \approx 8 \cdot 10^3$, while $m_h = 21$ at the crossing for $n = 1600$.

Theorem B.8 (local upper bound). *Let $n \geq 1000$ and $13\varepsilon(1 + H_n) \leq \frac{1}{8}$. Let $12 \leq m \leq n/2$ with $\lfloor 2m/3 \rfloor \geq \max(m_1(x), 16x \log m, e^2 x)$, and suppose that $\Delta(m) := \delta_c(\lfloor 2m/3 \rfloor) + \delta_\ell(m) \leq \frac{1}{16}$, where*

$$\delta_\ell(m) = \left[\frac{2x}{m+3} + \frac{xH_{m+1}}{m+1} + 4(m+3)e^{-m/9} + 13\varepsilon H_n \right] \mathbf{1}\{m < n/4\} \\ + \frac{x(17 + 33H_n + 65H_n^2) + 32 + 40.1 \log(11n)}{n}.$$

Then

$$f_n(m) \leq \frac{x}{m(m+1)} \left(1 + 3\sqrt{\Delta(m)} + \frac{3}{m} \right).$$

Proof. Step 0. Since $H_{\ell_0} \geq H_{124} > 5.4$, we have $m_1(x) \geq 11.8(\ell_0+1)$. With $m_1(x) \leq \lfloor 2m/3 \rfloor \leq n/3$ this gives $\ell_0 \leq n/35$, so Lemma B.7 applies and $m_h \leq m_1(x)$.

Step 1. Let $1 \leq k < m$ with $m - k \geq m_h$. By Lemma 3.4, f_n is nonincreasing on $[m_h, n]$, so $f_n(m) \leq f_n(\ell)$ for $m - k \leq \ell \leq m$. Hence

$$(k+1)f_n(m) \leq \mathbb{P}(M_n \geq m - k) - \mathbb{P}(M_n \geq m + 1).$$

Step 2. If $m' = m - k$ satisfies $6 \leq m' \leq n/2$ and $m' \geq 16x \log m'$, Theorem B.6 bounds the first term by $x(\frac{1}{m'} - \frac{1}{n}) + \frac{x}{m'} \delta_c(m')$.

Step 3. Let $r_1 = \lceil 8 + 40 \log(11n) \rceil$. For $n \geq 1000$ we have $381 \leq r_1 \leq n/2$. For every $\ell \in [m+1, n - r_1]$, Theorem B.5 gives $f_n(\ell) \geq \frac{x}{\ell(\ell+1)}(1 - e(\ell))$, also when $e(\ell) > 1$. Summing,

$$\mathbb{P}(M_n \geq m + 1) \geq x \left(\frac{1}{m+1} - \frac{1}{n} \right) - \text{I} - \text{II} - \text{III},$$

where $\text{III} = x(\frac{1}{n-r_1+1} - \frac{1}{n}) \leq \frac{2x(r_1-1)}{n^2} \leq \frac{2x(8+40 \log(11n))}{n^2}$, and I and II collect the terms $\frac{xe(\ell)}{\ell(\ell+1)}$ with $\ell \leq n/4$ and $\ell > n/4$. The sum I is empty unless $m < n/4$. For $\ell \geq m+1 \geq 13$ we have $\frac{2x}{\ell+2} \leq \frac{2x}{m+3}$, $\frac{xH_\ell}{\ell} \leq \frac{xH_{m+1}}{m+1}$ (as H_ℓ/ℓ decreases, because $H_\ell \geq \frac{\ell}{\ell+1}$), $4(\ell+2)e^{-(\ell-1)/9} \leq 4(m+3)e^{-m/9}$ (as this term decreases for $\ell \geq 7$), and $2\varepsilon H_{\ell+1} + 11\varepsilon H_{2\ell} \leq 13\varepsilon H_n$. With $\sum_{\ell > m} \frac{x}{\ell(\ell+1)} = \frac{x}{m+1}$ this bounds I by $\frac{x}{m+1}$ times the bracket in $\delta_\ell(m)$. In II we have $r = n - \ell \in [r_1, \frac{3}{4}n]$, $\frac{1}{\ell(\ell+1)} < \frac{16}{n^2}$ and $\sum_{\ell > n/4} \frac{1}{\ell(\ell+1)} < \frac{4}{n}$. The terms $\frac{2x}{\ell+2} + \frac{xH_\ell}{\ell} + \frac{3xH_n}{\ell+2} \leq \frac{4x(2+4H_n)}{n}$ contribute at most $\frac{16x^2(2+4H_n)}{n^2}$. The terms $\frac{8xH_n}{r}$ contribute at most $\frac{16x}{n^2} \cdot 8xH_n \sum_{r \leq n} \frac{1}{r} \leq \frac{128x^2H_n^2}{n^2}$. Finally $\tau(r) \leq n(0.7 + \frac{31}{381})e^{-(r-8)/40}$ and $\sum_{r \geq r_1} e^{-(r-8)/40} \leq \frac{e^{-(r_1-8)/40}}{1-e^{-1/40}} \leq \frac{40.51}{11n}$, so $\sum_{r \geq r_1} \tau(r) \leq 2.88$ and these terms contribute at most $\frac{46.1x}{n^2}$. Since $\frac{m+1}{x} \leq \frac{0.501n}{x}$,

$$\frac{m+1}{x}(\text{II} + \text{III}) \leq \frac{x(16.04 + 32.07H_n + 64.13H_n^2) + 23.1 + 1.002(8 + 40 \log(11n))}{n},$$

which is at most the last term of $\delta_\ell(m)$. So $\mathbb{P}(M_n \geq m + 1) \geq x(\frac{1}{m+1} - \frac{1}{n}) - \frac{x}{m+1} \delta_\ell(m)$.

Step 4. Steps 1 to 3 give

$$f_n(m) \leq \frac{x}{(m-k)(m+1)} + \frac{x}{k+1} \left[\frac{\delta_c(m-k)}{m-k} + \frac{\delta_\ell(m)}{m+1} \right].$$

Take $k = \lceil m\sqrt{\Delta(m)} \rceil \leq \frac{m}{4} + 1$. Then $m - k \geq \frac{3m}{4} - 1 \geq \lfloor \frac{2m}{3} \rfloor$ as $m \geq 12$. So $m - k \geq m_1(x) \geq m_h$ and $m - k \geq 16x \log m \geq 16x \log(m - k)$, and Steps 1 and 2 apply. Next, δ_c is nonincreasing on the integers $m' \geq \max(11, e^2 x)$. Indeed $\frac{\log m'}{m'}$, $\frac{1}{m'}$ and $m'e^{-0.093m'}$ decrease there, and $q(m') =$

$H_{m'} \log(m'/x)/m'$ satisfies $q(m' + 1) \leq q(m')$. For the last claim put $\lambda = \log(m'/x) \geq 2$ and $H = H_{m'} \geq H_{11} > 3$. Then $(m' + 1)H\lambda - m'(H + \frac{1}{m'+1})(\lambda + \frac{1}{m'}) \geq (H - 1)(\lambda - 1) - 2 \geq 0$. Since $\lfloor 2m/3 \rfloor \geq \max(e^2x, 16x \log 12) \geq 11$, we get $\delta_c(m - k) \leq \delta_c(\lfloor 2m/3 \rfloor)$. Finally $\frac{k}{m} \leq \sqrt{\Delta} + \frac{1}{m} \leq \frac{1}{3}$, so $\frac{m}{m-k} \leq 1 + \frac{3}{2}(\sqrt{\Delta} + \frac{1}{m})$ and $\frac{m+1}{m-k} \leq \frac{3}{2}(1 + \frac{1}{m})$. The bracket term is at most $\frac{x\Delta}{(k+1)(m-k)} \leq \frac{x\sqrt{\Delta}}{m(m-k)} \leq \frac{x}{m(m+1)} \cdot \frac{3}{2}\sqrt{\Delta}(1 + \frac{1}{m})$. Adding, $f_n(m) \leq \frac{x}{m(m+1)}(1 + 3\sqrt{\Delta} + \frac{3}{2m} + \frac{3\sqrt{\Delta}}{2m}) \leq \frac{x}{m(m+1)}(1 + 3\sqrt{\Delta} + \frac{3}{m})$. $\square$

B.5 Proof of Theorem 3.9 and the center

Proof of Theorem 3.9. Put $\Lambda = \log(x+1) \geq \log 2$, and recall $\mathbb{P}(A_n = n-m) = \frac{1}{2}[f_n(m) + f_n(n-m)]$. Since $x \log^2 n/n \rightarrow 0$, for large n we have $\varepsilon \leq \frac{1}{4}$, $n \geq 1000$ and $13\varepsilon(1 + H_n) \leq \frac{1}{8}$.

Step 1 (bookkeeping, uniform in $x \geq 1$). (i) Since $\log x \leq \Lambda$ and $1 \leq \Lambda/\log 2$, we have $\ell_0 + 1 \leq x(126 + 8\Lambda) \leq 190x\Lambda$. So $1 + 2H_{\ell_0} \leq 3 + 2\log \ell_0 \leq 13.5 + 2\log x + 2\log \Lambda \leq 13.5 + 4\Lambda \leq 23.5\Lambda$, and $m_1(x) \leq 4500x\Lambda^2$. (ii) Let $\omega \geq 16$ and $m \geq \omega x\Lambda^2$. Then $m \geq e^2x$ and $m \geq e^2$, since $16\log^2 2 > e^2$. On this range $\frac{\log m}{m}$ and $\frac{\log m \log(m/x)}{m}$ decrease in m , the second because $(\log m - 1)(\log \frac{m}{x} - 1) \geq 1$. Evaluating at $m = \omega x\Lambda^2$ and using $\log(\omega x\Lambda^2) \leq \log \omega + 3\Lambda$,

$$\frac{x \log m}{m} \leq \frac{\log \omega}{\omega \log^2 2} + \frac{3}{\omega \log 2}, \quad \frac{x \log m \log(m/x)}{m} \leq \frac{1}{\omega} \left(\frac{\log \omega}{\log 2} + 3 \right) \left(\frac{\log \omega}{\log 2} + 2 \right).$$

Also $\frac{x}{m} \leq \frac{1}{\omega \log^2 2}$ and $m \geq \omega \log^2 2$. As $\omega \rightarrow \infty$ these bounds tend to 0 and to ∞ , uniformly in $x \geq 1$. On the range of the theorem, $\lfloor 2m/3 \rfloor \geq \frac{2}{3}m - 1 \geq (\frac{2}{3}\omega_n - 3)x\Lambda^2$, because $x\Lambda^2 \geq \log^2 2 > \frac{1}{3}$. So (i) and (ii), with $\omega = \omega_n$ for m and $\omega = \frac{2}{3}\omega_n - 3$ for $\lfloor 2m/3 \rfloor$, show that once ω_n is large, $m \geq 12$ and $\lfloor 2m/3 \rfloor \geq \max(m_1(x), 16x \log m, e^2x)$ for every m in the range. (For $16x \log m$ note that $\log m \leq 2\log \lfloor 2m/3 \rfloor$.)

Step 2 (lower bound for $f_n(m)$). For $m \leq n/4$,

$$e_1(m) \leq \frac{2x}{m} + \frac{x(1 + \log m)}{m} + 13\varepsilon(1 + \log n) + 4(m+2)e^{-(m-1)/9},$$

which tends to 0 uniformly on the range by Step 1(ii) and $x \log n/n \rightarrow 0$. For $n/4 < m \leq n/2$ we have $r = n - m \geq n/2$ and $e_2(m) \leq \frac{8x}{n} + \frac{32xH_n}{n} + n(0.7 + \frac{62}{n})e^{-(n/2-8)/40} \rightarrow 0$.

Step 3 (lower bound for the mirror term). Let $\delta_n = (xH_n/n)^{1/2} \rightarrow 0$. If $m \geq \delta_n n$, we apply Theorem B.5(b) at $n-m \in [n/2, n-16]$, with $r = m$. Note that $\delta_n n = (xnH_n)^{1/2} \geq \sqrt{n} \geq 16$. Then $\frac{8xH_n}{m} \leq 8\delta_n$, $\tau(m) \leq n(0.7 + \frac{31}{\sqrt{n}})e^{-(\sqrt{n}-8)/40} \rightarrow 0$, and the other terms of $e_2(n-m)$ are $O(xH_n/n)$.

If $m < \delta_n n$, we use $f_n(n-m) \geq 0$. Then $m(m+1) \leq 2\delta_n^2 n^2$, so $\frac{1}{(n-m)(n-m+1)} \leq \frac{4}{n^2} \leq \frac{8\delta_n^2}{m(m+1)}$ and

$$\frac{1 - e(m)}{m(m+1)} \geq (1 - e(m) - 8\delta_n^2) \left[\frac{1}{m(m+1)} + \frac{1}{(n-m)(n-m+1)} \right].$$

In both cases the lower bound of the theorem holds with $\eta_n = \sup_m (e(m) + e_2(n-m)\mathbf{1}\{m \geq \delta_n n\}) + 8\delta_n^2$, the supremum taken over the range, and $\eta_n \rightarrow 0$ by Step 2 and the above.

Step 4 (upper bound). By Step 1 the hypotheses of Theorem B.8 other than $\Delta(m) \leq \frac{1}{16}$ hold at every m of the range for large n . We show that $\Delta(m) \rightarrow 0$ uniformly. At $m' = \lfloor 2m/3 \rfloor$ the first part of $\delta_c(m')$ tends to 0 by Step 1(ii) with $\omega = \frac{2}{3}\omega_n - 3$, using $H_{m'} \leq 1 + \log m'$ and $\log(m'/x) \leq \log m'$. Its exponential part is $ce^\varepsilon e^{x(0.11-0.093c)} \leq ce^{1/4} e^{0.11-0.093c} \rightarrow 0$, where $c = m'/x \geq (\frac{2}{3}\omega_n - 3)\log^2 2 \rightarrow \infty$ and we used $x \geq 1$. The bracket in $\delta_\ell(m)$ tends to 0 as in Step 2, and its last term is $O((xH_n^2 + \log n)/n) \rightarrow 0$. So Theorem B.8 gives $f_n(m) \leq \frac{x(1+o(1))}{m(m+1)}$

uniformly on the range. For the mirror term, $m_h \leq m_1(x) \leq 4500x \log^2(n+1) = o(n)$ by Lemma B.7 and Step 1(i). So $n - m \geq \lceil n/2 \rceil \geq m_h$, and $f_n(n - m) \leq f_n(\lceil n/2 \rceil)$ by Lemma 3.4. For even n , Theorem B.8 applies at $m = n/2$ for all large n , whatever ω_n is. Indeed, the estimates of Step 1 and of this step go through with $\omega = n/(2x\Lambda^2)$, which tends to infinity because $x\Lambda^2 \leq x \log^2(n+1) = o(n)$. This gives $f_n(n/2) \leq \frac{4x}{n(n+2)}(1+o(1))$. For odd n we have $\lfloor n/2 \rfloor \geq m_h$, so $f_n(\lceil n/2 \rceil) \leq f_n(\lfloor n/2 \rfloor) \leq \frac{4x}{(n-1)(n+1)}(1+o(1))$ in the same way. Both are $\frac{4x}{n^2}(1+o(1))$, and enlarging η_n gives the upper bounds of the theorem.

Step 5 (consequences). If $\delta'_n \rightarrow 0$ and $m \leq \delta'_n n$, then $\frac{4x}{n^2} \leq \frac{8\delta_n'^2 x}{m(m+1)}$, so the upper bound is $\frac{x(1+o(1))}{2m(m+1)}$. Also $m(m+1) = m^2(1+o(1))$, since $m \geq \omega_n \log^2 2 \rightarrow \infty$. For $m = un$ with $u \in [u_0, \frac{1}{2}]$, the hypothesis $\omega_n x \log^2(x+1) = o(n)$ puts m in the range for large n , and substituting $m = un$ gives the last claim. $\square$

Corollary B.9 (the center). *Let $n \geq 64$ be even. Then $C_n \geq \frac{4x}{n(n+2)}(1 - e_2(h))$. If the hypotheses of Theorem B.8 hold at $m = h$, then*

$$C_n \leq \frac{4x}{n(n+2)} \left(1 + 3\sqrt{\Delta(h)} + \frac{6}{n} \right).$$

If $x \log^2 n/n \rightarrow 0$, these hypotheses hold for all large even n , and $C_n = \frac{4x}{n^2}(1 + O(\sqrt{x \log^2 n/n}))$.

Proof. By Lemma 2.2, $C_n = f_n(h)$, and $h(h+1) = n(n+2)/4$. The 2 bounds are Theorems B.5(b) and B.8 at $m = h$. The hypotheses hold for large n by Step 4 of the proof of Theorem 3.9. Since $h \geq n/4$, $\delta_\ell(h)$ is its last term, which is $O(x \log^2 n/n)$. At $m' = \lfloor n/3 \rfloor$ the first part of $\delta_c(m')$ is $O(x \log^2 n/n)$ as well, and its exponential part is at most $n e^{1/4 + 0.11x - 0.093m'}$, which is smaller because $x = o(n)$. So $\Delta(h) = O(x \log^2 n/n)$. Also $e_2(h) = O(x \log n/n)$ and $\frac{4x}{n(n+2)} = \frac{4x}{n^2}(1 + O(\frac{1}{n}))$. $\square$

At the crossing (computed with $x = g_n$), the lower factor $1 - e_2(h)$ is positive from $n \approx 2.2 \cdot 10^3$ on. It is 0.67 at $n = 10^4$ and 0.95 at $n = 10^5$. At $m = h$ all hypotheses of Theorem B.8 except $\Delta(h) \leq \frac{1}{16}$ hold from $n = 10^5$ on, but $\Delta(h) \leq \frac{1}{16}$ first holds near $n = 1.107 \cdot 10^8$. There the upper factor is about 1.75, and it is 1.30 at $n = 10^9$ and 1.11 at $n = 10^{10}$.

B.6 Proof of Theorem 3.8

(a) Let Ψ_n be as in Section B.3, and let $\Psi_\infty(\theta) = \sum_{d \geq 1} \frac{e^{i\theta d} - 1}{d(d+1)} = \chi(e^{i\theta})$. Fourier inversion and the substitution $\theta = t/x$ give

$$x \text{CP}_n(m) = \frac{1}{2\pi} \int_{-\pi x}^{\pi x} e^{-i\lambda_m t} e^{A(t)} dt, \quad A(t) = x\Psi_n(t/x) - it \log x.$$

The Landau law has characteristic function $e^{B(t)}$, where $B(t) = (-it) \log(-it) = -it \log|t| - \frac{\pi}{2}|t|$ is the value of $u \log u$ at $u = -it$. So $f_L(\lambda) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-i\lambda t} e^{B(t)} dt$. We use 3 facts.

$$(1) |\Psi_n - \Psi_\infty| \leq \sum_{d \geq n} \frac{2}{d(d+1)} = \frac{2}{n}.$$

(2) For $0 < |\theta| \leq \pi$, $\Psi_\infty(\theta) = -i\theta \log(-i\theta) + R(\theta)$ with $|R(\theta)| \leq \theta^2(0.631|\log|\theta|| + 1.53)$. To see this, write $1 - e^{i\theta} = (-i\theta)u$ with $u = \text{sinc}(\theta/2) e^{i\theta/2}$, where $\text{sinc } y = \sin y/y > 0$ for $|y| \leq \frac{\pi}{2}$. Then $\log(1 - e^{i\theta}) = \log(-i\theta) + \log u$ for principal logarithms, since the 2 arguments add up to a number in $(-\frac{\pi}{2}, \frac{\pi}{2})$. As $\chi(s) = (1 - s) \log(1 - s)/s$,

$$R = -i\theta \log(-i\theta)(ue^{-i\theta} - 1) - i\theta ue^{-i\theta} \log u.$$

With $y = \theta/2$, $|ue^{-i\theta} - 1| = |\text{sinc } y - e^{iy}| \leq (1 - \text{sinc } y) + |1 - e^{iy}| \leq \frac{y^2}{6} + |y| \leq (\frac{1}{2} + \frac{\pi}{24})|\theta|$. Next, $|\log u|^2 = \log^2 \text{sinc}(\theta/2) + \theta^2/4$, and $-\log \text{sinc } y = \sum_{k \geq 1} \frac{\zeta(2k)}{k\pi^{2k}} y^{2k}$ has positive coefficients. So $|\log \text{sinc}(\theta/2)|/|\theta|$ increases in $|\theta|$, and $|\log u| \leq (\frac{1}{4} + \frac{\log^2(\pi/2)}{\pi^2})^{1/2}|\theta| \leq 0.53|\theta|$. Finally $|\log(-i\theta)| \leq |\log|\theta|| + \frac{\pi}{2}$ and $|u| \leq 1$. Since $\frac{1}{2} + \frac{\pi}{24} \leq 0.631$ and $0.631 \cdot \frac{\pi}{2} + 0.53 \leq 1.53$, the bound follows.

(3) For $|\theta| \leq \pi$, $-\text{Re } \Psi_\infty(\theta) = \sum_d \frac{1 - \cos \theta d}{d(d+1)} \geq \frac{|\theta|}{2\pi}$. Keep only $d \leq D := \lfloor \pi/|\theta| \rfloor$ and use $1 - \cos v \geq 2v^2/\pi^2$ for $|v| \leq \pi$. This gives at least $\frac{2\theta^2}{\pi^2} \sum_{d \leq D} \frac{d}{d+1} \geq \frac{\theta^2 D}{\pi^2} \geq \frac{|\theta|}{2\pi}$, since $D \geq \frac{\pi}{2|\theta|}$.

By (2), $x\Psi_\infty(t/x) - it \log x = B(t) + xR(t/x)$. So for $|t| \leq \pi x$, facts (1) and (2) and $|\log(|t|/x)| \leq |\log|t|| + \log x$ give

$$|A(t) - B(t)| \leq \frac{t^2}{x} (0.631(|\log|t|| + \log x) + 1.53) + \frac{2x}{n},$$

and facts (1) and (3) give $\max(\text{Re } A, \text{Re } B) \leq -\frac{|t|}{2\pi} + \frac{2x}{n}$. We use $|e^A - e^B| \leq |A - B|e^{\max(\text{Re } A, \text{Re } B)}$ on $|t| \leq \pi x$ and $|e^B| = e^{-\pi|t|/2}$ beyond. We have $\int_{\mathbb{R}} t^2 e^{-|t|/2\pi} dt = 992.20$, $\int_{\mathbb{R}} t^2 |\log|t|| e^{-|t|/2\pi} dt = 2739.54$ and $\int_{\mathbb{R}} e^{-|t|/2\pi} dt = 4\pi$. After division by 2π the coefficient of $\frac{\log x}{x}$ is $0.631 \cdot 992.20/2\pi = 99.64$, that of $\frac{1}{x}$ is $(0.631 \cdot 2739.54 + 1.53 \cdot 992.20)/2\pi = 516.73$, and that of $\frac{x}{n}$ is 4. The part beyond πx is $\frac{1}{2\pi} \int_{|t| > \pi x} e^{-\pi|t|/2} dt = \frac{2}{\pi^2} e^{-\pi^2 x/2}$. This proves (a).

(b) By Theorem 3.7, $|x f_n(m) - x \text{CP}_n(m)| \leq x \delta_B$, which gives the bound. For the limit, let U be uniform on $[0, 1)$ and independent of M_n . Then $(M_n + U - x \log x)/x$ has the density g with $g(\lambda) = x f_n(m)$ for $\lambda_m \leq \lambda < \lambda_m + \frac{1}{x}$. If $x \rightarrow \infty$ and $x^3 \log^2 n/n \rightarrow 0$, then $E_L + x \delta_B \rightarrow 0$. On a compact set of λ the relevant m are nonnegative once x is large, and $|\lambda - \lambda_m| < \frac{1}{x}$. As f_L is uniformly continuous, $g \rightarrow f_L$ uniformly on compact sets. By Scheffé's lemma $g \rightarrow f_L$ in L^1 , and since $U/x \rightarrow 0$ this gives the convergence in law. At the crossing $x_n \sim 2 \log n$ by Theorem 2.7, so both conditions hold.

(c) By the argument after Theorem 3.8, a constant κ_L exists if the computed sign $f_L''(\lambda_0) < 0$ is right. Note that f_L'' is bounded, since $t^2 e^{B(t)}$ is integrable. Let m' be the integer nearest to $x \log x + \lambda_0 x$, which is at least -0.23 , so $m' \geq 0$. Then $|\lambda_{m'} - \lambda_0| \leq \frac{1}{2x}$, and since $f_L'(\lambda_0) = 0$, $f_L(\lambda_{m'}) \geq f_L(\lambda_0) - \|f_L''\|_\infty/(8x^2)$. Let $\tilde{m}$ be a mode of f_n or $\tilde{m} = m^*$. We claim that $f_n(\tilde{m}) \geq f_n(m') - 16xH_n/n^2$. For a mode of f_n this is clear. For m^* we have $f_n(m^*) + f_n(n - m^*) = 2\mathbb{P}(A_n = n - m^*) \geq 2\mathbb{P}(A_n = n - m') \geq f_n(m')$. Here $n - m^* \geq \lceil n/2 \rceil \geq \lceil n/4 \rceil \geq m_h$. So by Lemma 3.4 f_n is nonincreasing on $[\lceil n/4 \rceil, n]$, and by Lemma B.2

$$f_n(n - m^*) \leq f_n(\lceil n/2 \rceil) \leq \frac{\mathbb{P}(M_n \geq \lceil n/4 \rceil)}{n/4} \leq \frac{16xH_n}{n^2}.$$

Now part (b), used at $\tilde{m}$ and at m' , gives $f_L(\lambda_{\tilde{m}}) \geq f_L(\lambda_{m'}) - 2(E_L + x \delta_B) - \frac{16x^2 H_n}{n^2} \geq f_L(\lambda_0) - 2E'$. Since $2E' < \kappa_L$, the defining property of κ_L gives $(\lambda_{\tilde{m}} - \lambda_0)^2 \leq 2E'/\kappa_L$, which proves (c).

At the crossing, $x_n \sim 2 \log n$ by Theorem 2.7. Then $x \delta_B$, $16x^2 H_n/n^2$ and $\|f_L''\|_\infty/(8x^2)$ are $o(\log x/x)$, so $E' = O(\log x/x)$. Also $m_h \leq m_1(x) \leq n/4$ for large n by Lemma B.7. Hence, once $E' < \kappa_L/2$, part (c) gives $m^* = x \log x + \lambda_0 x + O(\sqrt{x \log x})$. $\square$

B.7 Proof of Theorem 3.11(b)

Since $x \leq n^{1/4}$, we have $x \log^2 n/n \leq n^{-3/4} \log^2 n$, and every $o(1)$ below is bounded by a function of n alone. Let n_0 be so large that for $n \geq n_0$ the following hold. First, $\varepsilon \leq \frac{1}{4}$, $n \geq 1000$ and $13\varepsilon(1 + H_n) \leq \frac{1}{8}$. Second, $\ell_0 \leq m_1(x) \leq 4500x \log^2(x+1) \leq n/8$, using Step 1 of the proof of Theorem 3.9. So $m_h \leq m_1(x)$ and $e_1(\ell) \leq \frac{1}{2}$ for $\ell_0 \leq \ell \leq n/4$ by Lemma B.7. Third,

the hypotheses of Theorem B.8 hold at $m = h$, and Corollary B.9 gives $C_n \leq \frac{4x}{n(n+2)}(1 + \eta)$ with $\eta = 3\sqrt{\Delta(h)} + \frac{6}{n} \leq C\sqrt{x \log^2 n/n}$ for an absolute constant C . This follows from the proof of Corollary B.9, since for $x \leq n^{1/4}$ all its estimates hold with absolute constants and $m_1(x) \leq 4500n^{1/4} \log^2(n+1)$. By symmetry it suffices to take $a = n - m$ with $1 \leq m \leq h$.

Near the end, $m \leq m_h$. By Lemma 3.4, $f_n(m) \geq f_n(1)$. The proof of Theorem 3.5 shows $f_n(1) \geq \frac{x}{2(1-\varepsilon)}f_n(0)$, and $f_n(0) = 2E_n = 2C_n$ at the crossing. So $\mathbb{P}(A_n = n - m) \geq \frac{1}{2}f_n(m) \geq \frac{x}{2(1-\varepsilon)}C_n > C_n$, as $x \geq 2$.

The range $m_h \leq m \leq n/8$. By Lemma 3.4, $f_n(m) \geq f_n(\max(m, \ell_0))$. With $e_1 \leq \frac{1}{2}$ on $[\ell_0, n/8]$, Theorem B.5(a) gives $f_n(m) \geq \frac{x}{2(n/8)(n/8+1)}$. So $\mathbb{P}(A_n = n - m) \geq \frac{16x}{n(n+8)} > \frac{4x}{n(n+2)}(1 + \eta) \geq C_n$ for large n .

The range $n/8 < m \leq h$. Put $\Pi(m) = \frac{1}{2}[\frac{1}{m(m+1)} + \frac{1}{(n-m)(n-m+1)}]$, so $\Pi(h) = \frac{4}{n(n+2)} = \frac{1}{h(h+1)}$. Theorem B.5 at m and at $n-m$ gives $\mathbb{P}(A_n = n - m) \geq x\Pi(m)(1 - \bar{e})$ with $\bar{e} = \max(e(m), e(n-m))$. Since $m, n-m \geq n/8$, every term of $e(m)$ and $e(n-m)$ is $O(xH_n/n)$ except $4(m+2)e^{-(m-1)/9}$ and τ , which are smaller than any power of n . So $\bar{e} \leq C'x \log n/n$. Put $N = n+1$ and $t = h-m$. Then $m + \frac{1}{2} = \frac{N}{2} - t$ and $n-m + \frac{1}{2} = \frac{N}{2} + t$. Since $(k + \frac{1}{2})^{-2} \leq \frac{1}{k(k+1)}$ and $\frac{1}{h(h+1)} = (\frac{N}{2})^{-2}(1 - N^{-2})^{-1}$,

$$\frac{\Pi(m)}{\Pi(h)} \geq \frac{1}{2} \left[\left(1 - \frac{2t}{N}\right)^{-2} + \left(1 + \frac{2t}{N}\right)^{-2} \right] (1 - N^{-2}) \geq \left(1 + \frac{12t^2}{N^2}\right) (1 - N^{-2}),$$

because $\frac{1}{2}[(1-s)^{-2} + (1+s)^{-2}] = \sum_{k \geq 0} (2k+1)s^{2k} \geq 1 + 3s^2$ for $|s| < 1$. Let $b = \bar{e} + N^{-2}$, which is at most $\frac{1}{2}$ for large n . Then $\mathbb{P}(A_n = n - m) \geq x\Pi(h)(1 + \frac{12t^2}{N^2})(1 - b)$, and this exceeds $x\Pi(h)(1 + \eta) \geq C_n$ as soon as $\frac{12t^2}{N^2} > 2(\eta + b)$. Since $x \log^2 n/n \leq 1$, we have $\eta + b \leq C''\sqrt{x \log^2 n/n}$ with an absolute C'' . As $N \leq 2n$, the condition holds when $t \geq C_w n^{3/4}(x \log^2 n)^{1/4}$ with $C_w = \sqrt{C''}$, because then $\frac{12t^2}{N^2} \geq 3C''\sqrt{x \log^2 n/n}$. This proves part (b) of Theorem 3.11. $\square$

C Proofs for Section 4

We use the notation of Section 4, with $\tau_0 = 0$. For a first run D let $\tau_m^D = D + \tau'_{m-1}$, where (τ'_m) is an independent copy of (τ_m) , and $\tau_0^D = 0$. Then $u_m^D(k) = \mathbb{P}(\tau_m^D = k)$ and $q_m^D(k) = \mathbb{P}(\tau_{m-1}^D < k \leq \tau_m^D)$. Conditioning on the last run gives, for $m, k \geq 1$,

$$\begin{aligned} q_m(k) &= \sum_{j \geq 1} \bar{F}(j) u_{m-1}(k-j), & q_{m+1}^D(k) &= \sum_{j \geq 1} \bar{F}(j) u_m^D(k-j), \\ u_m^D(k) &= \sum_{d \geq 1} \mathbb{P}(D = d) u_{m-1}(k-d). \end{aligned} \tag{7}$$

Since τ_m and τ_m^D increase strictly, $\sum_m q_m(k) = \sum_m q_m^D(k) = 1$, $\sum_m u_m^D(k) \leq 1$ and $\sum_{m \leq m'} q_m(k) = \mathbb{P}(\tau_{m'} \geq k)$. We put $z_{m,k} = (k - m\mu)/a_m$ if $\alpha > 1$ and $z_{m,k} = k/a_m$ if $\alpha < 1$, and $\delta_m = \sup_k |a_m u_m(k) - g(z_{m,k})|$. We write ω for the modulus of continuity of g and $\bar{g}(x) = \sup_{|y| \geq x} g(y)$.

C.1 Proof of Lemma 4.3

(i) Summation by parts gives $\mathbb{E}(1 - x^L) = (1 - x) \sum_{k \geq 1} \bar{F}(k) x^{k-1}$, for complex $|x| \leq 1$ if $\alpha > 1$ (the sums converge absolutely since $\sum_k k p_k = \mu$) and for $0 \leq x < 1$ if $\alpha < 1$ (the terms are positive).

With $x = e^{-u}$ this reads

$$1 - \mathbb{E}e^{-uL} = (e^u - 1) \text{Li}_\alpha(e^{-u}), \quad \text{Li}_\alpha(z) = \sum_{k \geq 1} k^{-\alpha} z^k. \quad (8)$$

For non-integer α , $\text{Re } u \geq 0$ and $0 < |u| < 2\pi$, [22, 25.12.12] gives $\text{Li}_\alpha(e^{-u}) = \Gamma(1 - \alpha)u^{\alpha-1} + \sum_{k \geq 0} \zeta(\alpha - k)(-u)^k/k!$ with the principal branch. For $1 < \alpha < 2$ this gives $1 - \mathbb{E}e^{-uL} = \mu u + \Gamma(1 - \alpha)u^\alpha + O(|u|^2)$, and $u = -i\theta$ gives (i), since $K_\alpha = -\Gamma(1 - \alpha)$.

(ii) With $t = \theta/a_m$, part (i) gives $e^{-i\mu t} \mathbb{E}e^{itL} = 1 + K_\alpha(-i\theta)^\alpha/m + O(\theta^2 m^{-2/\alpha})$. Since $m^{1-2/\alpha} \rightarrow 0$, the characteristic function of $(\tau_m - m\mu)/a_m$ tends to $\exp(K_\alpha(-i\theta)^\alpha)$, whose modulus $\exp(K_\alpha \cos(\frac{\pi\alpha}{2})|\theta|^\alpha)$ is integrable. Lévy's continuity theorem, Fourier inversion and the Riemann-Lebesgue lemma give G and the stated properties of g . For real $u > 0$, part (i) reads $\mathbb{E}e^{-uL} = 1 - \mu u + K_\alpha u^\alpha + O(u^2)$, and the same computation gives $\mathbb{E}e^{-s(\tau_m - m\mu)/a_m} \rightarrow e^{K_\alpha s^\alpha}$ for $s \geq 0$. Applied with $2s$ in place of s , this shows that these variables are bounded in L^2 . So they are uniformly integrable, and $\mathbb{E}e^{-sG} = e^{K_\alpha s^\alpha}$.

(iii) Every p_k is positive, so L has maximal span 1. By (ii) and (v) the normalized sums converge to stable laws, and Gnedenko's local limit theorem [15, Theorem 4.2.1] gives $\delta_m \rightarrow 0$. Then $a_m u_m(k) \leq \|g\|_\infty + 1$ for large m , and $a_m u_m(k) \leq a_m$ for the other m , so $c_0 < \infty$.

(iv) If G' is an independent copy of G , then $|\mathbb{E}e^{i\theta G}|^2 = e^{-\sigma_\alpha |\theta|^\alpha}$ is the characteristic function of $G - G'$. By Plancherel's theorem, $I_\alpha = \frac{1}{2\pi} \int e^{-\sigma_\alpha |\theta|^\alpha} d\theta = \Gamma(1 + 1/\alpha)/(\pi \sigma_\alpha^{1/\alpha})$.

(v) Let $0 < \alpha < 1$. For real $u \downarrow 0$, (8) gives $1 - \mathbb{E}e^{-uL} = \Gamma(1 - \alpha)u^\alpha + O(u)$, with $\Gamma(1 - \alpha) > 0$. As $1/\alpha > 1$, $m(1 - \mathbb{E}e^{-sL/a_m}) \rightarrow \Gamma(1 - \alpha)s^\alpha$, so $\mathbb{E}e^{-s\tau_m/a_m} \rightarrow e^{-\Gamma(1-\alpha)s^\alpha}$, and the continuity theorem for Laplace transforms gives $\tau_m/a_m \Rightarrow G$. Both sides of $\mathbb{E}e^{-sG} = e^{-\Gamma(1-\alpha)s^\alpha}$ are continuous on $\text{Re } s \geq 0$ and analytic on $\text{Re } s > 0$, so the identity holds at $s = -i\theta$. Hence $|\mathbb{E}e^{i\theta G}| = \exp(-\Gamma(1 - \alpha) \cos(\frac{\pi\alpha}{2})|\theta|^\alpha)$ is integrable, and g exists as in (ii). Also $g = 0$ on $(-\infty, 0]$ since $G \geq 0$. For the tail, if $\mathbb{E}e^{-sG} = e^{-s^\alpha}$ then Pollard's series [16] writes the density as a series in the powers $y^{-\alpha k - 1}$, $k \geq 1$, that converges for $y > 0$, with first term $\frac{1}{\pi} \Gamma(1 + \alpha) \sin(\pi\alpha) y^{-1-\alpha}$. Scaling by $\Gamma(1 - \alpha)^{1/\alpha}$ and the reflection formula $\Gamma(1 + \alpha)\Gamma(1 - \alpha) = \pi\alpha/\sin(\pi\alpha)$ give $g(y) = \alpha y^{-1-\alpha} + O(y^{-1-2\alpha})$ as $y \rightarrow \infty$. $\square$

C.2 Two lemmas on renewal sums

A sum of m runs that is much larger than its typical size is most likely made by a single long run. The first lemma turns this into a local bound.

Lemma C.1. *There is a constant $C_B = C_B(\alpha)$ such that $u_m(k) \leq C_B m k^{-\alpha-1}$ for all $m \geq 1$ in each of the cases (a) $1 < \alpha < 2$ and $k \geq 2m\mu$, (b) $\alpha \geq 2$ and $k \geq 2m\mu$, (c) $0 < \alpha < 1$ and $k \geq a_m$.*

Proof. Here C depends only on α and may change from line to line. Since $\tau_m \geq m$ we may assume $m \leq k$. For $r > 2$, chosen below, put $y = k/r$.

One big jump. By the mean value theorem $p_\ell \leq \alpha \ell^{-\alpha-1}$. So for each i , $\mathbb{P}(\tau_m = k, L_i \geq y) = \sum_{\ell \geq y} p_\ell \mathbb{P}(\tau_m - L_i = k - \ell) \leq \alpha r^{\alpha+1} k^{-\alpha-1}$, and the m indices give at most $\alpha r^{\alpha+1} m k^{-\alpha-1}$.

No big jump. We may take $y \geq 1$, since otherwise $\max_i L_i < y$ is impossible. Let I be a set of indices, $\tau_I = \sum_{i \in I} L_i$, $x_0 > 0$ and $\lambda = s/y$ with $s \geq 0$. Markov's inequality and $e^x \leq 1 + x + \frac{x^2}{2} e^x$ for $x \geq 0$ give

$$\mathbb{P}(\tau_I \geq x_0, \max_{i \in I} L_i < y) \leq \exp\left(-\frac{s x_0}{y} + |I| \left(\frac{s}{y} \mathbb{E}[L; L < y] + \frac{s^2 e^s}{2y^2} \mathbb{E}[L^2; L < y] \right)\right). \quad (9)$$

In case (a) we take all m indices and $x_0 = k$, and use $\mathbb{E}[L; L < y] \leq \mu$, $\mathbb{E}[L^2; L < y] \leq \alpha \sum_{\ell < y} \ell^{1-\alpha} \leq Cy^{2-\alpha}$ and $k - m\mu \geq k/2$. The exponent is then at most $-sr/2 + Cr^\alpha s^2 e^s m/k^\alpha$. With $s = \frac{1}{2} \log(k^\alpha/m)$, which is positive since $m \leq k/2 < k^\alpha$, we have $e^s m/k^\alpha = e^{-s}$ and $s^2 e^{-s} \leq 1$. So the bound is $C(m/k^\alpha)^{r/4}$. With $r = 4 + 4/(\alpha - 1)$ and $m \leq k$ this is at most $C(m/k^\alpha)(k^{1-\alpha})^{1/(\alpha-1)} = Cmk^{-\alpha-1}$.

In case (b) we argue as in case (a). For $\alpha > 2$ we use $\mathbb{E}[L^2; L < y] \leq \mathbb{E}L^2 < \infty$, $s = \frac{1}{2} \log(k^2/m)$ and $r = 4\alpha$, which give $C(m/k^2)^\alpha \leq Cmk^{-\alpha-1}$. For $\alpha = 2$ we use $\mathbb{E}[L^2; L < y] \leq 2 \sum_{\ell < y} \ell^{-1} \leq C \log(ek)$, $r = 20$ and $s = \frac{1}{2} \log \frac{k^2}{m \log(ek)}$, which is positive since $m \log(ek) \leq k \log(ek)/(2\mu) < k^2$. This gives $C(m \log(ek)/k^2)^5 \leq Cmk^{-6} \log^5(ek) \leq Cmk^{-3}$.

In case (c) let $t = k^\alpha/m \geq 1$, so that $k = t^{1/\alpha} a_m$, and $r = 4 + 4/\alpha$. For $m = 1$ the claim is $p_k \leq \alpha k^{-\alpha-1}$. For $m \geq 2$ split the indices into $I_1 = \{1, \dots, \lfloor m/2 \rfloor\}$ and $I_2 = \{\lfloor m/2 \rfloor + 1, \dots, m\}$. On $\{\tau_m = k\}$ one of τ_{I_1}, τ_{I_2} is at least $k/2$, so by independence

$$\mathbb{P}(\tau_m = k, \max_i L_i < y) \leq \sum_{i=1,2} \mathbb{P}(\tau_{I_i} \geq k/2, \max_{I_i} L < y) \sup_j \mathbb{P}(\tau_{I_{3-i}} = j).$$

Each half has at least $\lfloor m/2 \rfloor \geq m/3$ indices, so by Lemma 4.3(iii) the supremum is at most $c_0/a_{\lfloor m/2 \rfloor} \leq 3^{1/\alpha} c_0/a_m$. In (9) with $x_0 = k/2$ we use $\mathbb{E}[L; L < y] \leq Cy^{1-\alpha}$, $\mathbb{E}[L^2; L < y] \leq Cy^{2-\alpha}$, $|I_i| \leq m$ and $my^{-\alpha} = r^\alpha/t$. The exponent is at most $-sr/2 + C(s + s^2 e^s)/t$, and $s = \frac{1}{2} \log t$ makes the second term bounded. So the first factor is at most $Ct^{-r/4}$, and $t^{-r/4}/a_m = t^{-1-1/\alpha}/a_m = mk^{-\alpha-1}$. $\square$

Lemma C.2. *Let $1 < \alpha < 2$ and let D be any a.s. finite first-run law. Then*

$$\sup_k \left| \frac{a_m}{\mu} q_m(k) - g(z_{m,k}) \right|, \quad \sup_k \left| a_m u_m^D(k) - g(z_{m,k}) \right|, \quad \sup_k \left| \frac{a_m}{\mu} q_{m+1}^D(k) - g(z_{m,k}) \right|$$

tend to 0 as $m \rightarrow \infty$. For $m \geq 2$, $\sup_k q_m(k) \leq \mu c_0/a_{m-1}$ and $\sup_k u_m^D(k) \leq c_0/a_{m-1}$.

Proof. By (7), q_m and u_m^D are mixtures of shifts of u_{m-1} with total weights μ and 1. This gives the bounds for $m \geq 2$.

Fix $J \geq 1$ and $Z > 0$, put $T_J = \sum_{j>J} \bar{F}(j)$ and write $z = z_{m,k}$. In the sum for $q_m(k)$ the terms with $j > J$ add up to at most $c_0 T_J/a_{m-1}$. For $j \leq J$ we have $|a_{m-1} u_{m-1}(k-j) - g(z_{m-1,k-j})| \leq \delta_{m-1}$ and $z_{m-1,k-j} = z a_m/a_{m-1} + (\mu - j)/a_{m-1}$. If $|z| \leq Z$, then $|z_{m-1,k-j} - z| \leq \eta_m = Z(a_m/a_{m-1} - 1) + (J + \mu)/a_{m-1}$. If $|z| > Z$ and $(J + \mu)/a_{m-1} \leq Z/2$, then $g(z)$ and $g(z_{m-1,k-j})$ both lie in $[0, \bar{g}(Z/2)]$. Since $\mu - \sum_{j \leq J} \bar{F}(j) = T_J$, we get, uniformly in k for large m ,

$$|a_{m-1} q_m(k) - \mu g(z)| \leq \mu(\delta_{m-1} + \omega(\eta_m) + \bar{g}(Z/2)) + (c_0 + \|g\|_\infty) T_J.$$

We let $m \rightarrow \infty$, then $J, Z \rightarrow \infty$, and use $a_m/a_{m-1} \rightarrow 1$. This gives the first limit. The second follows in the same way from the sum for u_m^D , with the weights $\mathbb{P}(D = d)$ and with $\mathbb{P}(D > J)$ in place of T_J . For the third we apply the argument to the sum for q_{m+1}^D , with $a_m u_m^D$ in place of $a_{m-1} u_{m-1}$. Then the second limit replaces $\delta_{m-1} \rightarrow 0$, the shift is $z_{m,k-j} - z_{m,k} = -j/a_m$, and the tail is at most $c_0 T_J/a_{m-1}$. $\square$

C.3 Proof of Theorem 4.4

We write the proof so that it also gives Theorem 4.6. It uses only the following facts. First, $a_m = a(m)$ for large m , where a is increasing and regularly varying with index in $(0, 1)$. Second,

$(\tau_m - m\mu)/a_m \Rightarrow G$, where G has a bounded, uniformly continuous density g that vanishes at $\pm\infty$, and $\delta_m \rightarrow 0$ and $c_0 < \infty$. Third, Lemma C.2 holds and $u_m(k) \leq C_B m k^{-\alpha-1}$ for $k \geq 2m\mu$. For $1 < \alpha < 2$ these hold with $a(x) = x^{1/\alpha}$, by Lemma 4.3 and Lemma C.1(a). Put $b_h = a(h/\mu)$, so that $b_h = (h/\mu)^{1/\alpha}$ for $1 < \alpha < 2$. We show that $C_n \sim 2b_h^{-1} \int g^2$. For $1 < \alpha < 2$ this is $2I_\alpha(\mu/h)^{1/\alpha}$, and the second form in Theorem 4.4 follows from $\sigma_\alpha = 2K_\alpha |\cos \frac{\pi\alpha}{2}|$ and $h = n/2$.

By regular variation, $b_h = o(h)$, $a(h/(8\mu)) \geq b_h/9$ and $a(h/(2\mu)) \geq b_h/3$ for large h , and $a_m/b_h \rightarrow 1$ uniformly in m with $|m\mu - h| \leq 2Mb_h$, for each fixed M . By Proposition 4.2, $C_n = \sum_{m \geq 1} [u_m(h)q_{m+1}^D(h) + u_m^D(h)q_m(h)]$. Fix $M > 1$, put $m_\pm = (h \pm Mb_h)/\mu$, and let $\Sigma_1, \dots, \Sigma_4$ be the parts of this sum over the regions

$$\mathcal{R}_1 = \{m \leq \frac{h}{4\mu}\}, \quad \mathcal{R}_2 = \{\frac{h}{4\mu} < m < m_-\}, \quad \mathcal{R}_3 = \{m_- \leq m \leq m_+\}, \quad \mathcal{R}_4 = \{m > m_+\}.$$

Region $\mathcal{R}_1$. Here $h \geq 4m\mu$, so $u_m(h) \leq C_B m h^{-\alpha-1} \leq C_B h^{-\alpha}/(4\mu)$. Also $q_1(h) = h^{-\alpha}$. For $m \geq 2$ we split the sum for $q_m(h)$ at $j = h/2$. The terms with $j \geq h/2$ add up to at most $(h/2)^{-\alpha}$. For $j < h/2$ we have $h - j > h/2 \geq 2\mu(m-1)$, so the other terms add up to at most $\mu C_B (m-1)(h/2)^{-\alpha-1} \leq 2^{\alpha-1} C_B h^{-\alpha}$. Since $\sum_m q_{m+1}^D(h) \leq 1$ and $\sum_m u_m^D(h) \leq 1$, we get $\Sigma_1 = O(h^{-\alpha})$. This is $o(1/b_h)$, since the index of a is less than $1 < \alpha$.

Region $\mathcal{R}_2$. For large h we have $m-1 \geq h/(8\mu)$ here, so $a_m \geq a_{m-1} \geq b_h/9$, and $u_m(h), u_m^D(h) \leq 9c_0/b_h$ by Lemma C.2. Let $m' = \lfloor m_- \rfloor$. Then

$$\sum_{m \in \mathcal{R}_2} q_m(h) \leq \mathbb{P}(\tau_{m'} \geq h), \quad \sum_{m \in \mathcal{R}_2} q_{m+1}^D(h) \leq \mathbb{P}(D + \tau'_{m'} \geq h) \leq \mathbb{P}(D \geq b_h) + \mathbb{P}(\tau_{m'} \geq h - b_h).$$

Since $m'\mu \leq h - Mb_h$ and $a_{m'} \leq b_h$, the event $\{\tau_{m'} \geq h - b_h\}$ implies $(\tau_{m'} - m'\mu)/a_{m'} \geq M - 1$. As D is a.s. finite and $(\tau_{m'} - m'\mu)/a_{m'} \Rightarrow G$, we get $\limsup_h b_h \Sigma_2 \leq 18c_0 \mathbb{P}(G \geq M - 1)$.

Region $\mathcal{R}_4$. For large h we have $m-1 \geq h/(2\mu)$ here, so $u_m(h), u_m^D(h) \leq 3c_0/b_h$. Let $k = \lfloor m_+ \rfloor$. Then $\sum_{m \in \mathcal{R}_4} q_m(h) = \mathbb{P}(\tau_k < h)$, and since $D \geq 1$, $\sum_{m \in \mathcal{R}_4} q_{m+1}^D(h) = \mathbb{P}(D + \tau'_k < h) \leq \mathbb{P}(\tau_k < h)$. Now $(h - k\mu)/a_k \leq (\mu - Mb_h)/a_k \rightarrow -M$, and the law of G is continuous, so $\limsup_h b_h \Sigma_4 \leq 6c_0 \mathbb{P}(G \leq -M)$.

Region $\mathcal{R}_3$. Here $m \rightarrow \infty$ and $a_m = b_h(1 + o(1))$ uniformly. By Lemma C.2 and $\delta_m \rightarrow 0$, uniformly in $m \in \mathcal{R}_3$,

$$u_m(h)q_{m+1}^D(h) + u_m^D(h)q_m(h) = \frac{2\mu}{a_m^2} (g(z_{m,h})^2 + o(1)).$$

Put $y_m = (h - m\mu)/b_h$. Then $|z_{m,h} - y_m| \leq M|b_h/a_m - 1| \rightarrow 0$, so $g(z_{m,h})^2 = g(y_m)^2 + o(1)$. The y_m with $m \in \mathcal{R}_3$ are the points of an arithmetic progression with step μ/b_h that lie in $[-M, M]$. Hence

$$b_h \Sigma_3 = 2(1 + o(1)) \frac{\mu}{b_h} \sum_{m \in \mathcal{R}_3} (g(y_m)^2 + o(1)) \rightarrow 2 \int_{-M}^M g^2.$$

Adding the 4 regions,

$$\limsup_{h \rightarrow \infty} \left| b_h C_n - 2 \int g^2 \right| \leq 2 \int_{|z| > M} g^2 + 18c_0 (\mathbb{P}(G \geq M - 1) + \mathbb{P}(G \leq -M)),$$

and letting $M \rightarrow \infty$ gives $C_n \sim 2b_h^{-1} \int g^2$. $\square$

C.4 Proof of Theorem 4.6

Let g be the standard normal density, so that $\int g^2 = 1/(2\sqrt{\pi})$. The proof of Lemma C.2 used only $\delta_m \rightarrow 0$, $c_0 < \infty$, the uniform continuity and decay of g , $T_J \rightarrow 0$, $a_{m-1} \rightarrow \infty$ and $a_m/a_{m-1} \rightarrow 1$. So that lemma holds for $\alpha \geq 2$ with the norming below. With Lemma C.1(b), the proof of Theorem 4.4 then gives $C_n \sim 1/(\sqrt{\pi} b_h)$, once we check its other facts.

(a) Let $\alpha > 2$. Summation by parts gives $\mathbb{E}L^2 = \sum_k (2k-1)\bar{F}(k) = 2\zeta(\alpha-1) - \zeta(\alpha)$, so v is as stated and finite. With $a(x) = \sqrt{vx}$ the central limit theorem gives $(\tau_m - m\mu)/a_m \Rightarrow \mathcal{N}(0,1)$, and Gnedenko's theorem [15, Theorem 4.2.1] gives $\delta_m \rightarrow 0$ and $c_0 < \infty$. So $C_n \sim (\pi v h/\mu)^{-1/2} = \sqrt{2\mu/(\pi v n)}$. For the stationary start Proposition 4.1 gives $E_n = n^{1-\alpha}(1 + O(1/n))/(2(\alpha-1)\mu)$, and dividing gives $R_n \sim C_2(\alpha)n^{3/2-\alpha}$.

(b) Let $\alpha = 2$, so that $\mu = \zeta(2)$ and $\mathbb{E}L^2 = \infty$. The truncated second moment $U(x) = \mathbb{E}[L^2; L \leq x] = \sum_{k \leq x} (2k+1)/(k+1)^2 = 2 \log x + O(1)$ is slowly varying. So L is in the domain of attraction of the normal law, and $(\tau_m - m\mu)/a_m \Rightarrow \mathcal{N}(0,1)$ for any a_m with $mU(a_m)/a_m^2 \rightarrow 1$ [15]. Centering the truncated moment at μ changes it only by $O(1)$. For $a(x) = \sqrt{x \log x}$ we have $mU(a(m))/a(m)^2 = (\log m + \log \log m + O(1))/\log m \rightarrow 1$. This a is increasing on $[3, \infty)$ and regularly varying of index $\frac{1}{2}$, and we take $a_m = a(m)$ for $m \geq 3$ and $a_1 = a_2 = 1$. Gnedenko's theorem again gives $\delta_m \rightarrow 0$ and $c_0 < \infty$. So $C_n \sim 1/(\sqrt{\pi} b_h)$ with $b_h^2 = (h/\mu) \log(h/\mu) \sim n \log n/(2\mu)$, that is, $C_n \sim \sqrt{2\mu/(\pi n \log n)}$. For the stationary start $E_n = n^{-1}(1 + O(1/n))/(2\mu)$ by Proposition 4.1, so $R_n \sim \sqrt{\pi \log n}/(8\mu^3 n)$.

In both cases a fresh start gives the smaller edge $E_n = \frac{1}{2}n^{-\alpha}$, so $R_n \rightarrow 0$ for either start. $\square$

C.5 Proof of Theorem 4.7

Let $0 < \alpha < 1$ with a fresh start. Then $u_m^D = u_m$ and $q_m^D = q_m$, so Proposition 4.2 gives $C_n = \sum_{m \geq 1} u_m(h)[q_m(h) + q_{m+1}(h)]$. Here $a_m = m^{1/\alpha}$, g is the density of Lemma 4.3(v), and $\delta_m \rightarrow 0$. Since g is bounded and $g(y) = O(y^{-1-\alpha})$, we have $\int_0^\infty yg(y)^2 dy < \infty$. Put $t_m = h/a_m$ and fix $M > 1$.

Step 1. Let $g_1(t) = \int_0^t w^{-\alpha} g(t-w) dw$. We claim that $mq_m(h) = g_1(t_m) + o(1)$ as $h \rightarrow \infty$, uniformly in m with $t_m \in [1/(2M), 2M]$. Such m tend to infinity. Since $|k/a_{m-1} - k/a_m| \leq 2M(a_m/a_{m-1} - 1) \rightarrow 0$ for $k \leq h$, the local limit theorem for u_{m-1} and the uniform continuity of g give $a_m u_{m-1}(k) = g(k/a_m) + o(1)$ uniformly in $0 \leq k \leq h$. As $mj^{-\alpha} = (j/a_m)^{-\alpha}$ and $u_{m-1}(0) = 0$, the sum for q_m in (7) becomes

$$mq_m(h) = \frac{1}{a_m} \sum_{j=1}^{h-1} \left(\frac{j}{a_m}\right)^{-\alpha} \left(g\left(t_m - \frac{j}{a_m}\right) + o(1)\right).$$

For $x > 0$ we have $\frac{1}{a_m} \sum_{j \leq xa_m} (j/a_m)^{-\alpha} \leq x^{1-\alpha}/(1-\alpha)$. With $x = t_m \leq 2M$ this shows that the $o(1)$ terms add up to $o(1)$. With $x = \eta$ it shows that the terms with $j \leq \eta a_m$ are $O(\eta^{1-\alpha})$, and so is $\int_0^\eta w^{-\alpha} g(t_m - w) dw$. The other terms form a Riemann sum with mesh $1/a_m$ of a function that is uniformly continuous on $\{\eta \leq w \leq t \leq 2M\}$. Letting $\eta \rightarrow 0$ proves the claim. Next, the Laplace transform of g_1 is $\Gamma(1-\alpha)s^{\alpha-1}e^{-\Gamma(1-\alpha)s^\alpha}$. This is $\frac{1}{\alpha}$ times $-\frac{d}{ds}e^{-\Gamma(1-\alpha)s^\alpha}$, the transform of $tg(t)$. Both functions are continuous on $(0, \infty)$, so $g_1(t) = tg(t)/\alpha$.

Step 2. If $t_m > M$, then $h > Ma_m$ and Lemma C.1(c) gives $u_m(h) \leq C_B m h^{-\alpha-1} \leq C_B M^{-\alpha}/h$. Since $\sum_m (q_m(h) + q_{m+1}(h)) \leq 2$, these m contribute at most $2C_B M^{-\alpha}/h$ to C_n . If $t_m < 1/M$, then $u_m(h) \leq c_0/a_m < c_0/(Mh)$, and these m contribute at most $2c_0/(Mh)$. For the remaining m we have $t_{m+1} = t_m(1 + o(1)) \in [1/(2M), 2M]$ and $(m+1)/m \rightarrow 1$, and g_1 is uniformly continuous

on $[1/(2M), 2M]$. So Step 1, $1/a_m = t_m/h$ and $\delta_m \rightarrow 0$ give, uniformly,

$$u_m(h)[q_m(h) + q_{m+1}(h)] = \frac{t_m}{h}(g(t_m) + o(1)) \frac{2}{m} \left(\frac{t_m g(t_m)}{\alpha} + o(1) \right) = \frac{2}{\alpha h m} (t_m^2 g(t_m)^2 + o(1)).$$

As m runs over these values, $\log m = \alpha \log(h/t_m)$ runs over an interval of length $2\alpha \log M + o(1)$ with steps $(1 + o(1))/m$. So the sum of $1/m$ is $O(\log M)$, the $o(1)$ terms contribute $o(1/h)$, and the rest is a Riemann sum in the variable $\log m$. It tends to $\frac{2}{\alpha h} \int_{1/M}^M t^2 g(t)^2 \frac{\alpha dt}{t}$. Hence

$$hC_n = 2 \int_{1/M}^M t g(t)^2 dt + O(M^{-\alpha} + M^{-1}) + o(1),$$

and letting $M \rightarrow \infty$ gives $nC_n = 2hC_n \rightarrow 4 \int_0^\infty t g(t)^2 dt$.

Step 3. Let G_1, G_2 be independent copies of G and $Y = G_1/(G_1 + G_2)$. The map $(G_1, G_2) \mapsto (Y, G_1 + G_2)$ has inverse $(y, r) \mapsto (yr, (1 - y)r)$ with Jacobian r . So Y has the density $f(y) = \int_0^\infty r g(yr) g((1 - y)r) dr$ on $(0, 1)$, and $f(\frac{1}{2}) = \int_0^\infty r g(r/2)^2 dr = 4 \int_0^\infty t g(t)^2 dt$. Hence $nC_n \rightarrow f(\frac{1}{2})$. Note that f is continuous at $\frac{1}{2}$. Indeed, for $y \in [\frac{1}{4}, \frac{3}{4}]$ both yr and $(1 - y)r$ are at least $r/4$, so by the tail in Lemma 4.3(v) the integrand is at most $r \min(\|g\|_\infty^2, Cr^{-2-2\alpha})$. This is integrable, and dominated convergence applies.

We now identify the law of Y . It does not change if G is multiplied by a constant, so we may take $\mathbb{E}e^{-sG} = e^{-s^\alpha}$. For $x > 0$ and $0 < p < 1$,

$$x^p = \frac{p}{\Gamma(1 - p)} \int_0^\infty s^{-p-1} (1 - e^{-sx}) ds, \quad x^{-p} = \frac{1}{\Gamma(p)} \int_0^\infty s^{p-1} e^{-sx} ds.$$

Taking expectations and substituting $w = s^\alpha$ gives $\mathbb{E}G^p = \Gamma(1 - p/\alpha)/\Gamma(1 - p)$ for $|p| < \alpha$ (see also [17]). Let $W = G_1/G_2$, so that $Y = W/(1 + W)$. By independence and $\Gamma(1 - z)\Gamma(1 + z) = \pi z / \sin(\pi z)$,

$$\mathbb{E}W^p = \frac{\Gamma(1 - \frac{p}{\alpha})\Gamma(1 + \frac{p}{\alpha})}{\Gamma(1 - p)\Gamma(1 + p)} = \frac{\sin \pi p}{\alpha \sin(\pi p/\alpha)}, \quad |p| < \alpha.$$

Let $\rho_\alpha(w) = \frac{\sin \pi \alpha}{\pi} \frac{w^{\alpha-1}}{w^{2\alpha} + 2w^\alpha \cos \pi \alpha + 1}$ for $w > 0$. We use the classical integral

$$\int_0^\infty \frac{v^\beta dv}{v^2 + 2v \cos \theta + 1} = \frac{\pi \sin \beta \theta}{\sin \theta \sin \pi \beta}, \quad |\beta| < 1, \quad 0 < \theta < \pi,$$

read as $\theta/\sin \theta$ at $\beta = 0$. For $-1 < \beta < 0$ it follows from partial fractions and $\int_0^\infty v^\beta (v + e^{\pm i\theta})^{-1} dv = -\pi e^{\pm i\beta\theta} / \sin(\pi\beta)$, and the substitution $v \mapsto 1/v$ shows that both sides are even in β . With $v = w^\alpha$, $\beta = p/\alpha$ and $\theta = \pi\alpha$ it gives $\int_0^\infty w^p \rho_\alpha(w) dw = \mathbb{E}W^p$ for $|p| < \alpha$. At $p = 0$ this shows that ρ_α is a probability density. So $\log W$ and the logarithm of a variable with density ρ_α have the same moment generating function near 0, and W has density ρ_α . Then Y has density $\rho_\alpha(\frac{y}{1-y})(1-y)^{-2} = \ell_\alpha(y)$. Both f and ℓ_α are densities of Y that are continuous at $\frac{1}{2}$, so

$$f(\frac{1}{2}) = \ell_\alpha(\frac{1}{2}) = \frac{2 \sin \pi \alpha}{\pi(1 + \cos \pi \alpha)} = \frac{2}{\pi} \tan \frac{\pi \alpha}{2}.$$

Dividing $E_n = \frac{1}{2}n^{-\alpha}$ by C_n gives the formula for R_n .

Step 4. Put $c = \cos \frac{\pi \alpha}{2} > 0$, so that $1 + \cos \pi \alpha = 2c^2$ and $\sin(\pi \alpha)/(\pi c^2) = \frac{2}{\pi} \tan \frac{\pi \alpha}{2}$. For $y = \frac{1}{2} + \eta$ the binomial series give $(y(1 - y))^{\alpha-1} = 4^{1-\alpha}(1 - 4(\alpha - 1)\eta^2 + O(\eta^4))$ and

$$y^{2\alpha} + 2y^\alpha(1 - y)^\alpha \cos \pi \alpha + (1 - y)^{2\alpha} = 4^{1-\alpha}c^2 \left(1 + \frac{4\alpha(\alpha - c^2)}{c^2} \eta^2 + O(\eta^4) \right).$$

Hence

$$\ell_\alpha(\tfrac{1}{2} + \eta) = \frac{2}{\pi} \tan \frac{\pi\alpha}{2} \left[1 + \frac{4(\cos^2 \frac{\pi\alpha}{2} - \alpha^2)}{\cos^2 \frac{\pi\alpha}{2}} \eta^2 + O(\eta^4) \right].$$

The function $\alpha \mapsto \cos \frac{\pi\alpha}{2} - \alpha$ decreases strictly on $(0, 1)$ from 1 to -1 and vanishes at α_c . So the coefficient of η^2 is positive for $\alpha < \alpha_c$ and negative for $\alpha > \alpha_c$, and $y = \frac{1}{2}$ is a strict local minimum of ℓ_α in the first case and a strict local maximum in the second. This agrees with [10, eq. (7.6)]. $\square$

D Proof of the aging crossing

We use the setting of Remark 4.11. So $0 < c < 2$, X_1 is a fair sign, $\mathbb{P}_+$ is the law given $X_1 = +1$, and the switch indicators $\xi_k = \mathbf{1}\{X_k \neq X_{k-1}\}$, $2 \leq k \leq n$, are independent Bernoulli(c/k) variables, independent of X_1 . Put

$$\rho(a, b) = \prod_{k=a+1}^b \left(1 - \frac{c}{k}\right) = \frac{\Gamma(b+1-c)\Gamma(a+1)}{\Gamma(b+1)\Gamma(a+1-c)}, \quad \omega_t = \frac{c}{t-c}, \quad \lambda_{t,n} = \frac{c}{t} \rho(t, n),$$

so $\lambda_{t,n}$ is the probability of a switch at time t and none in $(t, n]$. Let T_s^+ be the time of the s -th step equal to $+1$. Let B be a Beta(c, c) variable, $d_c = 4^{1-c}/B(c, c)$ its density at $\frac{1}{2}$, and $\Theta(c) = \log(2\Gamma(2-c)d_c)$. For even n let $h = n/2$, $C_n = \mathbb{P}(A_n = h)$ and $E_n = \mathbb{P}(A_n = n)$. Changing the sign of every step preserves the law of the walk and sends A_n to $n - A_n$. We call this flip symmetry. Gautschi's inequality [21] says that $x^{1-s} < \Gamma(x+1)/\Gamma(x+s) < (x+1)^{1-s}$ for $x > 0$ and $0 < s < 1$. We apply it with $x = y$ and $s = 1 - c$ if $c < 1$. If $c > 1$ we apply it with $s = 2 - c$ and use $\Gamma(y+1-c) = \Gamma(y+2-c)/(y+1-c)$ and $1 \leq y/(y+1-c) \leq 1 + 2/y$. This gives

$$\frac{\Gamma(y+1-c)}{\Gamma(y+1)} = y^{-c}(1+\theta) \quad \text{with} \quad -\frac{1}{y} \leq \theta \leq \frac{2}{y} \quad (y \geq 2), \quad (10)$$

and $\theta = 0$ if $c = 1$.

D.1 Path weights and the exact law at $c = 1$

Proposition D.1 (path weights). *Let $\mathcal{T}(x) = \{2 \leq t \leq n : x_t \neq x_{t-1}\}$ be the set of switch times of $x \in \{\pm 1\}^n$. Then*

$$\mathbb{P}(X = x) = \frac{1}{2} \rho(1, n) \prod_{t \in \mathcal{T}(x)} \omega_t. \quad (11)$$

So $E_n = \frac{1}{2} \rho(1, n)$. For $1 \leq s \leq n-1$ the ratios $\mathbb{P}(A_n = s)/E_n$ and $\mathbb{P}_+(A_n = s)/\mathbb{P}_+(A_n = n)$ are continuous and strictly increasing in $c \in (0, 2)$, and they tend to 0 as $c \downarrow 0$.

Proof. The first step contributes the factor $\frac{1}{2}$, and step $k \geq 2$ contributes c/k if $k \in \mathcal{T}(x)$ and $1 - c/k$ otherwise. Factoring out $\rho(1, n)$ leaves $(c/t)/(1 - c/t) = \omega_t$ for each switch. Only the path with every step $+1$ has $A_n = n$, and it has no switch. So $\mathbb{P}(A_n = s)/E_n$ is the sum of $\prod_{t \in \mathcal{T}(x)} \omega_t$ over the paths x with s steps equal to $+1$. For $1 \leq s \leq n-1$ each of these paths switches at least once, and each ω_t is positive, strictly increasing in c and tends to 0 as $c \downarrow 0$. Under $\mathbb{P}_+$ the factor $\frac{1}{2}$ disappears and only paths with $x_1 = +1$ occur. $\square$

Proposition D.2 (exact law at $c = 1$). *Let $c = 1$, $n \geq 2$ and $1 \leq s \leq n$. Then $\mathbb{P}_+(A_n = s, X_n = +1) = \frac{s-1}{n(n-1)}$ and $\mathbb{P}_+(A_n = s, X_n = -1) = \frac{n-s}{n(n-1)}$, so $\mathbb{P}_+(A_n = s) = \frac{1}{n}$. With a fair first step, $\mathbb{P}(A_n = s) = \frac{1}{n}$ for $1 \leq s \leq n-1$ and $E_n = \frac{1}{2n}$.*

Proof. Let $u_n^\pm(s) = \mathbb{P}_+(A_n = s, X_n = \pm 1)$, which is 0 unless $1 \leq s \leq n$. At $c = 1$ step $n + 1$ switches with probability $1/(n + 1)$, so $(n + 1)u_{n+1}^+(s + 1) = n u_n^+(s) + u_n^-(s)$ and $(n + 1)u_{n+1}^-(s) = u_n^+(s) + n u_n^-(s)$ for every s . We use induction on n . For $n = 2$ we have $u_2^+(2) = u_2^-(1) = \frac{1}{2}$ and the other values vanish, as claimed. If the claim holds for n , the identities $n(s - 1) + (n - s) = s(n - 1)$ and $(s - 1) + n(n - s) = (n - 1)(n + 1 - s)$ give it for $n + 1$. Summing over X_n gives $\mathbb{P}_+(A_n = s) = \frac{1}{n}$. By flip symmetry $\mathbb{P}(A_n = s) = \frac{1}{2}(\mathbb{P}_+(A_n = s) + \mathbb{P}_+(A_n = n - s))$. $\square$

Given $X_1 = +1$ and $c = 1$, the steps $X_2, \dots, X_n$ start with a fair sign and then switch at their own step $j \geq 2$ with probability $1/(j + 1)$. So after a shift of one step the uniform law above is the exercise in footnote 1 of [25]. Read the other way, it says that a walk of N steps with a fair first step that switches at step $k \geq 2$ with probability $1/(k + 1)$ has exactly $N + 1$ equally likely values $0, \dots, N$ for its number of $+1$ steps.

Corollary D.3 (crossings). *Let $n \geq 2$ and $1 \leq s \leq n - 1$. (a) The ratio $\mathbb{P}_+(A_n = s)/\mathbb{P}_+(A_n = n)$ is strictly increasing in c and equals 1 at $c = 1$. So given $X_1 = +1$, bin n is more likely than each of the bins $1, \dots, n - 1$ when $c < 1$ and less likely when $c > 1$, while bin 0 has probability 0. (b) With a fair first step there is exactly one $c_n(s) \in (0, 2)$ with $\mathbb{P}(A_n = s) = E_n$, and $c_n(s) < 1$. For even n the crossing of Remark 4.11 is $c_n = c_n(h)$.*

Proof. Part (a) follows from Propositions D.1 and D.2. In (b) the ratio $\mathbb{P}(A_n = s)/E_n$ is continuous and strictly increasing, tends to 0 as $c \downarrow 0$, and equals 2 at $c = 1$. So it equals 1 exactly once, at some $c < 1$. $\square$

D.2 Averaging at the center and a bracket

Lemma D.4 (averaging at the center). *Let $n \geq 2$ be even and $T = T_h^+$. Then*

$$C_n = 2 \sum_{r=h}^{n-1} \mathbb{P}(T = r) \lambda_{r+1,n} \quad \text{and} \quad \sum_{r=h}^{n-1} \mathbb{P}(T = r) = \frac{1}{2}. \quad (12)$$

If $0 < c \leq 1$, then $c/n \leq C_n \leq \lambda_{h+1,n}$.

Proof. On the event $\{A_n = h, X_n = -1\}$ the last step equal to $+1$ is the h -th one, at a time $T = r \leq n - 1$, and it is followed by a switch at $r + 1$ and no switch in $(r + 1, n]$. Conversely, these conditions give $A_n = h$ and $X_n = -1$. As $\{T = r\}$ depends only on $X_1, \xi_2, \dots, \xi_r$, we get $\mathbb{P}(A_n = h, X_n = -1) = \sum_{r=h}^{n-1} \mathbb{P}(T = r) \lambda_{r+1,n}$, and flip symmetry gives the same value for $X_n = +1$. Next, $T \leq n - 1$ exactly when $A_{n-1} \geq h$, which has probability $\frac{1}{2}$ because $n - 1$ is odd and A_{n-1} is symmetric about $(n - 1)/2$. Finally $\lambda_{t+1,n}/\lambda_{t,n} = t/(t + 1 - c) \leq 1$ for $c \leq 1$. So $\lambda_{n,n} = c/n \leq \lambda_{t,n} \leq \lambda_{h+1,n}$ for $h + 1 \leq t \leq n$, and by (12) C_n is a weighted average of these values. $\square$

For $N \geq 2$ put $\eta_N(c) = N\rho(1, N)/(2c)$. Since $\rho(1, N)$ decreases in c and equals $1/N$ at $c = 1$, η_N decreases strictly on $(0, 1]$, from ∞ at 0 to $\eta_N(1) = \frac{1}{2}$. Let $c^{(N)} \in (0, 1)$ be the root of $\eta_N = 1$.

Theorem D.5 (bracket). *The numbers $c^{(N)}$ increase with N , and $1 - c^{(N)} = \frac{\log 2}{\log N + 1 + \gamma} + O((\log N)^{-3})$. For every even $n \geq 2$ we have $c^{(h+1)} \leq c_n \leq c^{(n)}$. In particular $(1 - c_n) \log n \rightarrow \log 2$, and $c_n \geq c^{(3)} = (9 - \sqrt{57})/2 > 0.725$ for $n \geq 4$.*

Proof. For $c < 1$ we have $\eta_{N+1}/\eta_N = (N+1-c)/N > 1$. So $\eta_{N+1}(c^{(N)}) > 1$ and $c^{(N+1)} > c^{(N)}$. By Corollary D.3(b) it is enough to compare C_n with E_n for $0 < c < 1$. If $c > c^{(n)}$, then $\eta_n(c) < 1$, that is $E_n = \frac{1}{2}\rho(1, n) < c/n \leq C_n$ by Lemma D.4. So $c_n \leq c^{(n)}$. If $c < c^{(h+1)}$, then $\eta_{h+1}(c) > 1$, and with $\rho(1, n) = \rho(1, h+1)\rho(h+1, n)$ this gives $E_n > \frac{c}{h+1}\rho(h+1, n) = \lambda_{h+1, n} \geq C_n$. So $c_n \geq c^{(h+1)}$.

By (10) with $y = N$, the equation $\eta_N(c) = 1$ reads $(1-c)\log N = \Theta_0(c) + O(1/N)$ with $\Theta_0(c) = \log(2c\Gamma(2-c))$, uniformly for $0 < c \leq 1$. For $N \geq 3$ we have $c^{(N)} \in [c^{(3)}, 1)$, where Θ_0 is bounded. So $1 - c^{(N)} = O(1/\log N)$, and we may expand at $c = 1$. With $\Theta_0(1) = \log 2$, $\Theta'_0(1) = 1 - \psi(1) = 1 + \gamma$ and $e = 1 - c^{(N)}$, Taylor's theorem gives $e(\log N + 1 + \gamma) = \log 2 + O(e^2) + O(1/N)$, which is the expansion. The limit of $(1 - c_n)\log n$ follows since $\log(h+1) \sim \log n$. Finally $c^{(3)}$ solves $3\rho(1, 3) = (2-c)(3-c)/2 = 2c$. $\square$

The reason for the log 2 is already visible at $c = 1$. There every interior bin is $1/n$ and the end is $1/(2n)$. The factor 2 comes from the fair start under switching probability c/k , since only the start +1 can reach bin n . With switching probability $c/(k+1)$ and a fair start, the end and the interior bins are equal at $c = 1$ by the paragraph after Proposition D.2. For c a little below 1 the end is of order n^{-c} while the center stays of order $1/n$. So the 2 are equal when n^{1-c} is about 2.

D.3 The continuum occupation and a local limit at the center

Proposition D.6 (continuum occupation). *Let Π be a Poisson process on $(0, \infty)$ with intensity $c dt/t$, and let $Y(1)$ be a fair sign independent of Π . Put $Y(t) = Y(1)(-1)^{\Pi((1, t])}$ for $t \geq 1$ and $Y(t) = Y(1)(-1)^{\Pi((t, 1])}$ for $0 < t < 1$. Then for every $r > 0$ the fraction $Z_r = \frac{1}{r} \int_0^r \mathbf{1}\{Y(t) = +1\} dt$ has exactly the Beta(c, c) law.*

This agrees with [25, Remark 6.2], which obtains the Beta(c, c) one-dimensional marginals of the zigzag process of [25, Definition 6.1] from a scaling limit and mentions a direct proof by E. Crane that is not written there. We give a direct proof.

Proof. The points $\log t, t \in \Pi$, form a Poisson process $\tilde{\Pi}$ of rate c on $\mathbb{R}$, and $W(u) = Y(e^u)$ satisfies $W(v) = W(u)(-1)^{\tilde{\Pi}((u, v])}$ for $u < v$. For each a , $W(a)$ is $W(0)$ times a function of $\tilde{\Pi}$, so it is a fair sign independent of $\tilde{\Pi}$. As $\tilde{\Pi} - a$ has the law of $\tilde{\Pi}$, the pair $(W(a), \tilde{\Pi} - a)$ has the law of $(W(0), \tilde{\Pi})$, and so $W(a + \cdot)$ has the law of W . Hence $Z_r = \int_0^1 \mathbf{1}\{Y(rt) = +1\} dt$ has the law of $Z_1 = \int_0^\infty e^{-s} \mathbf{1}\{V_s = +1\} ds$, where $V_s = W(-s)$ starts at a fair sign and switches at rate c . Let $\mu_\pm$ be the law of Z_1 given $V_0 = \pm 1$, let τ be the first switch of V , and let $U = e^{-\tau}$. Then $\mathbb{P}(U \leq u) = u^c$, so U is Beta($c, 1$). After τ the process starts afresh from $-V_0$, independently of τ . Splitting the integral at τ shows that (μ_+, μ_-) is a fixed point of the map Ψ that sends a pair (ν_+, ν_-) of laws on $[0, 1]$ to the laws of $1 - U + UZ_-$ and UZ_+ , where $Z_\pm \sim \nu_\pm$ are independent of U . Coupling $Z_\pm$ optimally with $Z'_\pm \sim \nu'_\pm$, independently of U , gives $\mathbb{E}|UZ_\pm - UZ'_\pm| = \mathbb{E}U \mathbb{E}|Z_\pm - Z'_\pm|$. So Ψ contracts the larger of the 2 Wasserstein distances by the factor $\mathbb{E}U = c/(c+1)$, and it has only one fixed point.

Let G_a, G_b, G_d be independent Gamma variables with shapes a, b, d and unit scale. A change of variables shows that $G_a/(G_a + G_b)$ is Beta(a, b) and independent of $G_a + G_b$, hence of the pair $(G_a + G_b, G_d)$ and of $(G_a + G_b)/(G_a + G_b + G_d)$, which is Beta($a + b, d$). Their product $G_a/(G_a + G_b + G_d)$ is Beta($a, b + d$). With $(a, b, d) = (c, 1, c)$ this says that UZ is Beta($c, c + 1$) whenever Z is Beta($c + 1, c$) and independent of U . Applying it to $Z = Z_+$ and to $Z = 1 - Z_-$, and using $1 - U + UZ_- = 1 - U(1 - Z_-)$, we see that Ψ maps the pair Beta($c + 1, c$), Beta($c, c + 1$) to itself. So Z_1 is the equal mixture of these 2 laws. As $B(c + 1, c) = B(c, c + 1) = \frac{1}{2}B(c, c)$, the mixture has density $x^{c-1}(1-x)^{c-1}(x+1-x)/B(c, c)$, which is the Beta(c, c) density. $\square$

Lemma D.7 (coupling). *Let $J \subset (0, 2)$ be compact and $\kappa_J = \min_{c \in J} \frac{\min(c, 1)}{1 + \min(c, 1)}$. There is C_J such that for $c \in J$ and every integer $r \geq 2$, $\sup_{y \in \mathbb{R}} |\mathbb{P}(A_r \geq y) - \mathbb{P}(rB \geq y)| \leq C_J (\log r) r^{-\kappa_J}$.*

Proof. Take Π and Y as in Proposition D.6 and an integer $1 \leq K \leq r$. The variables $\xi'_k = \mathbf{1}\{\Pi((k-1, k]) \text{ is odd}\}$, $k > K$, are independent of each other and of $Y(K)$, and $\mathbb{P}(\xi'_k = 1) = p'_k = \frac{1}{2}(1 - (1 - 1/k)^{2c})$. By Taylor's theorem $|p'_k - c/k| = \frac{1}{2}c|2c - 1|(1 - \vartheta)^{2c-2}k^{-2}$ for some $0 < \vartheta < 1/k$. This is at most $3k^{-2}$. Indeed $\frac{1}{2}c|2c - 1| < 3$ and the power of $1 - \vartheta$ is at most 1 if $c \geq 1$, while for $c < 1$ the power is at most 4 and $\frac{1}{2}c|2c - 1| \leq \frac{1}{2}$. For $K < k \leq r$ we couple $\xi_k \sim \text{Bernoulli}(c/k)$ with ξ'_k so that $\mathbb{P}(\xi_k \neq \xi'_k) = |p'_k - c/k|$, using independent extra randomness. We put $X_K = Y(K)$ and $X_k = X_{k-1}(-1)^{\xi_k}$ for $K < k \leq r$, and we draw $X_1, \dots, X_{K-1}$ from the law of the walk given X_K , again with independent randomness. In the walk X_K is a fair sign and $(\xi_k)_{k > K}$ is independent of $(X_1, \dots, X_K)$, so this is the law of the walk.

On the event $\mathcal{A}$ that $\xi_k = \xi'_k$ for $K < k \leq r$ we have $X_k = Y(k)$ for $K \leq k \leq r$, and $\mathbb{P}(\mathcal{A}^c) \leq \sum_{k > K} 3k^{-2} \leq 3/K$. If also $\Pi((k-1, k]) = 0$, then $Y = Y(k)$ on $(k-1, k]$, so step k adds the same amount to A_r as the interval $(k-1, k]$ adds to rZ_r . The other steps $k \leq r$ number at most $K + \Pi((K, r])$, and each changes the difference by at most 1. Hence $|A_r - rZ_r| \leq K + \Pi((K, r])$ on $\mathcal{A}$. The variable $\Pi((K, r])$ is Poisson with mean $c \log(r/K) \leq 2 \log r$, so it exceeds K with probability at most $2 \log r / K$. With $b_K = (3 + 2 \log r)/K$ we get $\mathbb{P}(rZ_r \geq y + 2K) - b_K \leq \mathbb{P}(A_r \geq y) \leq \mathbb{P}(rZ_r \geq y - 2K) + b_K$, and rZ_r has the law of rB . So the left side of the lemma is at most $b_K + \sup_x \mathbb{P}(x \leq B \leq x + 2K/r)$.

For $c \leq 1$ the density of B is at most $2^{1-c}(x^{c-1} + (1-x)^{c-1})/B(c, c)$, and x^{c-1} has integral at most δ^c/c over an interval of length δ . For $c \geq 1$ the density is at most d_c . So with $a_J = \min(\min J, 1)$, and as the mass is also at most 1, an interval of length δ has mass at most $C'_J \delta^{a_J}$ for all $\delta > 0$ and $c \in J$. Take $K = \lceil r^{\kappa_J} \rceil$. Then $b_K \leq (3 + 2 \log r)r^{-\kappa_J}$, $2K/r \leq 4r^{\kappa_J-1}$, and $(r^{\kappa_J-1})^{a_J} = r^{-\kappa_J}$ because $\kappa_J = a_J/(1 + a_J)$. $\square$

Theorem D.8 (local limit at the center). *For each compact $J \subset (0, 2)$ there is C_J such that $|nC_n - d_c| \leq C_J (\log n) n^{-\kappa_J}$ for $c \in J$ and even $n \geq 4$.*

Proof. Let $T = T_h^+$, $F(r) = \mathbb{P}(T \leq r) = \mathbb{P}(A_r \geq h)$, $\Phi(r) = \mathbb{P}(rB \geq h)$ and $\Lambda_r = n\lambda_{r+1, n}$. Then $F(h-1) = \Phi(h-1) = \Phi(h) = 0$, and Lemma D.7 gives $|F(r) - \Phi(r)| \leq \delta_n := C_J (\log n) n^{-\kappa_J}$ for $h \leq r \leq n-1$, after enlarging C_J . By Lemma D.4, $nC_n = 2 \sum_{r=h}^{n-1} (F(r) - F(r-1))\Lambda_r$. The ratio $\lambda_{t+1, n}/\lambda_{t, n} = t/(t+1-c)$ shows that Λ_r is monotone in r . Also $\Lambda_{n-1} = c$ and $0 < \Lambda_h < 2c$, for $c \leq 1$ because $\rho \leq 1$ and $n < 2(h+1)$, and for $c > 1$ because Λ_r increases. So summation by parts shows that replacing F by Φ changes nC_n by at most $2\delta_n(\Lambda_{n-1} + |\Lambda_{n-1} - \Lambda_h|) \leq 4c\delta_n$. For $h < r \leq n-1$ we have $\Phi(r) - \Phi(r-1) = \mathbb{P}(h/r \leq B < h/(r-1))$, and there $\lceil h/B \rceil = r$. So the new main term is $2\mathbb{E}[\Lambda_{\lceil h/B \rceil}; B \geq h/(n-1)]$.

By (10) with $y = n$ and $y = r+1$, $\Lambda_r = c((r+1)/n)^{c-1}(1 + O(1/n))$ for $h \leq r \leq n-1$. On the event above $(r+1)/n$ lies in $[\frac{1}{2}, 2]$ and within $2/n$ of $1/(2B)$, and x^{c-1} is 4-Lipschitz on $[\frac{1}{2}, 2]$. So $\Lambda_{\lceil h/B \rceil} = c(2B)^{1-c} + O(1/n)$. Also $\mathbb{P}(\frac{1}{2} \leq B < h/(n-1)) = O_J(1/n)$, since the density of B is bounded near $\frac{1}{2}$ uniformly on J . So nC_n is within $4c\delta_n$ of

$$2c\mathbb{E}[(2B)^{1-c}; B \geq \frac{1}{2}] + O_J(1/n) = \frac{2c2^{1-c}}{B(c, c)} \int_{1/2}^1 (1-b)^{c-1} db + O_J(1/n) = d_c + O_J(1/n). \quad \square$$

D.4 The expansion of the crossing

Theorem D.5 gives the first term of the expansion in Remark 4.11. The constant in M needs the center to a relative error $o(1/\log n)$, which Theorem D.8 provides.

Theorem D.9 (the aging crossing). *For even n , with $M = \log n + \Theta'(1) = \log n + \gamma + 2 - 2\log 2$,*

$$c_n = 1 - \frac{\log 2}{M} + O(M^{-3}).$$

Proof. Let $J = [c^{(3)}, 1]$. Then $\kappa_J = c^{(3)}/(1+c^{(3)}) > 0.42$, and d_c is bounded below on J by a positive constant. By Proposition D.1 and (10) with $y = n$, $\log E_n = -\log 2 - \log \Gamma(2-c) - c \log n + O(1/n)$. By Theorem D.8, $\log C_n = \log d_c - \log n + O(n^{-0.4})$ for large n . Both hold uniformly for $c \in J$, so

$$\log C_n - \log E_n = (c-1) \log n + \Theta(c) + O(n^{-0.4}) \quad (c \in J). \quad (13)$$

Here $\Theta(1) = \log 2$ because $d_1 = 1$, and $\Theta'(c) = -\psi(2-c) - \log 4 - 2\psi(c) + 2\psi(2c)$ with $\psi(1) = -\gamma$ and $\psi(2) = 1 - \gamma$ gives $\Theta'(1) = \gamma + 2 - 2\log 2$. By Theorem D.5, $c_n \in J$ for $n \geq 4$ and $e = 1 - c_n = O(1/\log n)$. The left side of (13) vanishes at $c = c_n$. As Θ is smooth on J , Taylor's theorem gives $eM = \log 2 + O(e^2) + O(n^{-0.4}) = \log 2 + O(M^{-2})$. So $e = \log 2/M + O(M^{-3})$. $\square$

With one more term in Taylor's theorem and $e^2 = (\log 2)^2/M^2 + O(M^{-4})$, the same argument adds the term $-\Theta''(1)(\log 2)^2/(2M^3)$ to the expansion, with error $O(M^{-4})$. Here $\Theta''(1) = \psi'(1) - 2\psi'(1) + 4\psi'(2) = \pi^2/2 - 4$, since $\Theta''(c) = \psi'(2-c) - 2\psi'(c) + 4\psi'(2c)$, $\psi'(1) = \zeta(2)$ and $\psi'(2) = \zeta(2) - 1$. At $n = 200$ the bracket of Theorem D.5 is $[0.88749, 0.89878]$, Theorem D.9 gives 0.89319, and with the M^{-3} term we get 0.89236. The high-precision value is $c_{200} = 0.89253$ (Section 5). The convergence is slow, since $(1 - c_{200}) \log 200 = 0.569$ while the limit is $\log 2 = 0.693$. The constants C_J and the constants in the O -terms of this appendix are not computed.

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